IV Robustness Value Project

Beniamino Green

Table of contents

Introduction 2
Analysis of IV Papers 2
Corpus 2
Distribution of Robustness Values: 2
Associations with the Robustness Value: 3
Contour Plots 4
IV contour Plots 6
Identifying Predictors of IV Robustness Value via Simulation 9
Bounding the RV_{IV} With the first-stage F Statistic 12
Assuming the Exclusion Restriction Holds: 12
Appendix 14
Table of Study Point Estimates 14
Contour Figures 17
IV Contour Plots 52
Bibliography 120

Introduction

This report is a summary of the work I have done so far to characterize the distribution of IV robustness values across empirical papers in Political Science, and to understand the behavior of the robustness values via simulation. The `ivrobustness` package developed for this project can be found here. The repository for the analytics, including the source for this report can be found here

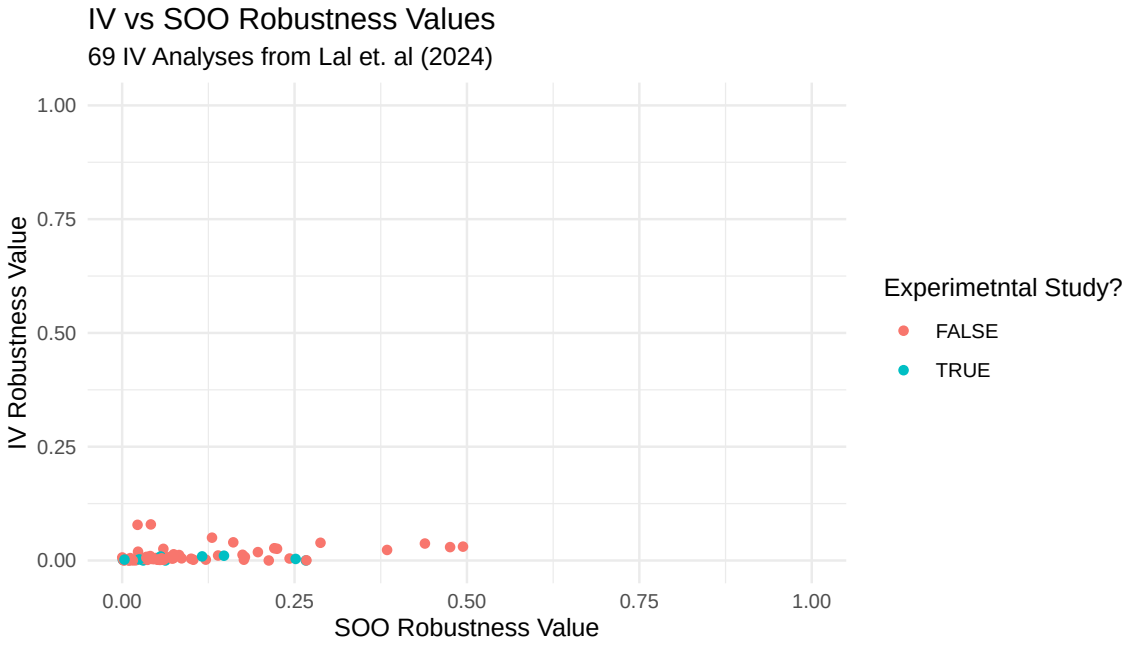
Analysis of IV Papers

Corpus

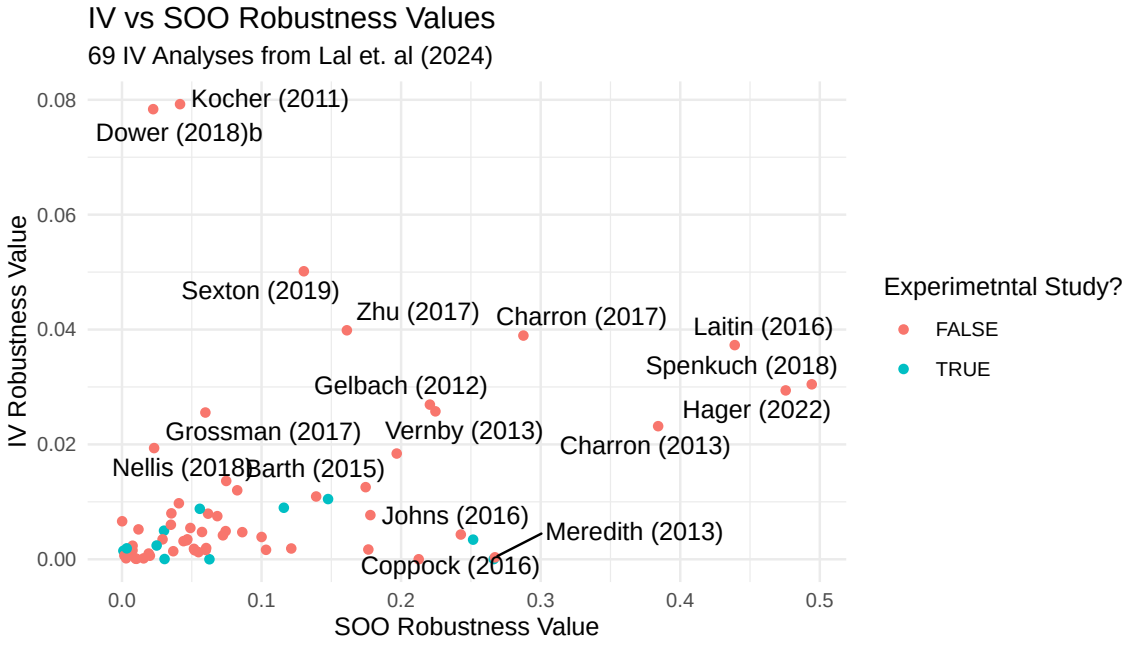
I started with the corpus of studies collected by A. Lal, M. Lockhart, Y. Xu, and Z. Zu [1], which contains 67 instrumental variable studies. Using their replication materials, I extracted every standardized dataset from the replication materials, yielding 69 instrument, treatment, and outcome triples. This number is larger than the headline 67 number of studies because some studies had multiple IV analyses. Some studies had specifications with a single instrument. These were dropped from the analysis.

Distribution of Robustness Values:

I begin by examining the distributions of the IV and SOO robustness values. A scatterplot of the SOO and IV robustness values can be seen in the figure below. Each datapoint represents a study. The position on the x-axis corresponds to the SOO robustness value for that study, while the position on the Y corresponds to it's IV robustness values. For all experimental studies, I calculate the robustness values while imposing exogeneity, as we know that the instrument is exogenous in this context.

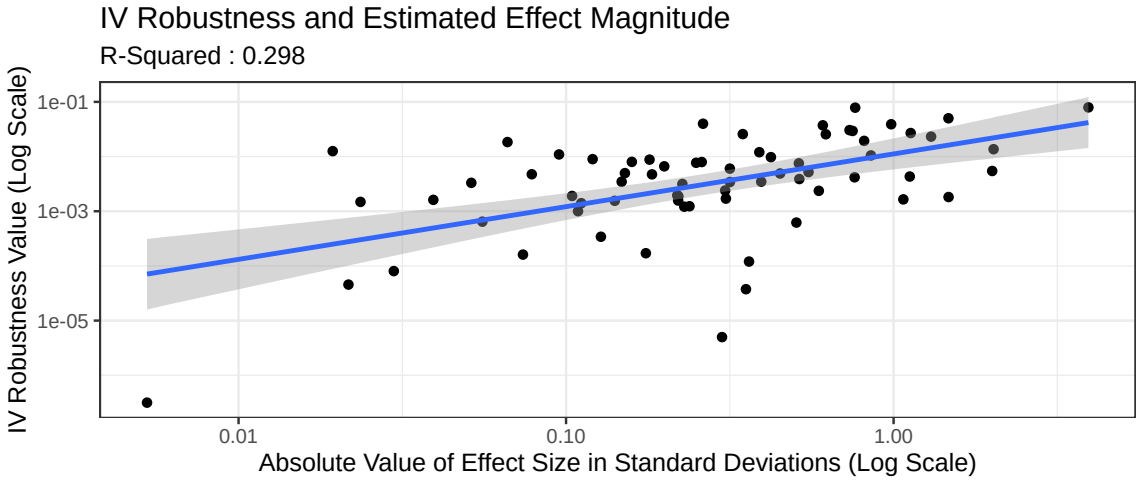


Almost all of the points are compressed along the x-axis, suggesting that the Robustness values for IV studies is very low. The SOO values are almost uniformly higher than the IV values; there are only four studies in which the SOO Robustness value exceeds the IV Robustness value. These four studies are also some of the least robust; all robustness values for these studies are less than .1. The figure below displays the same information but scales up variation on the Y axis such that points can be more easily distinguished.

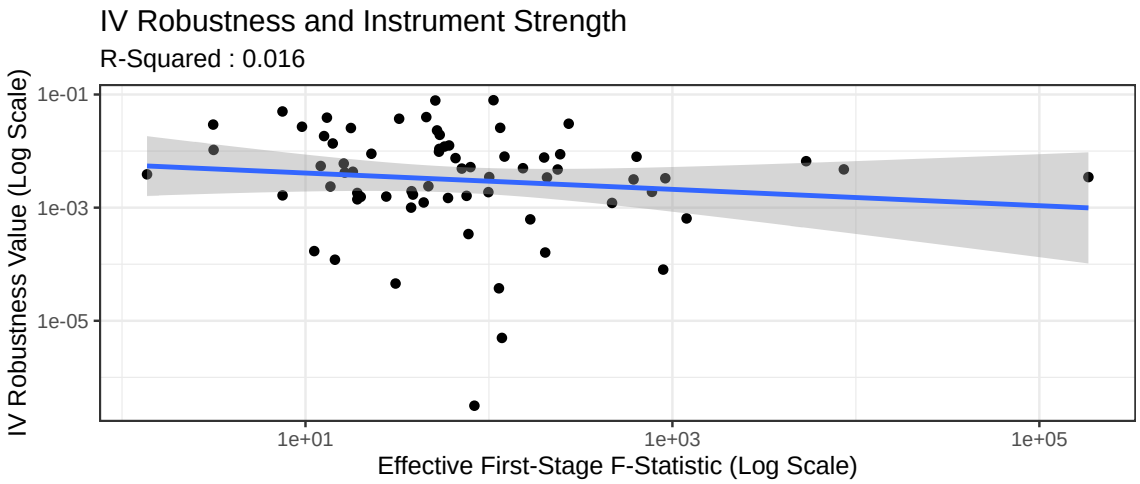


Associations with the Robustness Value:

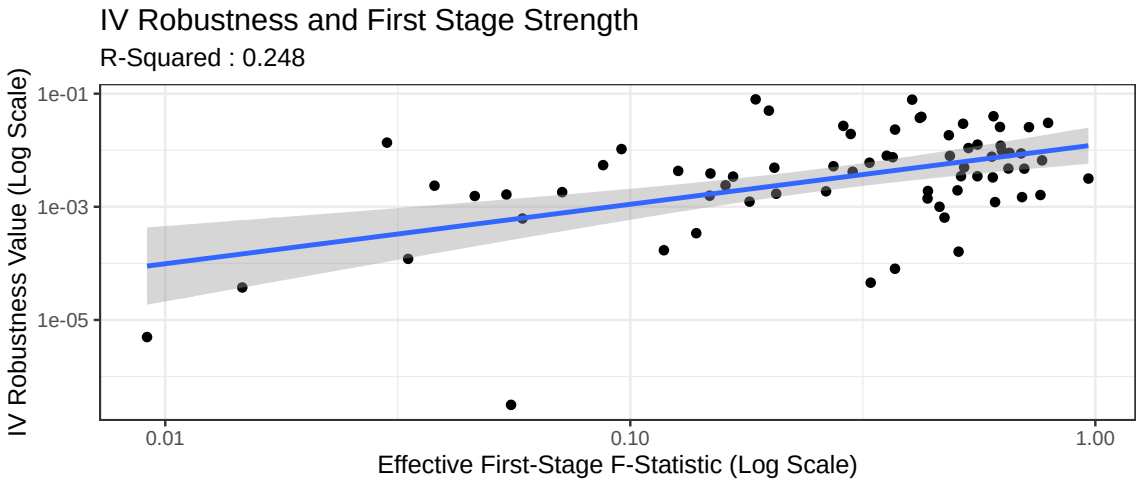
I then plot the distribution of IV Robustness Values against the effect size estimated by 2SLS. There appears to be some relationship; stronger effects are generally associated with higher robustness values.



I investigate whether the first-stage F-statistic is associated with a more robust analysis, but find little relationship between it and the Robustness Value:



This is likely because the F-statistic is a function of both the first-stage strength and the sample size, so a small first-stage will result in a large F-statistic as the sample size increases. Accordingly, I also plot the RV against the first-stage coefficient, finding a significant relationship:

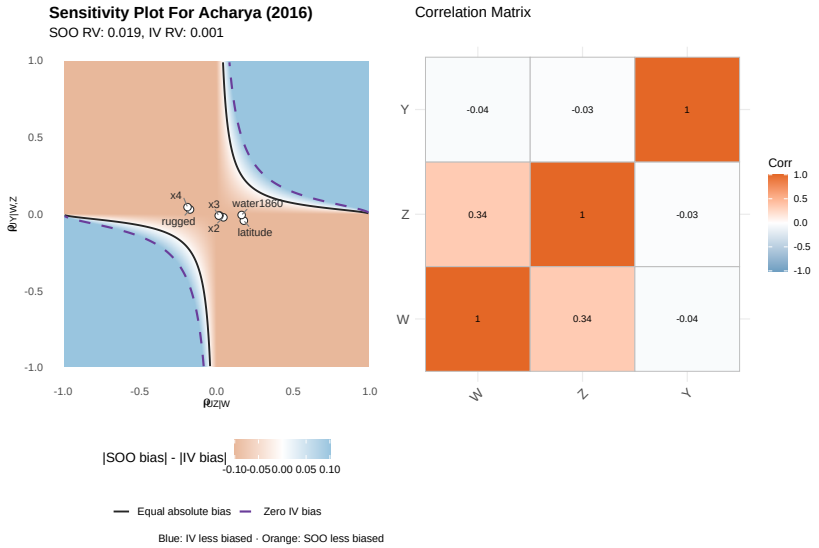


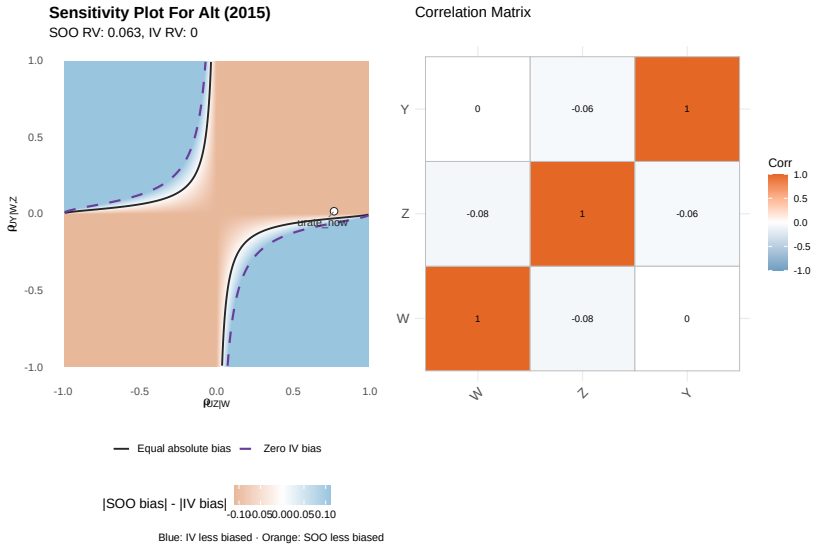
Contour Plots

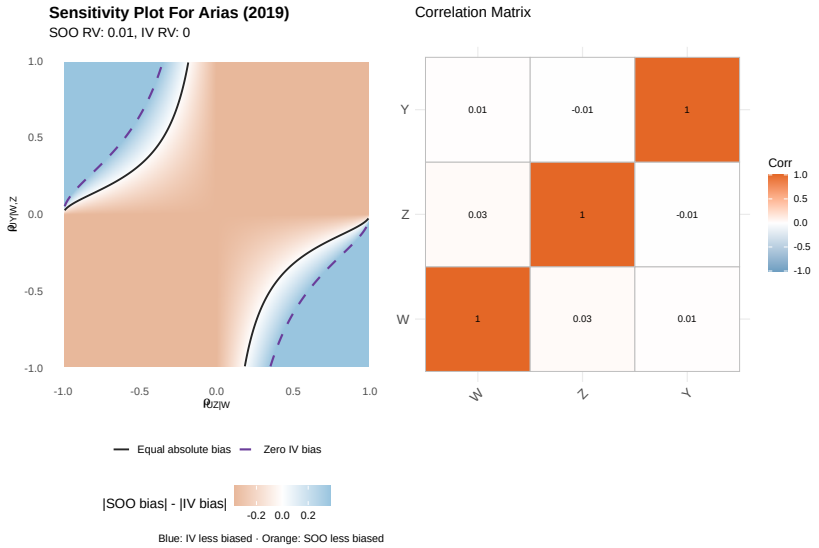
Next, I created contour plots as described in section 5.2 of M. Huang and C. McCartan [2], which plot contours of the relative bias of SOO vs IV for different levels of violations in the SOO identifying assumptions. These plots also allow us to judge how extreme a confounder would have to be relative to the controls included in the regression specification. I plot the first 3 of these contour plots below, all 69 plots can be found in the appendix.

The controls included in the plots are almost always well within the "equal absolute bias" contour lines, suggesting that in order for IV to have less bias than SOO in most settings, we would have to

believe that there is a confounder that violates the SOO assumptions significantly more severely than any of the control variables.

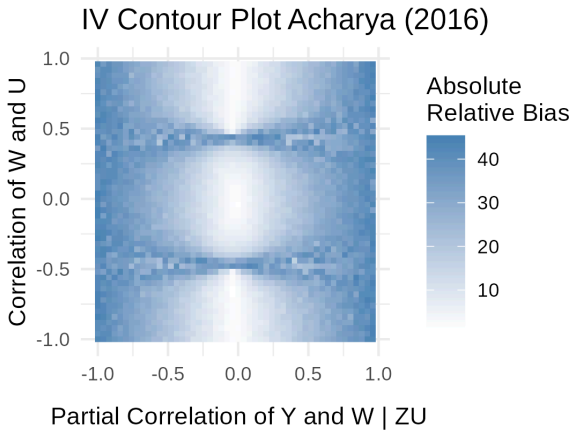


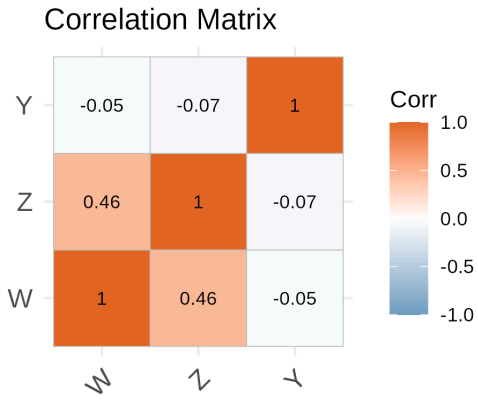


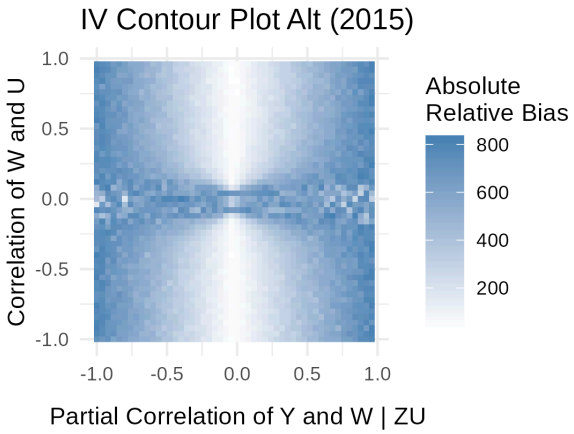


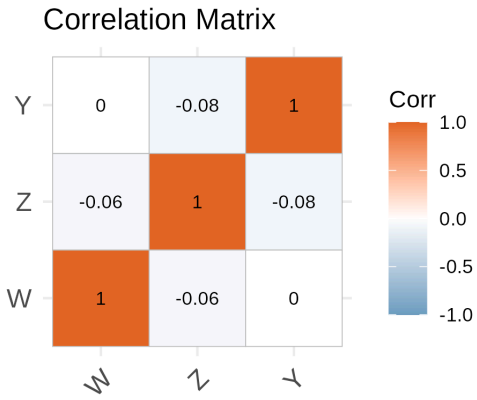
IV contour Plots

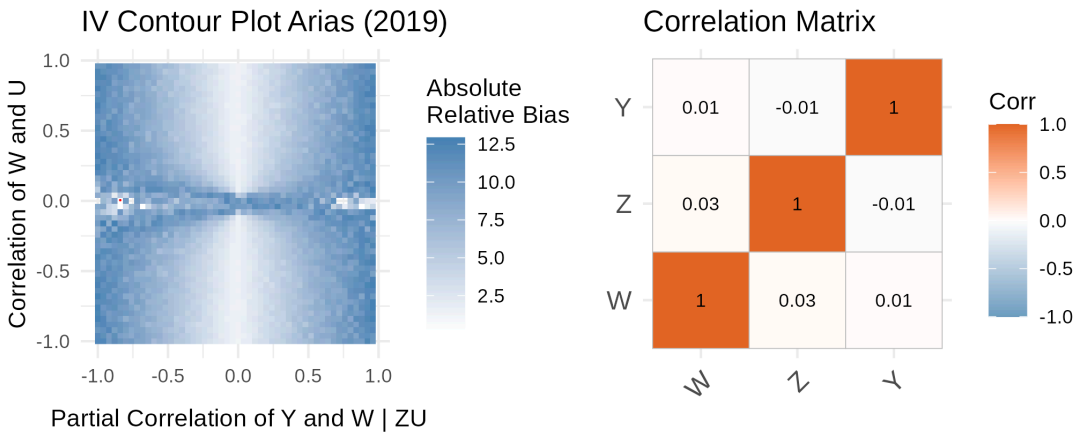
I have also created IV contour plots for the IV analyses which display the bias as a function of violations in the exclusion restriction and exogeneity. These are still a work-in-progress. The red contour lines in these diagrams corresponds to an absolute bias enough to overturn the point estimate. As with the previous contour plots, I include the plot for every study in the appendix.


IV Contour Plot Acharya (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
40
30
20
10


Correlation Matrix
Y
Z
W
-0.05
-0.07
1
0.46
1
-0.07
1
0.46
-0.05
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

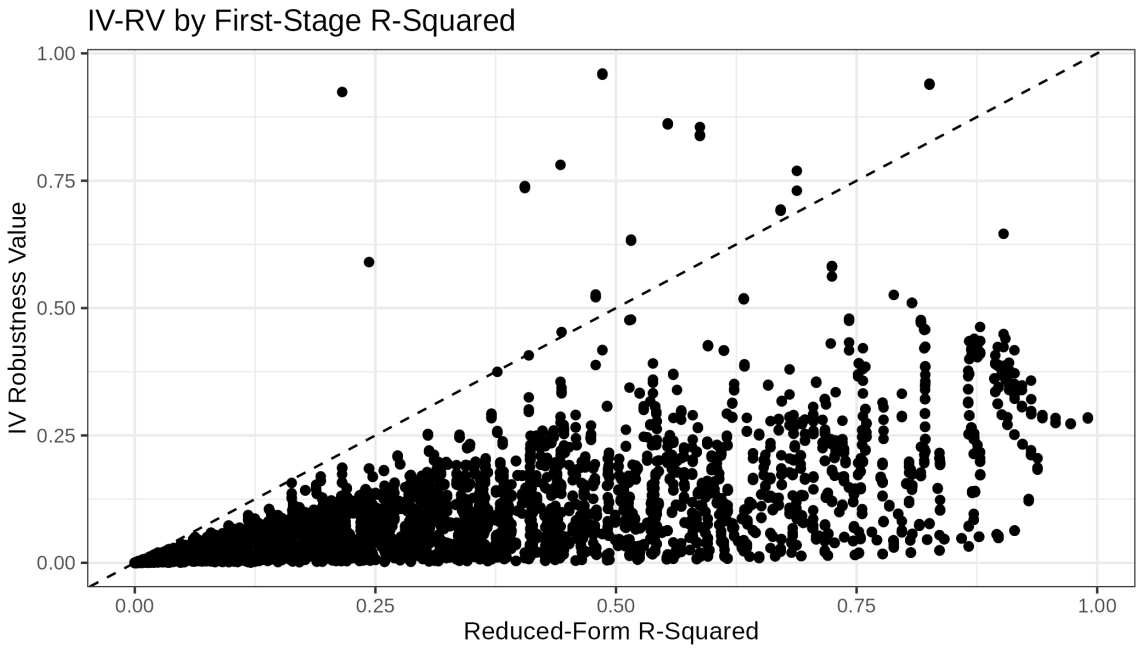

IV Contour Plot Alt (2015)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
800
600
400
200


Correlation Matrix
Y
Z
W
0
-0.08
1
-0.06
1
-0.08
1
-0.06
0
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

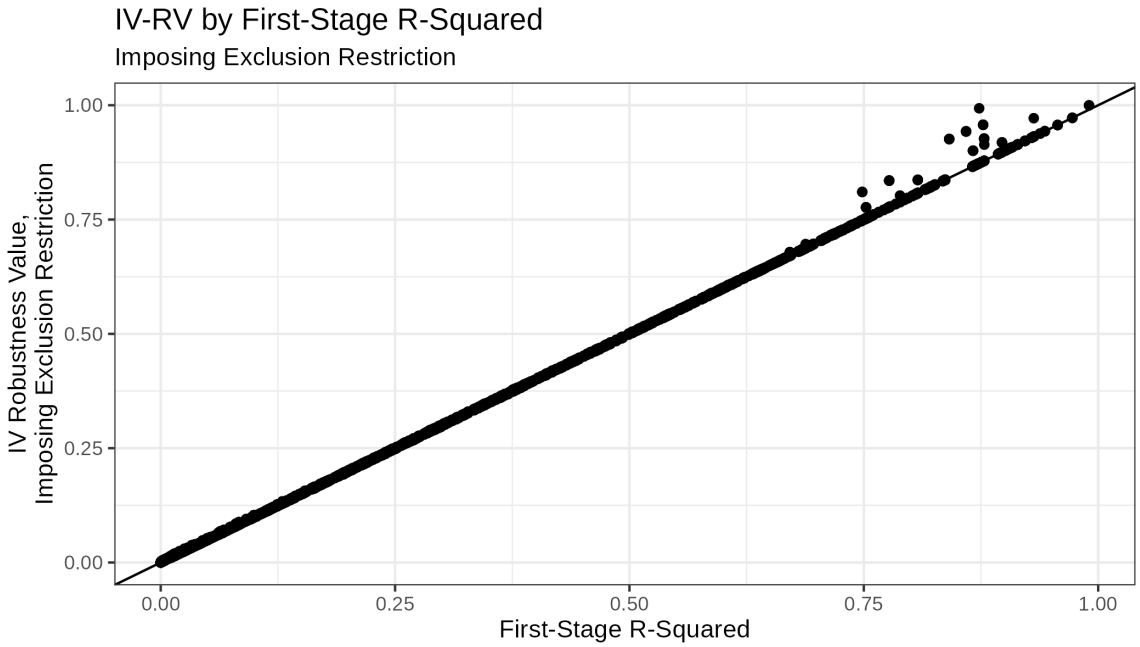


Identifying Predictors of IV Robustness Value via Simulation

I have also explored the characteristics of the IV Robustness value via simulation, trying to understand what factors predict high (or low) robustness values. The principal conclusion from this exercise is that the First-Stage R-squared value provides a bound on the IV robustness value, unless the treatment is very correlated with the outcome. Below, I plot the relationship between the First-Stage R-squared and the IV robustness value. We can see that almost all points lie below the $y = x$ line which means that in almost all cases, the IV robustness value is less than the reduced-form R-squared value.

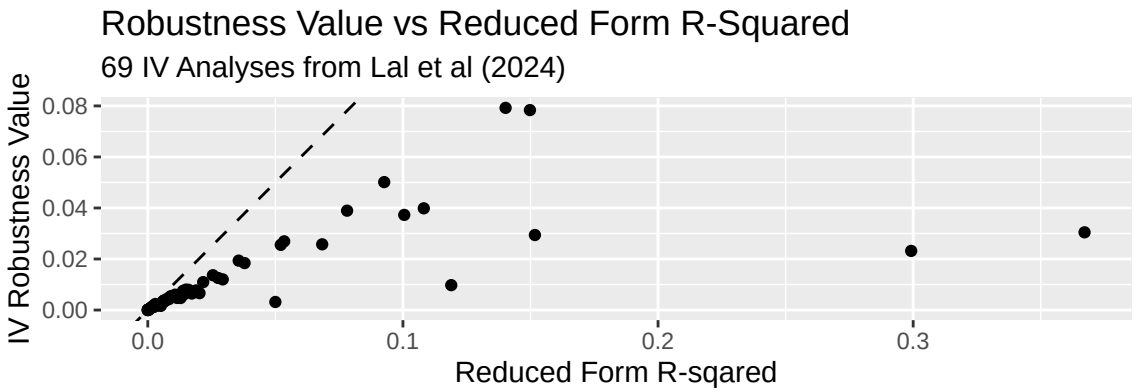


This lead me to hypothesize that the RV is bounded above by the R^2 from the reduced-form regression of Y on W. I show this is the case in the next section, so long as the outcome and treatment are not too correlated. Before this, I show the relationship between the IV Robustness Value and the reduced-from R-squared while imposing the exclusion restriction holds exactly:



Almost all of the points lie on the $y = x$ line, which suggests that the reduced-form r-squared value gives the IV robustness value when the exclusion restriction is imposed.

Finally, I plot the distribution of IV robustness values versus the reduced form R-squared among the 69 analyses we analyze. The R-squared bounds the robustness value in every case, suggesting that the R-squared provides a valid bound for the types of studies examined in political science[1]. The median robustness value is 52% of the bound, which suggests that the bound is a reasonable heuristic, although there are some points that lie far below the y=x line. For these points the R-squared bound overstates the robustness of the study.



[1]I assume RCTs can have exogeneity violations for the purposes of this analysis

Bounding the RV_{IV} With the first-stage F Statistic

I start with the formula given in Theorem 3.1 of the original paper, which I replicate after removing references to W^Y:

$$\text{bias}(\tau_{\text{IV}}) = \underbrace{\frac{1}{R_{Z\sim W}}\frac{\text{sd}(W)}{\text{sd}(Z)}}_{\text{Observable Scaling Factor}} \left\{ \underbrace{\beta_{w_z}}_{\text{Exclusion Restriction}} + \beta_u \underbrace{\varphi_u}_{\text{Exogeneity}} \frac{\text{Var}(U)}{\text{Var}(W)} \right\}$$

I then substitute in $\frac{\text{Cov}(Z,W)}{\text{sd}(Z)\,\text{sd}(W)}$ for $R_{Z\sim W}$ and simplify:

$$\text{bias}(\tau_{\text{IV}}) = \frac{\text{sd}(Z)\,\text{sd}(W)}{\text{Cov}(Z,W)}\frac{\text{sd}(W)}{\text{sd}(Z)}\left\{\beta_{w_z} + \beta_u\varphi_u\frac{\text{Var}(U)}{\text{Var}(W)}\right\}$$

$$\text{bias}(\tau_{\text{IV}}) = \frac{\text{Var}(W)}{\text{Cov}(Z,W)}\left\{\beta_{w_z} + \beta_u\varphi_u\frac{\text{Var}(U)}{\text{Var}(W)}\right\}$$

Generally, we are interested in computing the minimal violations in the IV / RV assumptions that are enough to drive the true IV effect to zero (a bias of size τ_{IV}). This corresponds to finding the minimum values of $R_{W\sim U}$, $R_{Y\sim W \mid Z,U}$ that satisfy the following equality:

$$\left|\underbrace{\frac{\text{Cov}(W,Y)}{\text{Cov}(W,Z)}}_{\tau_{\text{IV}}}\right| = \left|\frac{\text{Var}(W)}{\text{Cov}(Z,W)}\left\{\beta_{w_z} + \beta_u\varphi_u\frac{\text{Var}(U)}{\text{Var}(W)}\right\}\right|$$

Assuming the Exclusion Restriction Holds:

$$|\tau_{\text{IV}}| = \left|\frac{\text{Var}(W)}{\text{Cov}(Z,W)}\left\{\underbrace{\cancel{\beta_{w_z}}}_{=0} + \beta_u\varphi_u\frac{\text{Var}(U)}{\text{Var}(W)}\right\}\right| \text{(Assume Exclusion Restiction)}$$

$$|\tau_{\text{IV}}| = \left|\frac{\text{Var}(W)}{\text{Cov}(Z,W)}\frac{\text{Var}(U)}{\text{Var}(W)}\beta_u\varphi_u\right| \text{(Commute)}$$

$$|\tau_{\text{IV}}| = \left|\frac{\text{Var}(U)}{\text{Cov}(Z,W)}\beta_u\varphi_u\right| \text{(Cancel Var}(W)\text{)}$$

I then use $\varphi_u = R_{W\sim U}\frac{\text{sd}(W)}{\text{sd}(U)}$ to write:

$$\left|\frac{\text{Cov}(W,Y)}{\text{Cov}(W,Z)}\right| = \left|\frac{\text{Var}(U)}{\text{Cov}(Z,W)}\beta_u R_{W\sim U}\frac{\text{sd}(W)}{\text{sd}(U)}\right| \qquad \text{(Substitute in } \varphi_u\text{)}$$

$$\left(\frac{\text{Cov}(W,Y)}{\text{Cov}(W,Z)}\right)^2 = \left(\frac{\text{Var}(U)}{\text{Cov}(Z,W)}\beta_u R_{W\sim U}\frac{\text{sd}(W)}{\text{sd}(U)}\right)^2 \quad \text{(Square Both Sides)}$$

$$\text{Cov}(W,Y)^2 = \text{Var}(U)^2 {\beta_u}^2 R^2_{W\sim U}\frac{\text{Var}(W)}{\text{Var}(U)} \qquad \text{(power of a product)}$$

$$\text{Cov}(W,Y)^2 = \text{Var}(U)\ \text{Var}(W){\beta_u}^2 R^2_{W\sim U}$$

$$\frac{\text{Cov}(W,Y)^2}{\text{Var}(U)\ \text{Var}(W){\beta_u}^2} = R^2_{W\sim U}$$

Where the right hand side is the first component of the IV robustness value. Assuming both exclusion restriction and no true effect, we have ${\beta_u}^2\ \text{Var}(U) = \text{Var}(Y)R^2_{Y\sim U}$. Performing this substitution, we have:

$$\frac{\text{Cov}(W,Y)^2}{\text{Var}(Y)R^2_{Y\sim U}\ \text{Var}(W)} = R^2_{W\sim U}$$

Finally, we can take the Infimum over the left hand side to bound the RV on the right. Sending $R^2_{Y\sim U} \to 1$, we have:

$$\inf\left(\frac{\text{Cov}(W,Y)^2}{\text{Var}(Y)R^2_{Y\sim U}\ \text{Var}(W)}\right) = \inf(R^2_{W\sim U})$$

$$\frac{\text{Cov}(W,Y)^2}{\text{Var}(Y)\ \text{Var}(W)} > \inf(R^2_{W\sim U})$$

$$R^2_{W\sim Y} > \inf(R^2_{W\sim U})$$

The right hand side is the reduced-form R-squared, so we have just proved that the $\text{IV}_{\{RV\}}$ is bounded above by the reduced-form R-squared value. Importantly however, in the last steps, we assumed that we could send $R^2_{Y\sim U} \to 1$ without violating the treatment confounding cap that $R^2_{Z\sim U\ |\ W} < .95$. This cap will not be an issue in settings where the treatment and outcome are not heavily correlated, but if the treatment and outcome are very correlated, then it will not be possible to send $R^2_{Y\sim U} \to 1$ without also breaking the treatment confounding cap. Specifically, this cap will never be an issue if $R^2_{Z\sim Y\ |\ W} < .95$).

Appendix

Table of Study Point Estimates

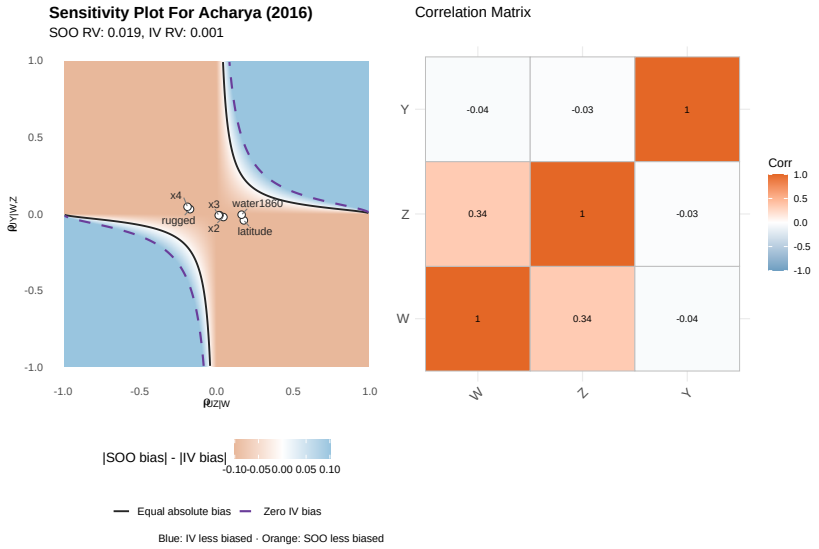
Paper Name	F-Stat	OLS Coef.	IV Coef.	RF Coef.	FS Coef.	N
Acharya (2016)	37.6527	−0.0318	−0.2766	−0.1128	0.4079	1120
Alt (2015)	83.3152	−0.0131	−0.0347	0.0270	−0.9354	5705
Arias (2019)	14.4763	−0.0215	0.8282	0.2760	0.3332	2745
Baccini (2021)	468.6180	−0.0127	−0.0433	−4.5381	104.8786	3065
Barth (2015)	12.6294	−0.7755	−1.5822	0.0257	−0.0162	117
Bhavnani (2018)	236.8206	0.0195	0.0220	0.0121	0.5504	569
Blair (2022)	60.6442	−1.3155	−1.8768	1.8050	−0.9617	847
Blattman (2014)	59.8414	0.0007	0.0015	0.0213	14.2500	1900
Carnegie (2017)	7.5007	0.1903	1.7054	0.2632	0.1543	1792
Charron (2013)	52.1369	0.0176	0.0205	0.3174	15.4663	58
Charron (2017)	13.0893	−0.0886	−0.1466	−0.0013	0.0090	179
Chong (2019)	30.9749	0.0715	0.0509	0.0125	0.2449	3350
Cirone (2018)	168.0023	0.0305	0.6341	−0.0052	−0.0083	8147
Colantone (2018)a	613.9804	12.0854	12.2993	1.5671	0.1274	167
Colantone (2018)b	19.1709	0.6442	1.3096	0.0514	0.0392	7782
Coppock (2016)	113.3680	0.3126	0.3728	0.0187	0.0502	773556
Croke (2016)	71.1356	−0.0204	−0.0980	−0.0657	0.6708	1842
Delao (2013)	153.2854	0.0222	0.1563	0.0532	0.3401	417
Dietrich (2015)	1.3690	0.0576	0.2659	0.0267	0.1003	370
Digiuseppe (2022)	12.1084	0.0443	0.8158	0.1792	0.2196	2368
Dower (2018)a	14.0828	−3.8696	−32.7701	−3.9279	0.1199	361
Dower (2018)b	51.0176	−4.2492	−42.4545	−11.7823	0.2775	365
Dube (2015)	185092.5288	0.1503	0.3149	1.1155	3.5422	16606
Feigenbaum (2015)	75.5233	−0.1080	−0.6976	−0.5863	0.8405	862
Flores (2013)	53.6345	0.0191	0.0456	0.1086	2.3803	592
Gelbach (2012)	9.5714	0.1304	0.3956	−3.3385	−8.4401	99
Gerber (2010)	3.1563	0.5658	3.8231	0.2742	0.0717	411
Goldstein (2017)	16.3711	0.0648	0.4673	1.3319	2.8506	1262

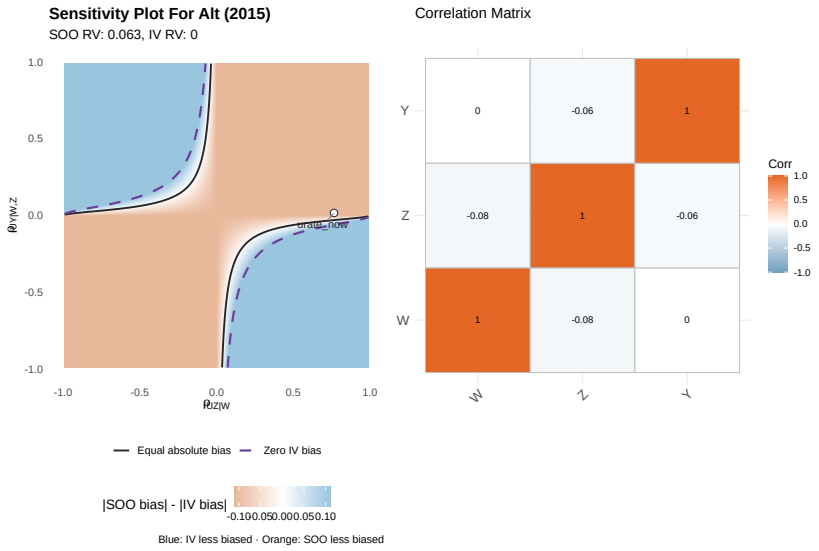
Paper Name	F-Stat	OLS Coef.	IV Coef.	RF Coef.	FS Coef.	N
Grossman (2017)	17.6860	0.0364	0.4348	−7.9151	−18.2051	518
Hager (2019)_apsr	637.5699	−0.2335	−0.5200	0.1011	−0.1943	878
Hager (2019)ajps_a	79.2985	0.0072	0.1363	−0.0001	−0.0009	3850
Hager (2019)ajps_b	100.3609	0.0150	0.0513	−0.0005	−0.0105	866
Hager (2022)	3.1403	0.1494	0.1765	0.4736	2.6827	297
Healy (2013)	244.4704	0.0451	0.1706	0.0832	0.4876	279
Henderson (2016)_a	13.6778	0.0170	−2.7068	0.0296	−0.0109	6275
Henderson (2016)_b	20.0070	0.0170	−1.8174	0.0269	−0.0148	6275
Hong (2022)	44.0880	0.0151	0.0476	0.0289	0.6073	1300
Johns (2016)	18.1200	0.0190	−0.0809	−0.0137	0.1692	2462
Kapoor (2018)	19.1635	−0.0256	0.4864	−1.2770	−2.6256	4297
Kim (2019)	914.6461	0.0170	3.9287	1.9490	0.4961	2749
Kocher (2011)	105.8858	0.0249	1.4793	0.3582	0.2422	9707
Kriner (2014)	121.5394	−0.0314	−0.1262	−0.0350	0.2777	636
Kuipers (2022)	27.5670	−0.0319	−0.1689	−0.0254	0.1504	1686
Laitin (2016)	32.4040	−1.3676	−1.3815	−0.0001	0.0001	137
Lei (2022)	5359.1714	0.2760	0.4776	0.4833	1.0119	148
Lelkes (2017)	11.1632	0.0041	0.0316	0.0033	0.1042	114803
Lerman (2017)	1194.6594	0.0093	0.0459	0.0202	0.4401	4389
Lorentzen (2014)	53.4100	−2.4789	−6.3664	−3.4227	0.5376	112
McClendon (2014)	207.1798	0.2823	0.3149	0.0247	0.0786	3647
Meredith (2013)	77.2953	0.2634	0.1634	0.0062	0.0379	14550
Nellis (2018)	53.9103	−0.0150	−0.6603	−0.5965	0.9034	437
Pianzola (2019)	46.7612	0.0805	0.7550	0.1032	0.1367	1775
Ritter (2016)	117.7002	0.1885	0.2635	0.0001	0.0004	1258733
Rueda (2017)	8598.3264	−0.6750	−0.9835	−0.7826	0.7957	4352
Schleiter (2016)	57.1949	3.0828	5.0282	0.3124	0.0621	191
Schubiger (2021)	38.5348	0.0702	0.2736	0.0172	0.0629	7295
Sexton (2019)	7.4923	−1.3472	−15.0645	4.3552	−0.2891	72
Spenkuch (2018)	272.0825	−0.2504	−0.2608	−15.6450	59.9780	982

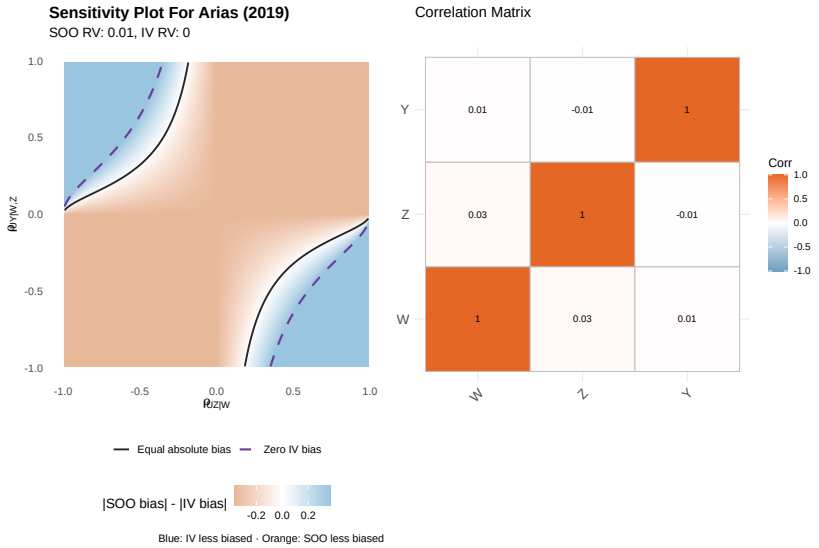
Paper Name	F-Stat	OLS Coef.	IV Coef.	RF Coef.	FS Coef.	N
Stewart (2017)	99.3150	0.8030	1.1929	−7.0905	−5.9438	466
Stokes (2016)	65.7306	−0.0203	−0.0770	−0.0585	0.7602	708
Tajima (2013)	202.6374	−0.0024	−0.0041	−0.0019	0.4470	51913
Trounstine (2016)	890.7053	−0.9265	−2.2644	−0.0259	0.0114	21145
Urpelainen (2022)	16.1654	0.0063	0.0296	0.9095	30.6883	1144
Vernby (2013)	115.1803	8.9328	10.2662	11.8895	1.1581	183
Webster (2022)	773.5894	0.0024	0.0108	0.0310	2.8585	3410
West (2017)	37.8652	0.0358	0.2073	0.1407	0.6784	2283
Wood (2022)	22.8647	0.2369	0.2869	0.1377	0.4800	435
Zhu (2017)	45.5515	0.3531	0.4855	0.5590	1.1514	61
Ziaja (2020)	199.9166	0.8746	0.8726	0.0599	0.0686	2367

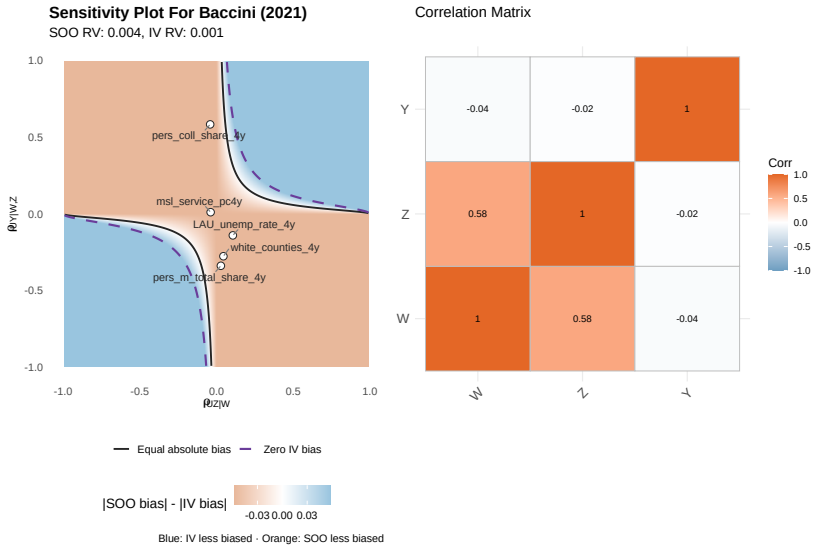
Contour Figures

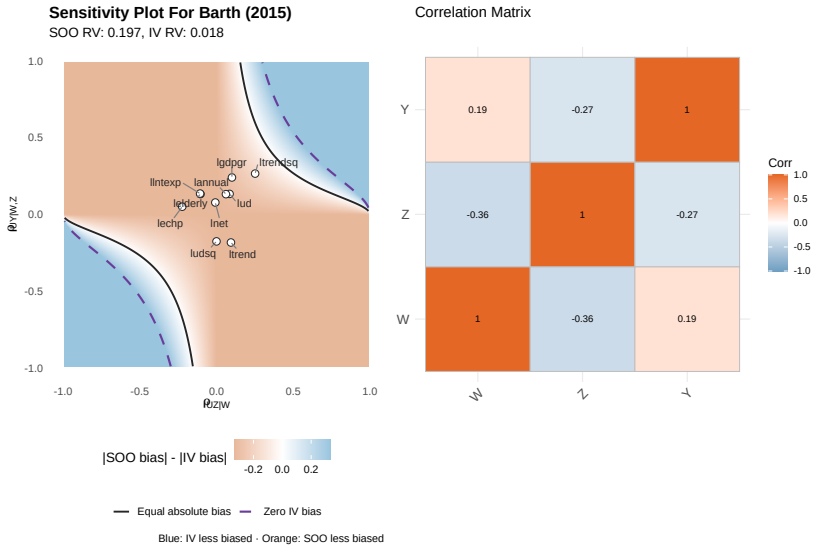
This section reports the contour plots for the 69 analyses we re-analyze. M. Huang and C. McCartan [2].

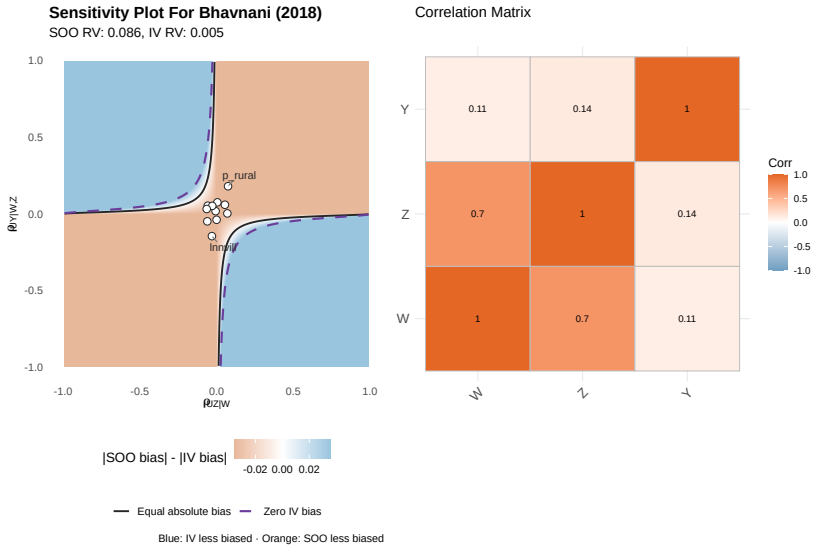


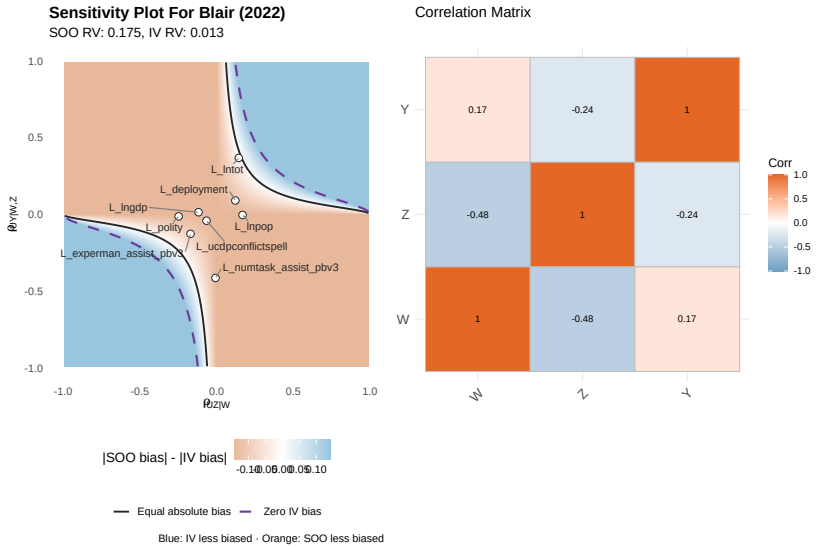


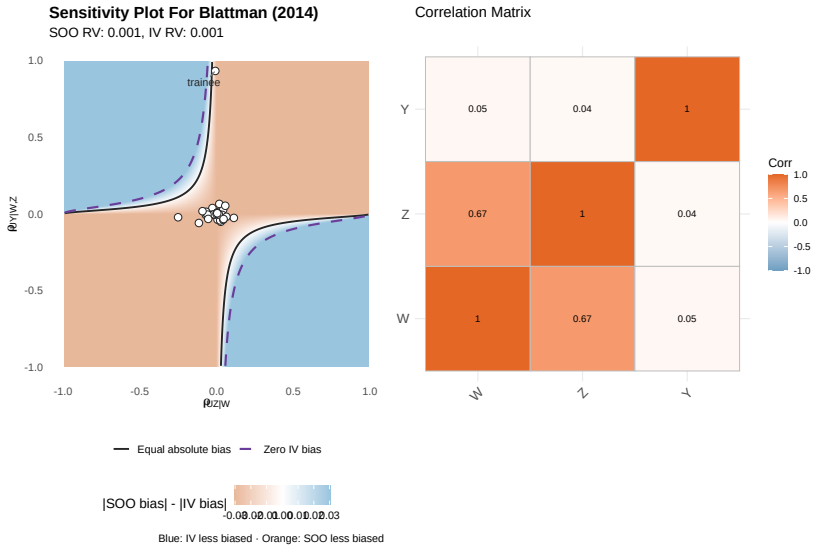

Sensitivity Plot For Arias (2019)
SOO RV: 0.01, IV RV: 0
Correlation Matrix
Y
Z
W
0.01
-0.01
1
0.03
1
-0.01
1
0.03
0.01
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.2 0.0 0.2
Blue: IV less biased · Orange: SOO less biased

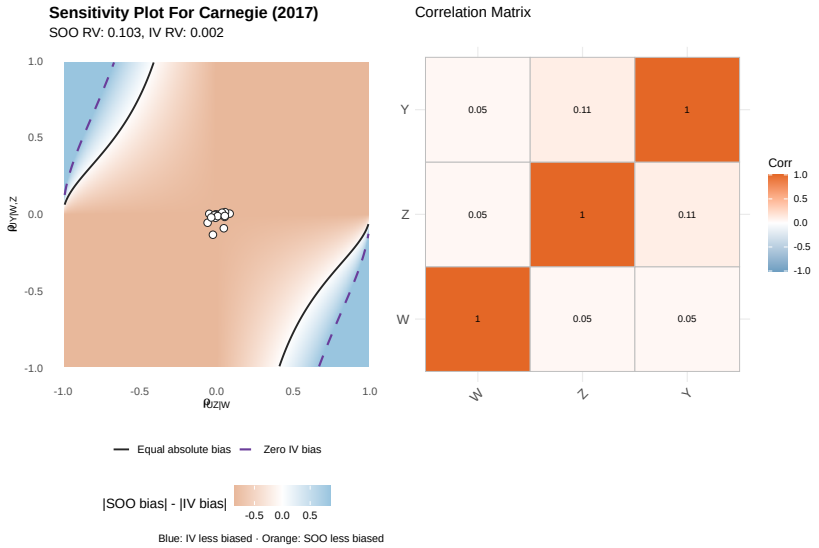

Sensitivity Plot For Baccini (2021)
SOO RV: 0.004, IV RV: 0.001
Correlation Matrix
pers_coll_share_4y
msl_service_pc4y
LAU_unemp_rate_4y
white_counties_4y
pers_m_total_share_4y
Y
Z
W
-0.04
-0.02
1
0.58
1
-0.02
1
0.58
-0.04
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.03 0.00 0.03
Blue: IV less biased · Orange: SOO less biased

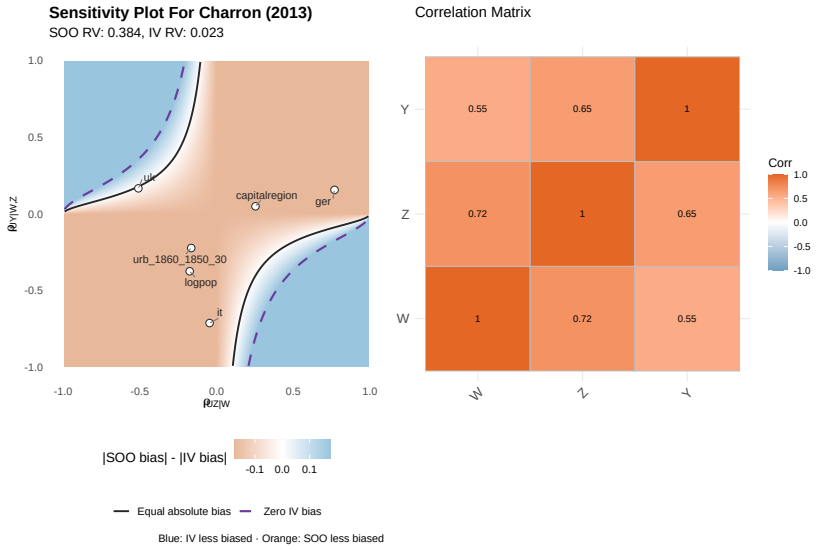

Sensitivity Plot For Barth (2015)
SOO RV: 0.197, IV RV: 0.018
Correlation Matrix
lgdpgr
ltrendsq
llntexp
lannual
lelderly
lud
lechp
lnet
ludsq
ltrend
Y
Z
W
0.19
-0.27
1
-0.36
1
-0.27
1
-0.36
0.19
Corr
|SOO bias| - |IV bias|
-0.2 0.0 0.2
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

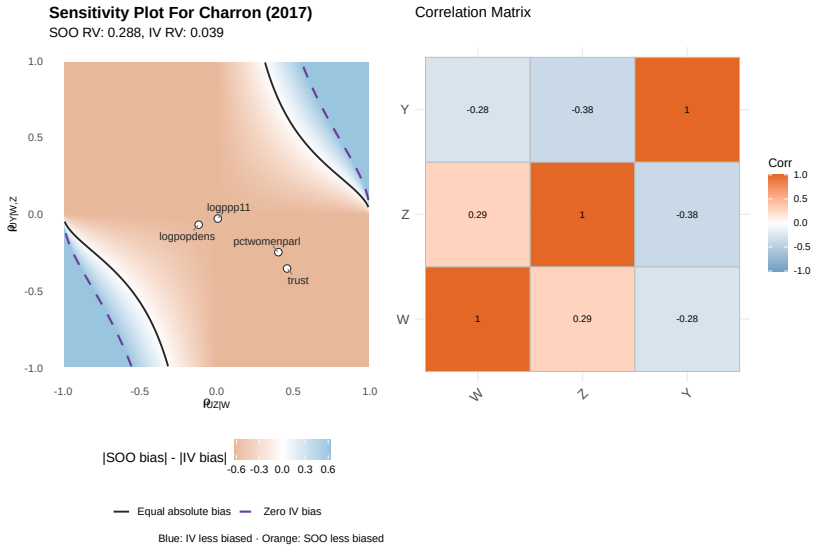

Sensitivity Plot For Bhavnani (2018)
SOO RV: 0.086, IV RV: 0.005
Correlation Matrix
p_rural
lnnnjjl
Y
Z
W
0.11
0.14
1
0.7
1
0.14
1
0.7
0.11
Corr
|SOO bias| - |IV bias|
-0.02 0.00 0.02
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

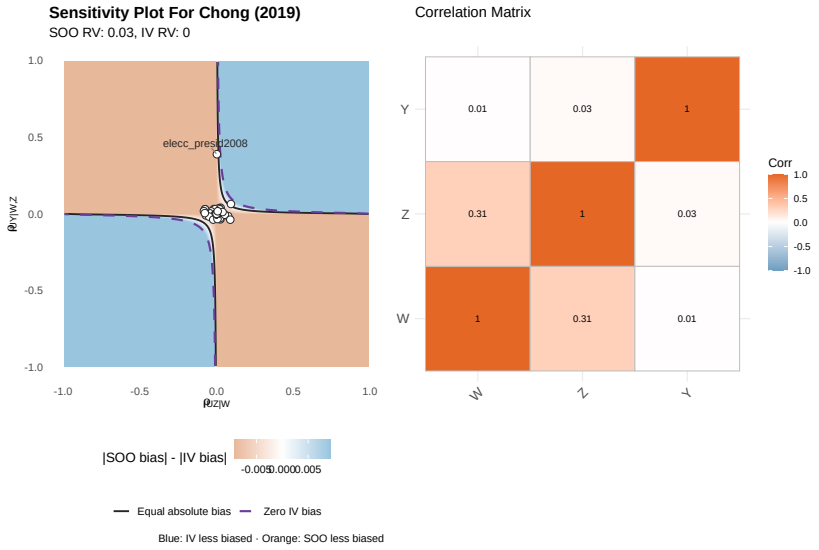

Sensitivity Plot For Blair (2022)
SOO RV: 0.175, IV RV: 0.013
L_lntot
L_deployment
L_lngdp
L_polity
L_lnpop
L_ucdpconflictspell
L_experman_assist_pbv3
L_numtask_assist_pbv3
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

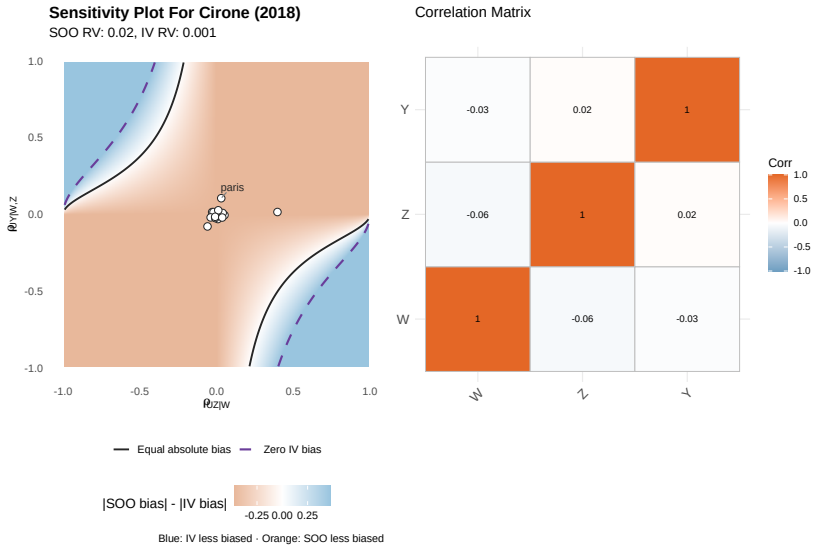

Sensitivity Plot For Blattman (2014)
SOO RV: 0.001, IV RV: 0.001
trainee
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

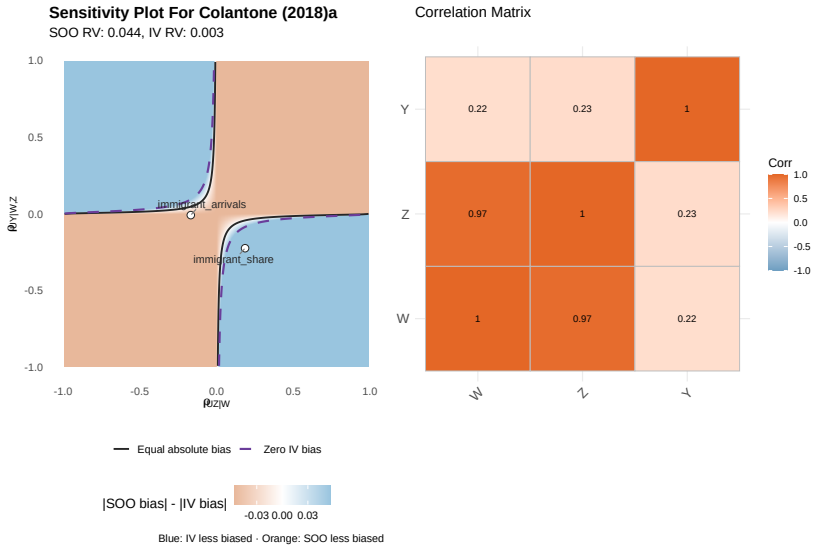

Sensitivity Plot For Carnegie (2017)
SOO RV: 0.103, IV RV: 0.002
Correlation Matrix
Y
Z
W
0.05
0.11
1
0.05
1
0.11
1
0.05
0.05
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.5 0.0 0.5
Blue: IV less biased · Orange: SOO less biased

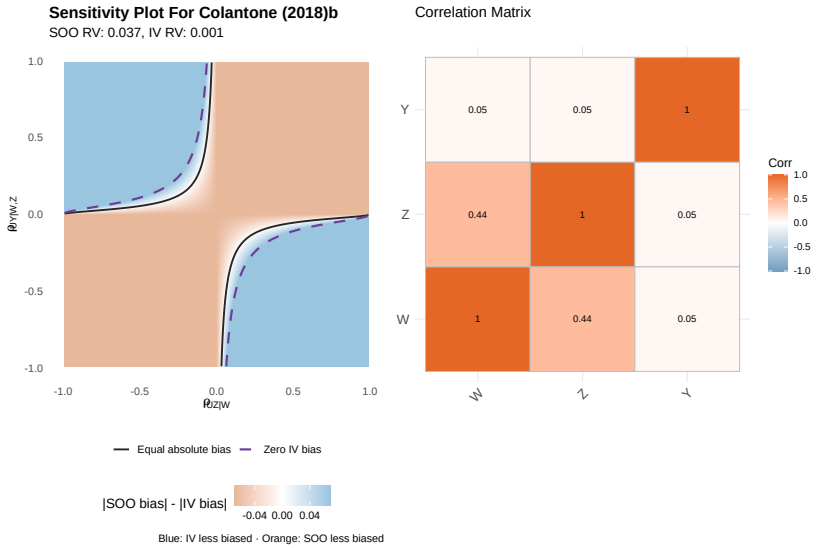

Sensitivity Plot For Charron (2013)
SOO RV: 0.384, IV RV: 0.023
Correlation Matrix
Y
Z
W
0.55
0.65
1
0.72
1
0.65
1
0.72
0.55
Corr
1.0
0.5
0.0
-0.5
-1.0
uk
capitalregion
ger
urb_1860_1850_30
logpop
it
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.1 0.0 0.1
Blue: IV less biased · Orange: SOO less biased

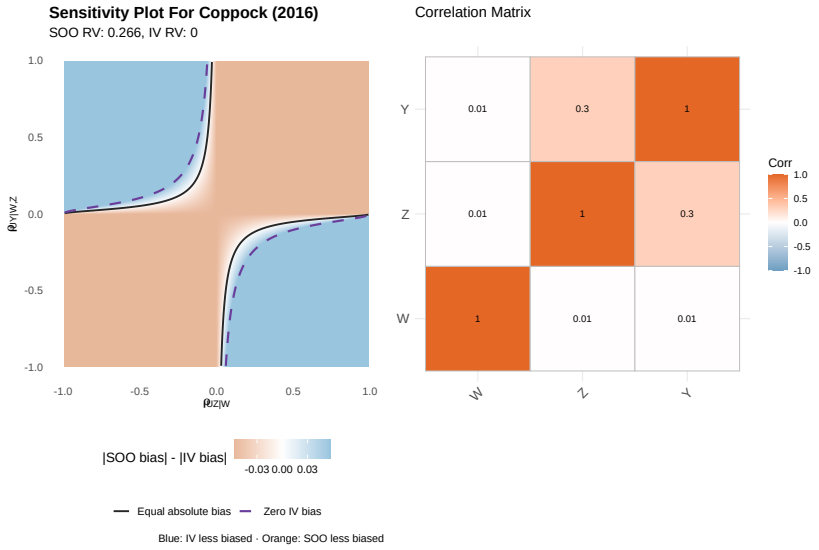

Sensitivity Plot For Charron (2017)
SOO RV: 0.288, IV RV: 0.039
Correlation Matrix
logpppp11
logpopdens
pctwomenparl
trust
Y
Z
W
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

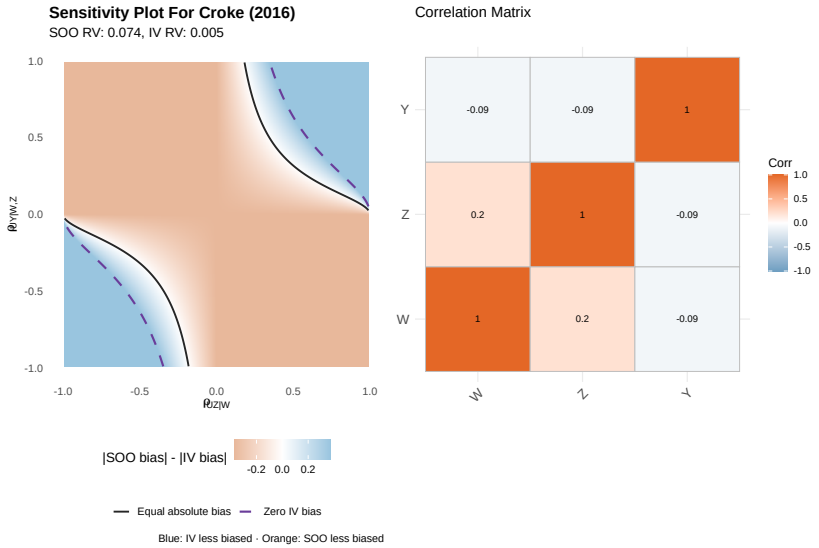

Sensitivity Plot For Chong (2019)
SOO RV: 0.03, IV RV: 0
Correlation Matrix
elecc_presid2008
Y
Z
W
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

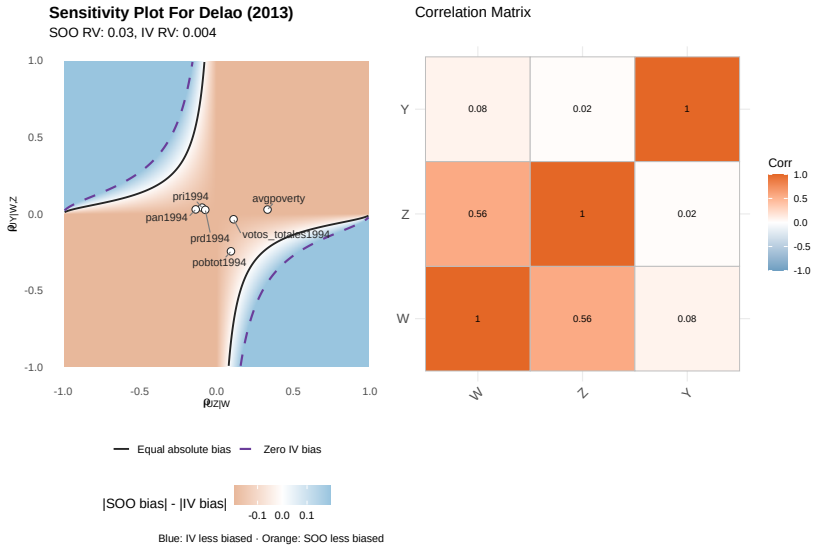

Sensitivity Plot For Cirone (2018)
SOO RV: 0.02, IV RV: 0.001
paris
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.25 0.00 0.25
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Y
Z
W
-0.03
0.02
1
-0.06
1
0.02
1
-0.06
-0.03
Corr
1.0
0.5
0.0
-0.5
-1.0

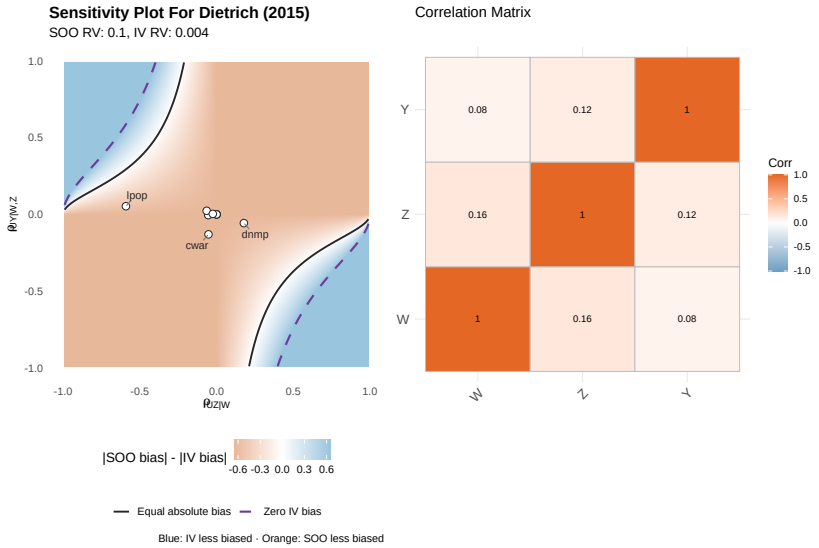

Sensitivity Plot For Colantone (2018)a
SOO RV: 0.044, IV RV: 0.003
immigrant_arrivals
immigrant_share
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.03 0.00 0.03
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Y
Z
W
0.22
0.23
1
0.97
1
0.23
1
0.97
0.22
Corr
1.0
0.5
0.0
-0.5
-1.0

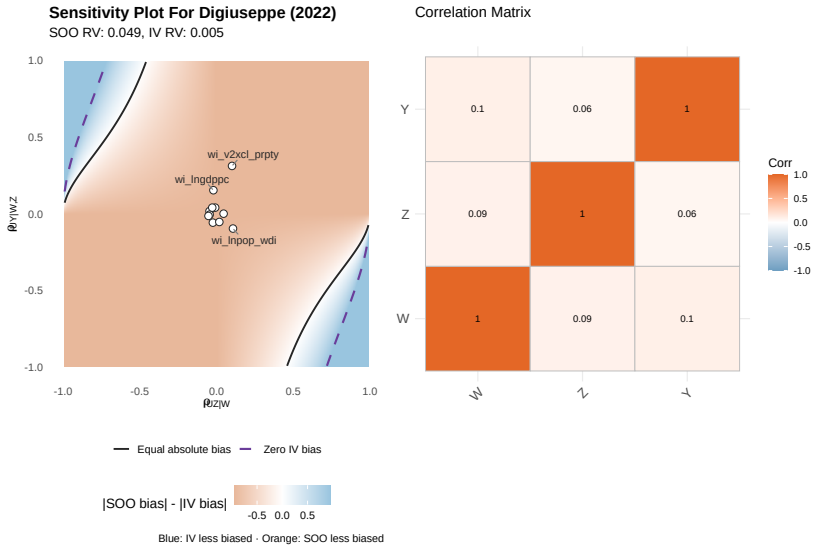

Sensitivity Plot For Colantone (2018)b
SOO RV: 0.037, IV RV: 0.001
Correlation Matrix
Y
Z
W
0.05
0.05
1
0.44
1
0.05
1
0.44
0.05
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.04 0.00 0.04
Blue: IV less biased · Orange: SOO less biased

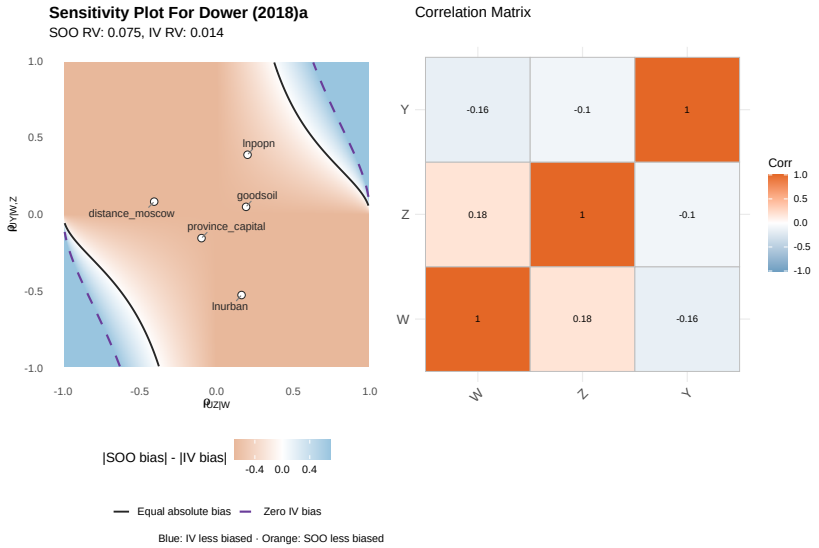

Sensitivity Plot For Coppock (2016)
SOO RV: 0.266, IV RV: 0
Correlation Matrix
Y
Z
W
0.01
0.3
1
0.01
1
0.3
1
0.01
0.01
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.03 0.00 0.03
Blue: IV less biased · Orange: SOO less biased

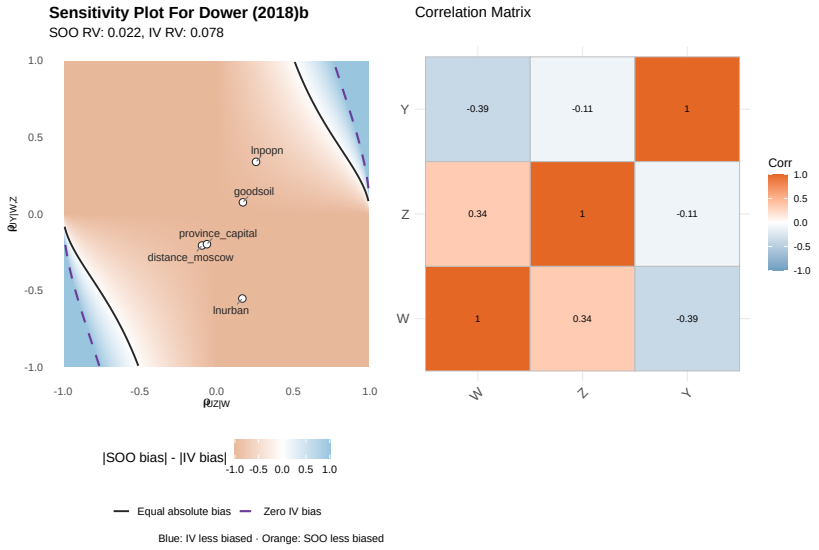

Sensitivity Plot For Croke (2016)
SOO RV: 0.074, IV RV: 0.005
Correlation Matrix
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
ρ_YW|W,Z
ρ_ZW|W
Y
Z
W
-0.09
-0.09
1
0.2
1
-0.09
1
0.2
-0.09
Corr
1.0
0.5
0.0
-0.5
-1.0
|SOO bias| - |IV bias|
-0.2 0.0 0.2
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

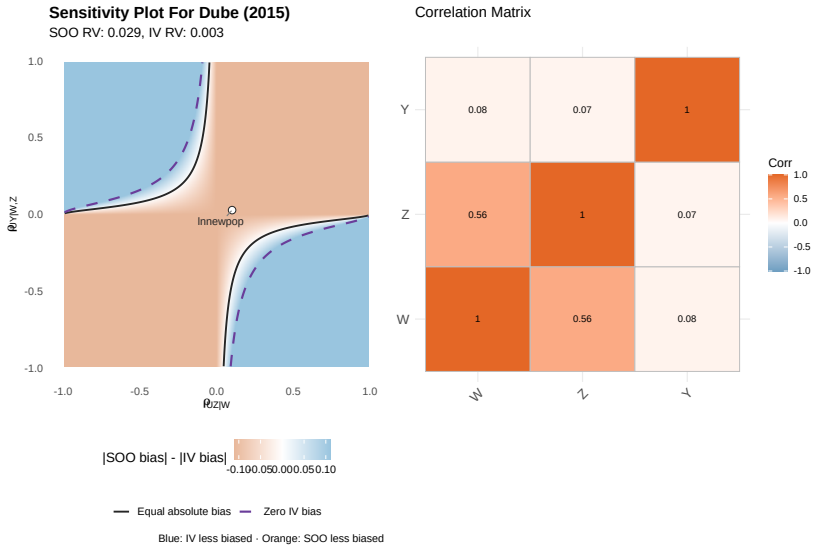

Sensitivity Plot For Delao (2013)
SOO RV: 0.03, IV RV: 0.004
Correlation Matrix
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
ρ_YW|W,Z
ρ_ZW|W
pri1994
avgpoverty
pan1994
votos_totales1994
prd1994
pobtot1994
Y
Z
W
0.08
0.02
1
0.56
1
0.02
1
0.56
0.08
Corr
1.0
0.5
0.0
-0.5
-1.0
|SOO bias| - |IV bias|
-0.1 0.0 0.1
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

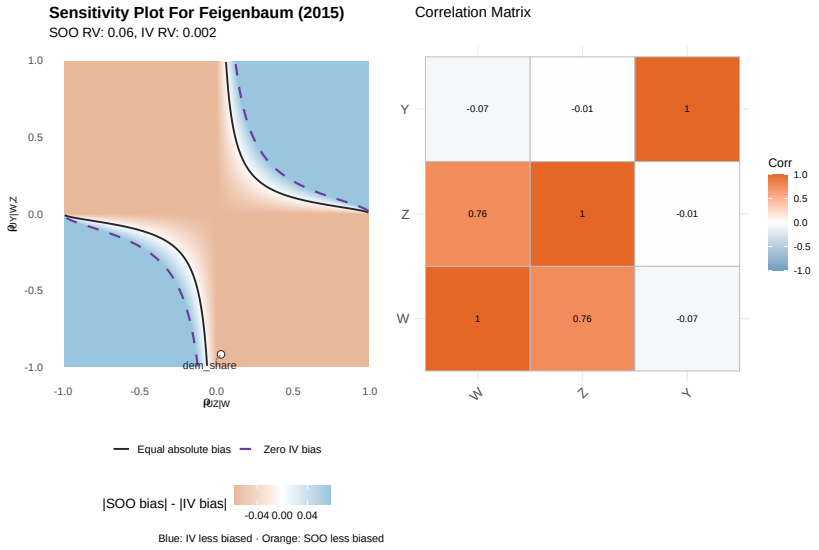

Sensitivity Plot For Dietrich (2015)
SOO RV: 0.1, IV RV: 0.004
Correlation Matrix
lpop
dnmp
cwar
Y
Z
W
0.08
0.12
1
0.16
1
0.12
1
0.16
0.08
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

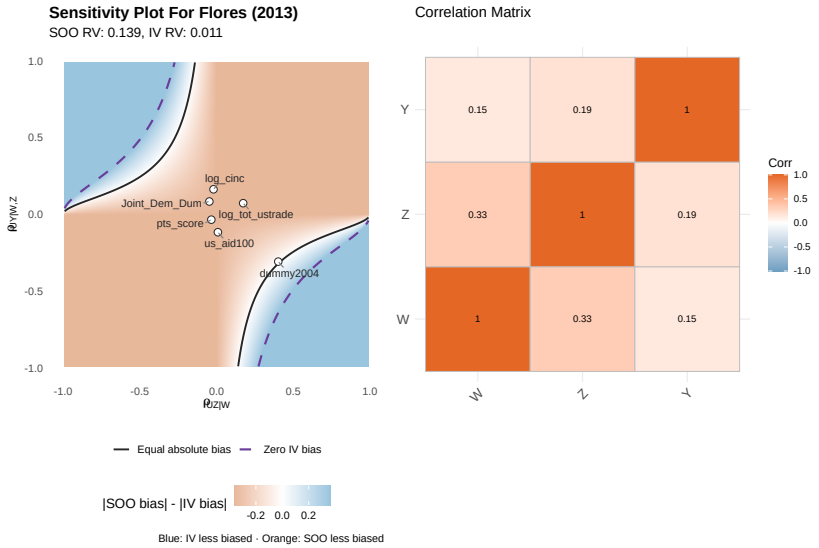

Sensitivity Plot For Digiuseppe (2022)
SOO RV: 0.049, IV RV: 0.005
Correlation Matrix
wi_v2xcl_prpty
wi_lngdppc
wi_lnpop_wdi
Y
Z
W
0.1
0.06
1
0.09
1
0.06
1
0.09
0.1
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

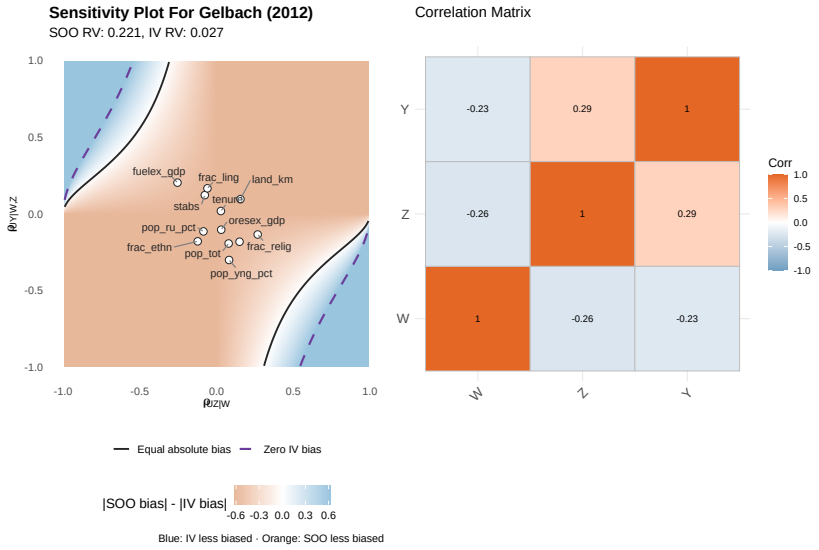

Sensitivity Plot For Dower (2018)a
SOO RV: 0.075, IV RV: 0.014
lnpopn
goodsoil
distance_moscow
province_capital
lnurban
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
-0.16
-0.1
1
0.18
1
-0.1
1
0.18
-0.16

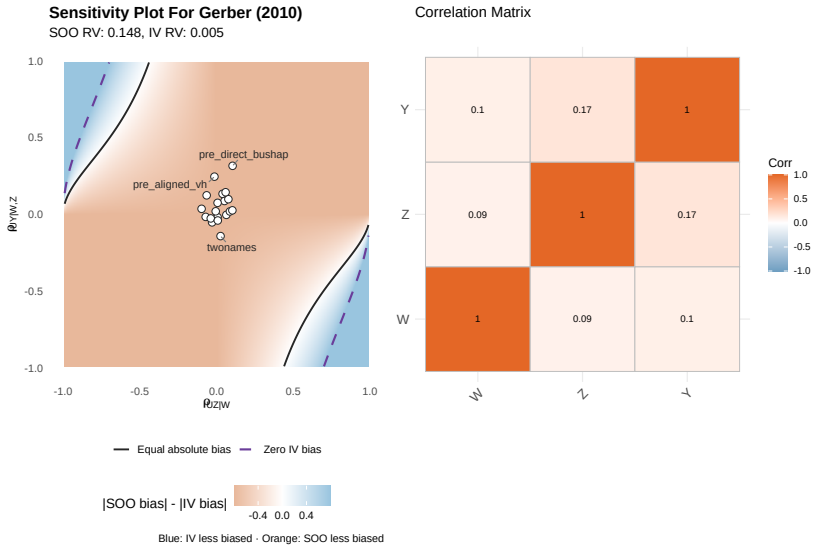

Sensitivity Plot For Dower (2018)b
SOO RV: 0.022, IV RV: 0.078
lnpopn
goodsoil
province_capital
distance_moscow
lnurban
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
-0.39
-0.11
1
0.34
1
-0.11
1
0.34
-0.39

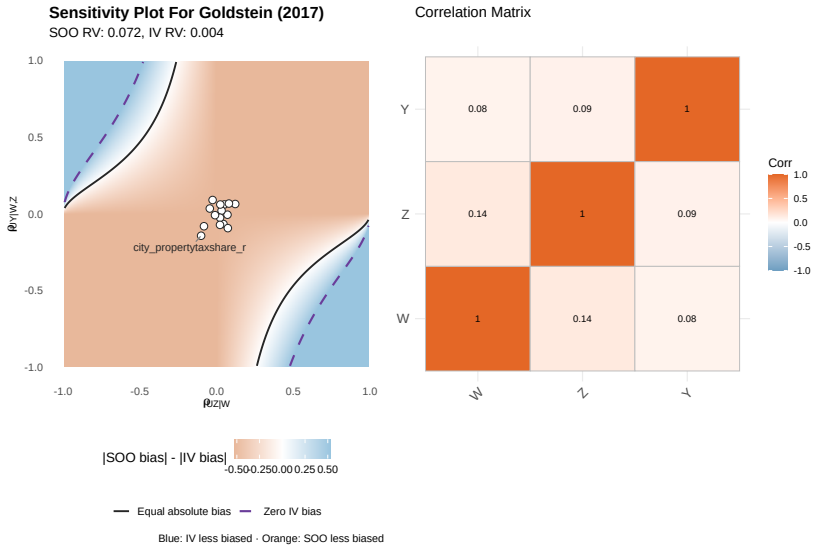

Sensitivity Plot For Dube (2015)
SOO RV: 0.029, IV RV: 0.003
Correlation Matrix
lnnewpop
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Corr
Y
Z
W
0.08
0.07
1
0.56
1
0.07
1
0.56
0.08

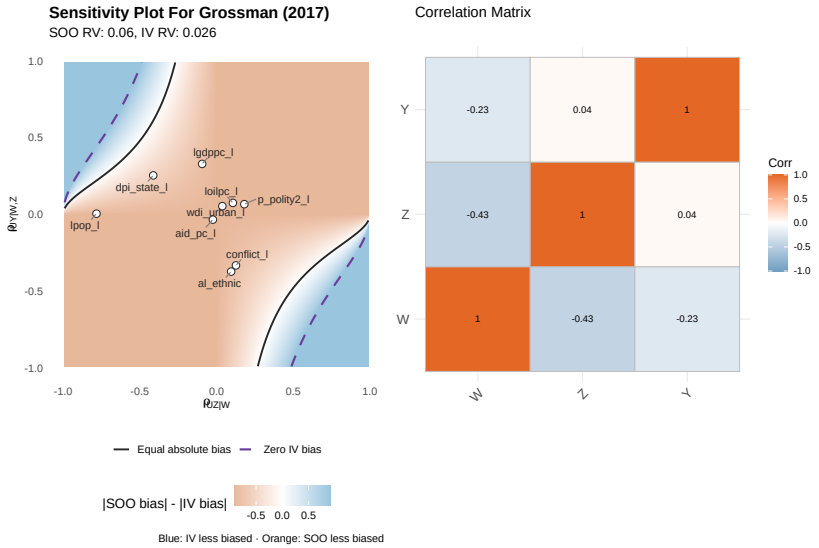

Sensitivity Plot For Feigenbaum (2015)
SOO RV: 0.06, IV RV: 0.002
Correlation Matrix
dem_share
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Corr
Y
Z
W
-0.07
-0.01
1
0.76
1
-0.01
1
0.76
-0.07

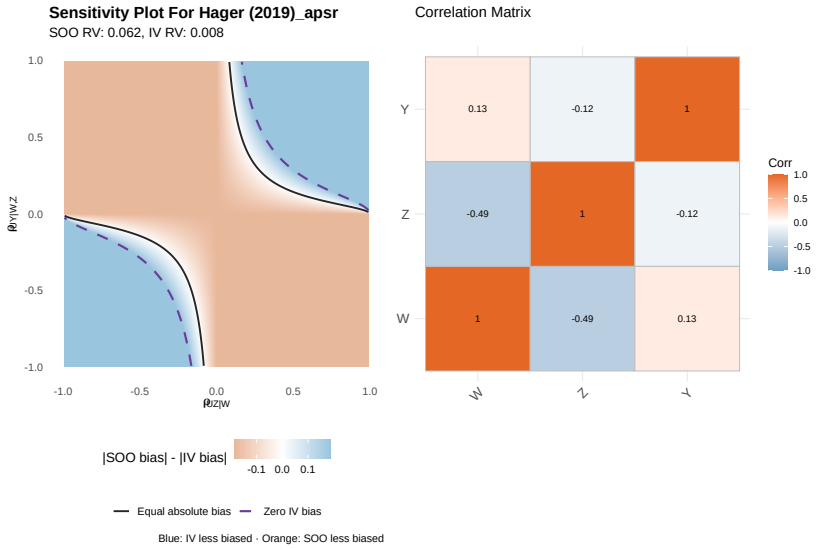

Sensitivity Plot For Flores (2013)
SOO RV: 0.139, IV RV: 0.011
log_cinc
Joint_Dem_Dum
log_tot_ustrade
pts_score
us_aid100
dummy2004
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
0.15
0.19
1
0.33
1
0.19
1
0.33
0.15

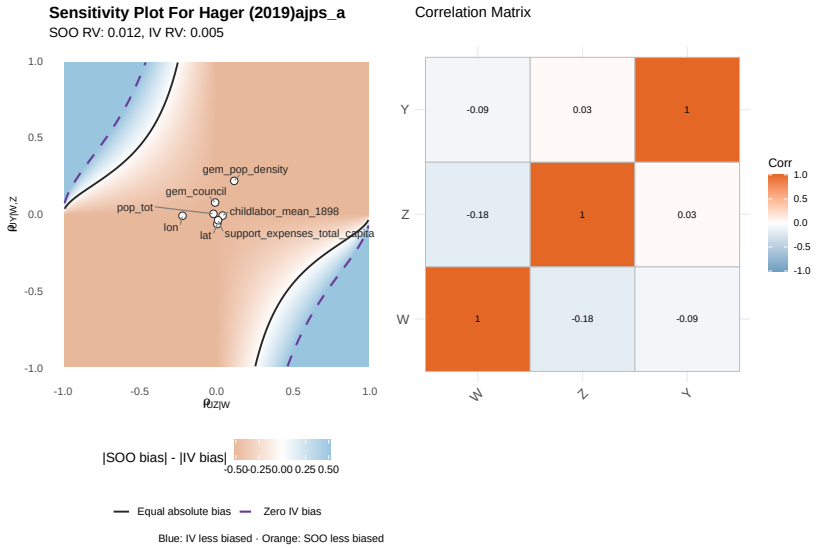

Sensitivity Plot For Gelbach (2012)
SOO RV: 0.221, IV RV: 0.027
fuelex_gdp
frac_ling
land_km
tenure
stabs
oresex_gdp
pop_ru_pct
frac_ethn
frac_relig
pop_tot
pop_yng_pct
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
-0.23
0.29
1
-0.26
1
0.29
1
-0.26
-0.23

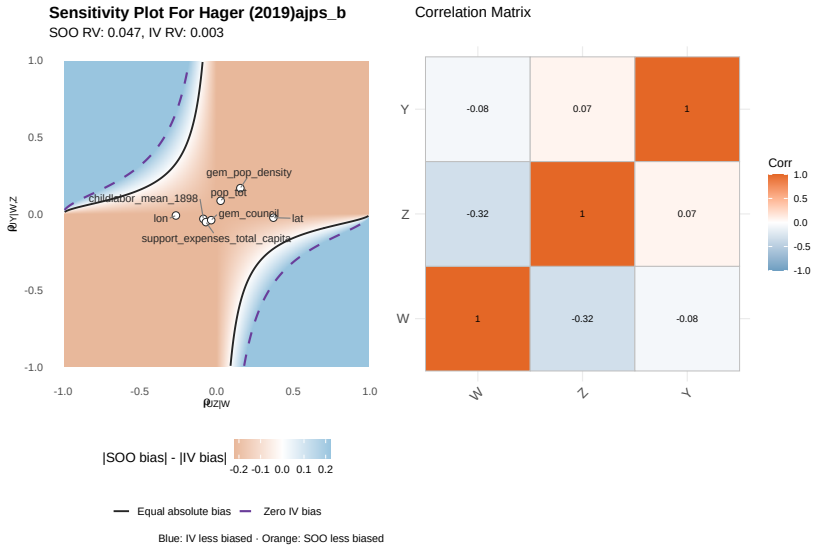

Sensitivity Plot For Gerber (2010)
SOO RV: 0.148, IV RV: 0.005
pre_direct_bushap
pre_aligned_vh
twonames
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

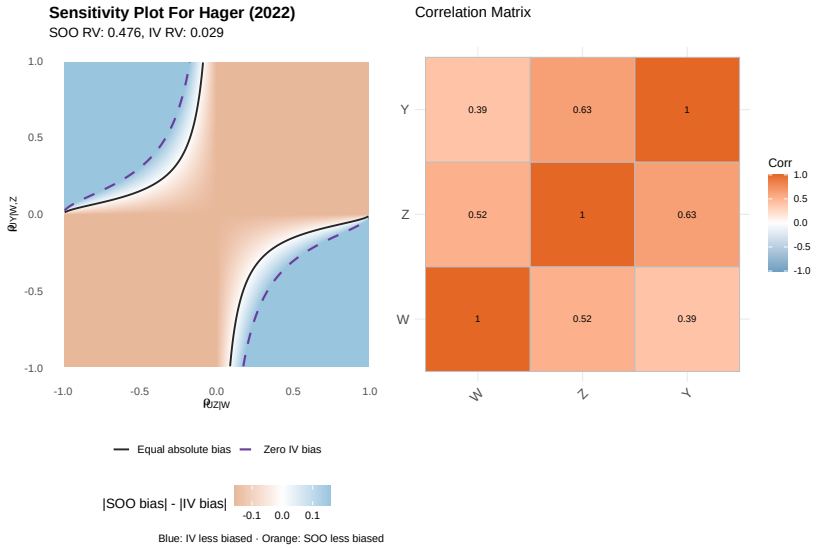

Sensitivity Plot For Goldstein (2017)
SOO RV: 0.072, IV RV: 0.004
city_propertytaxshare_r
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

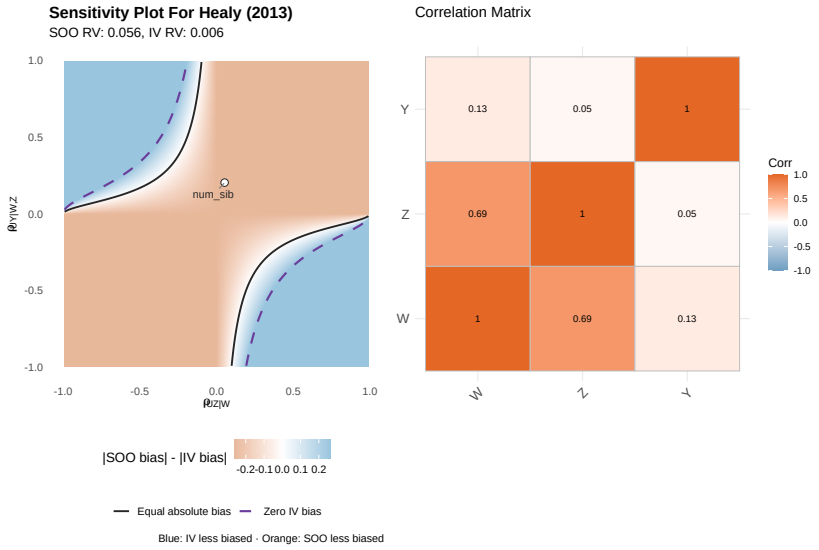

Sensitivity Plot For Grossman (2017)
SOO RV: 0.06, IV RV: 0.026
lgdppc_l
dpi_state_l
loilpc_l
p_polity2_l
wdi_urban_l
lpop_l
aid_pc_l
conflict_l
al_ethnic
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

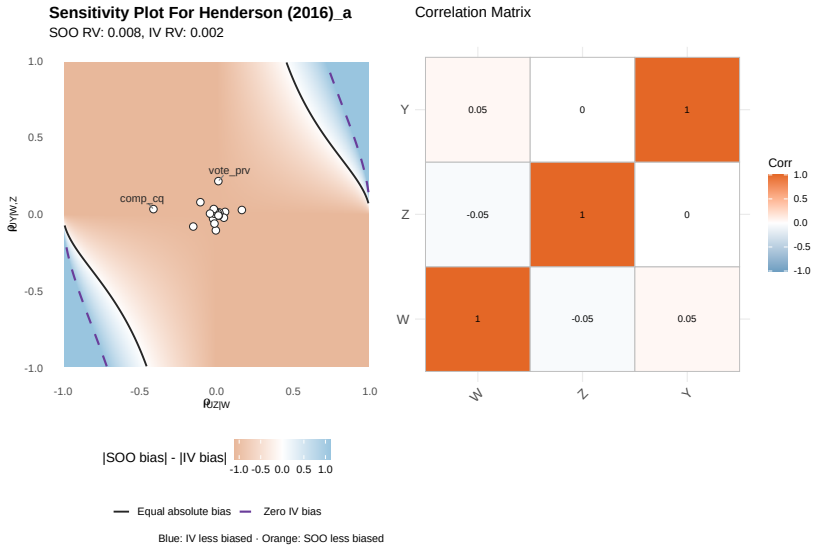

Sensitivity Plot For Hager (2019)_apsr
SOO RV: 0.062, IV RV: 0.008
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

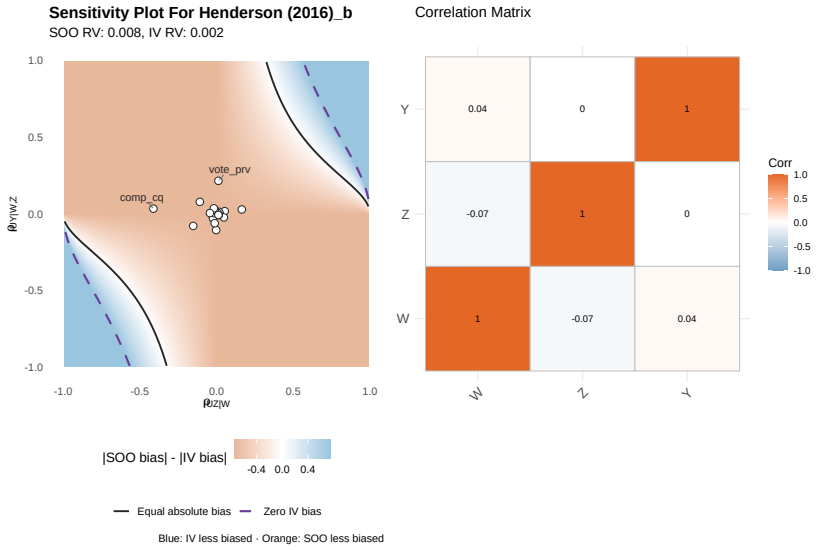

Sensitivity Plot For Hager (2019)ajps_a
SOO RV: 0.012, IV RV: 0.005
gem_pop_density
gem_council
pop_tot
childlabor_mean_1898
lon
lat
support_expenses_total_capita
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
-0.09
0.03
1
-0.18
1
0.03
1
-0.18
-0.09

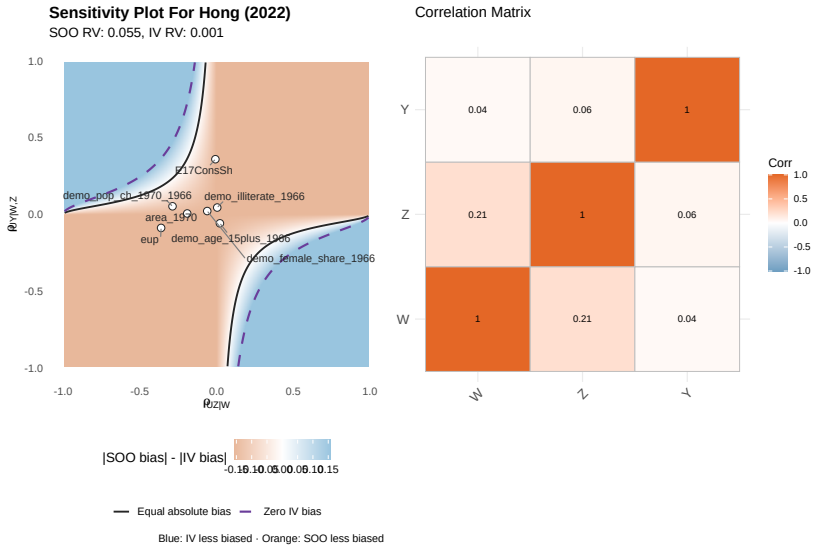

Sensitivity Plot For Hager (2019)ajps_b
SOO RV: 0.047, IV RV: 0.003
gem_pop_density
pop_tot
childlabor_mean_1898
lon
gem_council
lat
support_expenses_total_capita
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
-0.08
0.07
1
-0.32
1
0.07
1
-0.32
-0.08

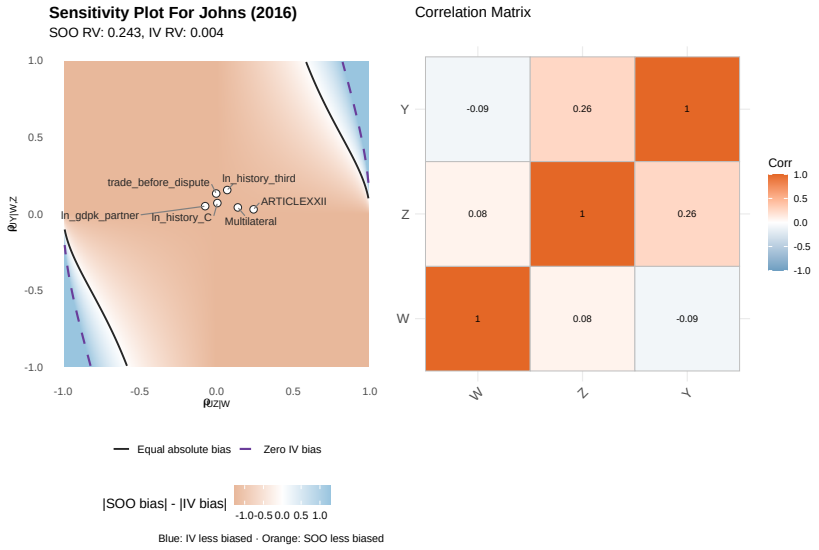

Sensitivity Plot For Hager (2022)
SOO RV: 0.476, IV RV: 0.029
Correlation Matrix
Y
Z
W
0.39
0.63
1
0.52
1
0.63
1
0.52
0.39
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.1 0.0 0.1
Blue: IV less biased · Orange: SOO less biased

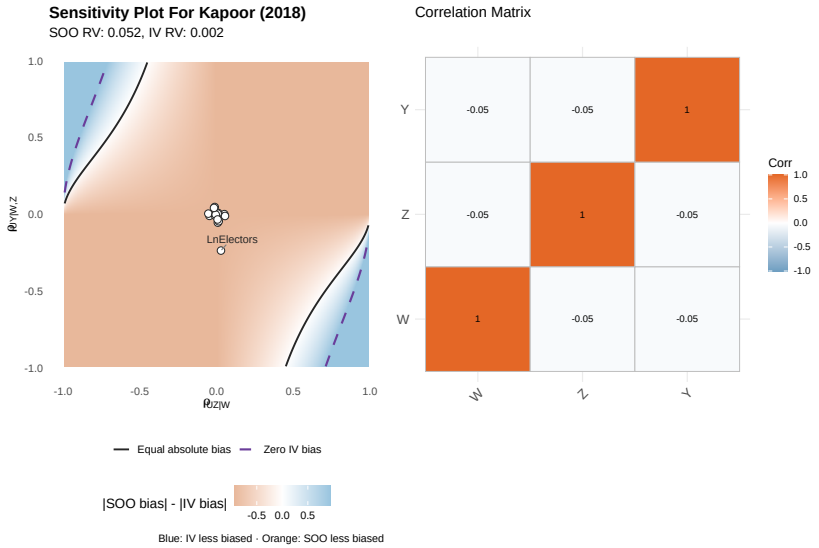

Sensitivity Plot For Healy (2013)
SOO RV: 0.056, IV RV: 0.006
Correlation Matrix
num_sib
Y
Z
W
0.13
0.05
1
0.69
1
0.05
1
0.69
0.13
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.2 -0.1 0.0 0.1 0.2
Blue: IV less biased · Orange: SOO less biased

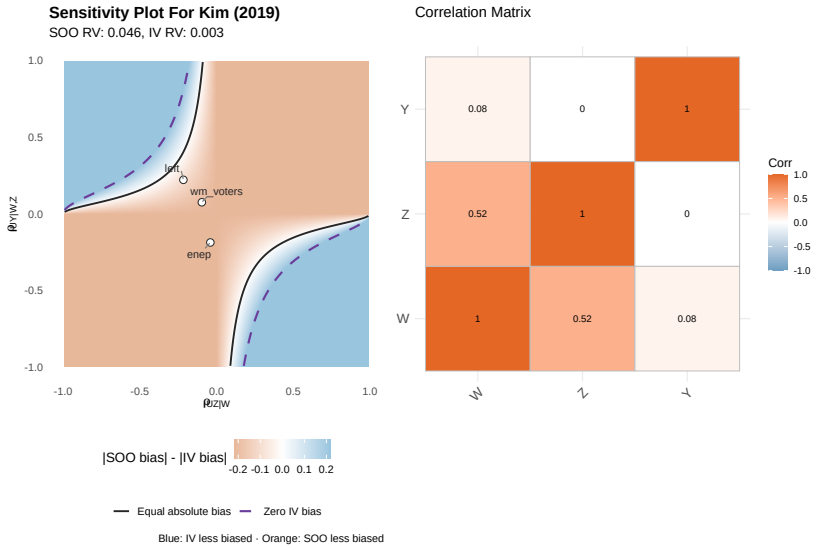

Sensitivity Plot For Henderson (2016)_a
SOO RV: 0.008, IV RV: 0.002
Correlation Matrix
vote_prv
comp_cq
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Corr
Y
Z
W
0.05
0
1
-0.05
1
0
1
-0.05
0.05

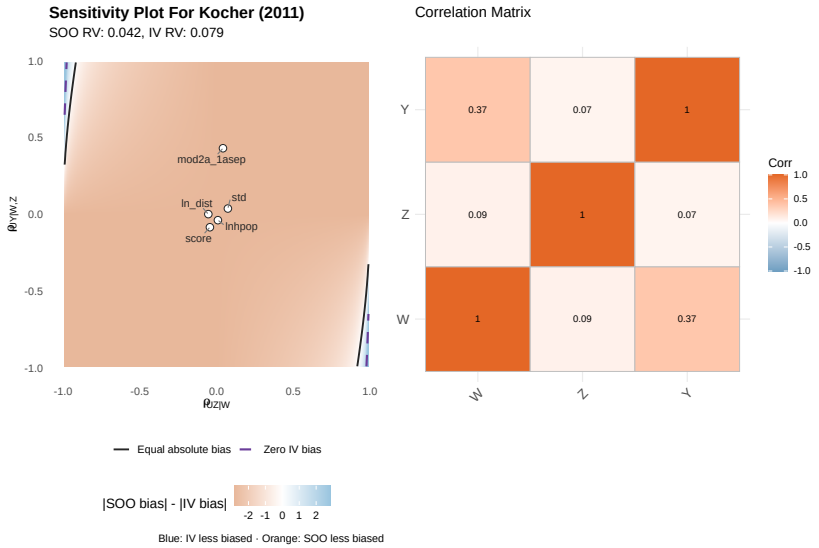

Sensitivity Plot For Henderson (2016)_b
SOO RV: 0.008, IV RV: 0.002
Correlation Matrix
vote_prv
comp_cq
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Corr
Y
Z
W
0.04
0
1
-0.07
1
0
1
-0.07
0.04

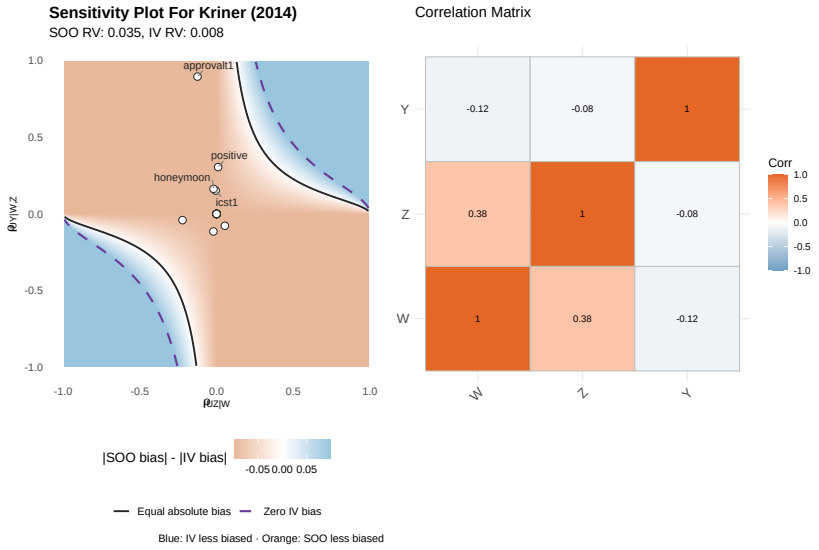

Sensitivity Plot For Hong (2022)
SOO RV: 0.055, IV RV: 0.001
Correlation Matrix
F17ConsSh
demo_pop_ch_1970_1966
demo_illiterate_1966
area_1970
eup
demo_age_15plus_1996
demo_female_share_1966
0.04
0.06
1
0.21
1
0.06
1
0.21
0.04
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

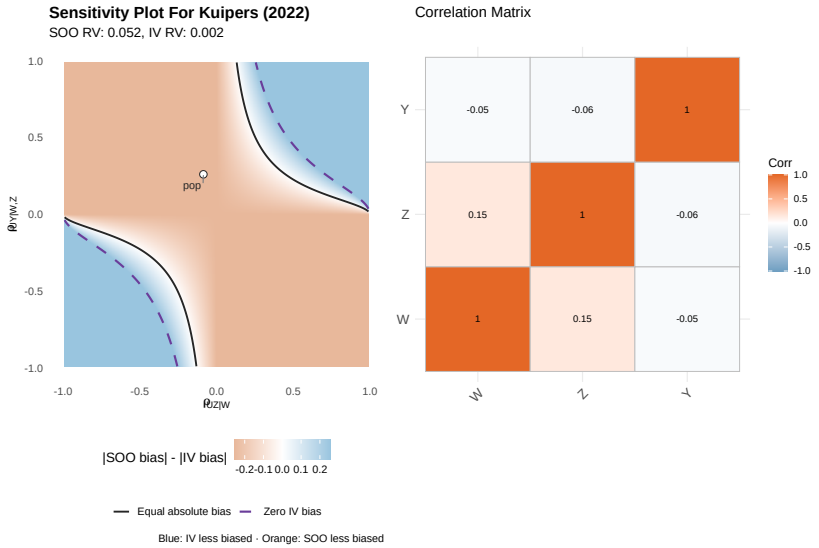

Sensitivity Plot For Johns (2016)
SOO RV: 0.243, IV RV: 0.004
Correlation Matrix
trade_before_dispute
ln_history_third
ln_gdpk_partner
ln_history_C
ARTICLEXXII
Multilateral
-0.09
0.26
1
0.08
1
0.26
1
0.08
-0.09
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

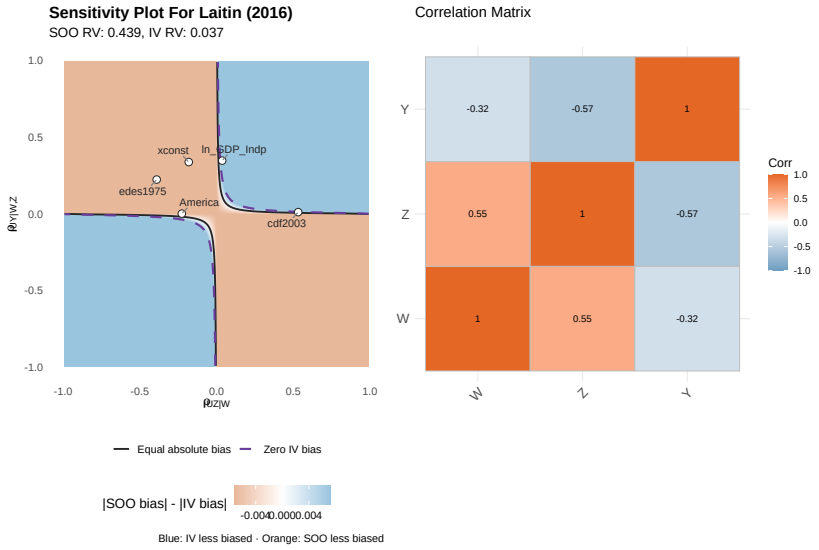

Sensitivity Plot For Kapoor (2018)
SOO RV: 0.052, IV RV: 0.002
LnElectors
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

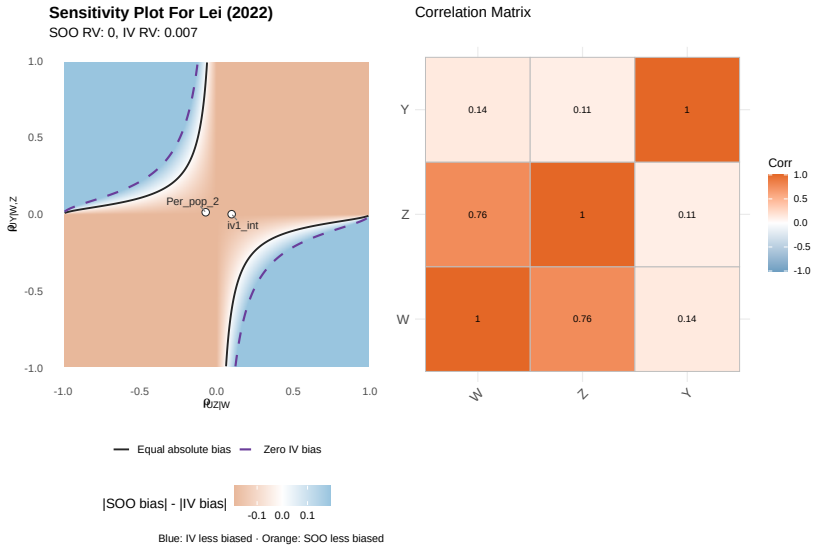

Sensitivity Plot For Kim (2019)
SOO RV: 0.046, IV RV: 0.003
lenft
wm_voters
enep
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

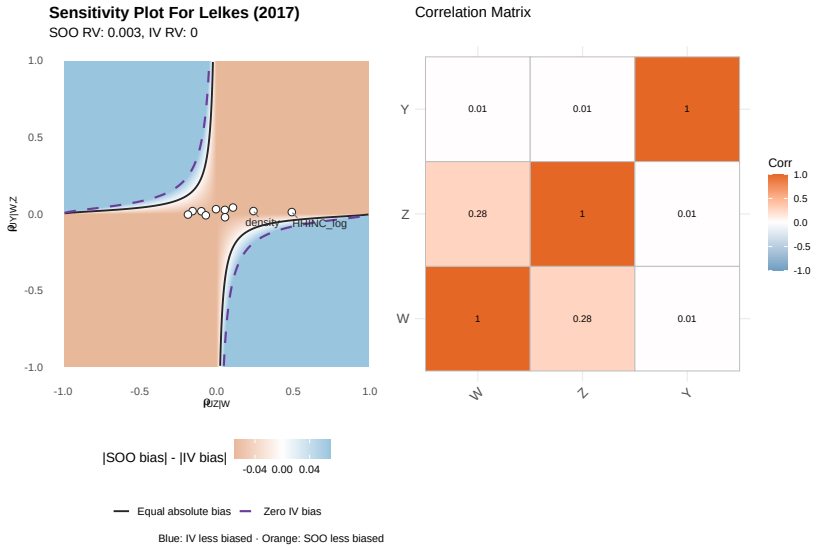

Sensitivity Plot For Kocher (2011)
SOO RV: 0.042, IV RV: 0.079
mod2a_1asep
ln_dist
std
lnhpop
score
Correlation Matrix
Y
Z
W
0.37
0.07
1
0.09
1
0.07
1
0.09
0.37
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

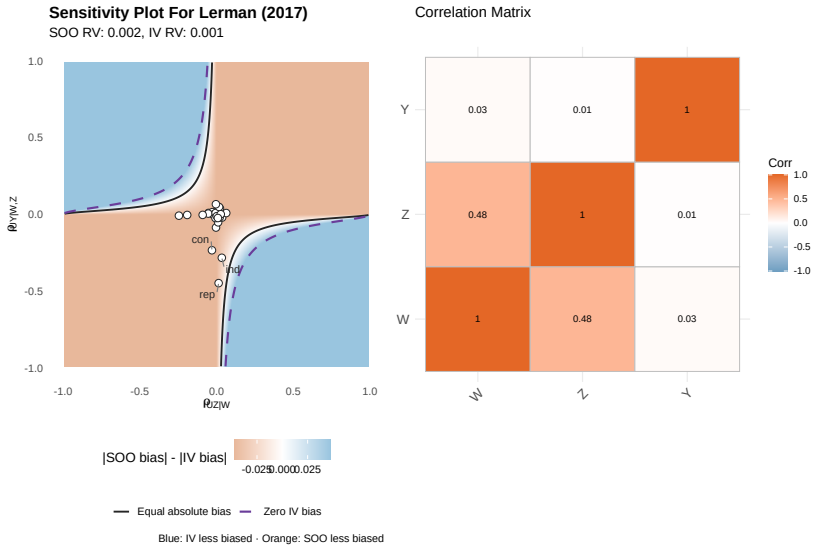

Sensitivity Plot For Kriner (2014)
SOO RV: 0.035, IV RV: 0.008
approvalt1
positive
honeymoon
icst1
Correlation Matrix
Y
Z
W
-0.12
-0.08
1
0.38
1
-0.08
1
0.38
-0.12
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

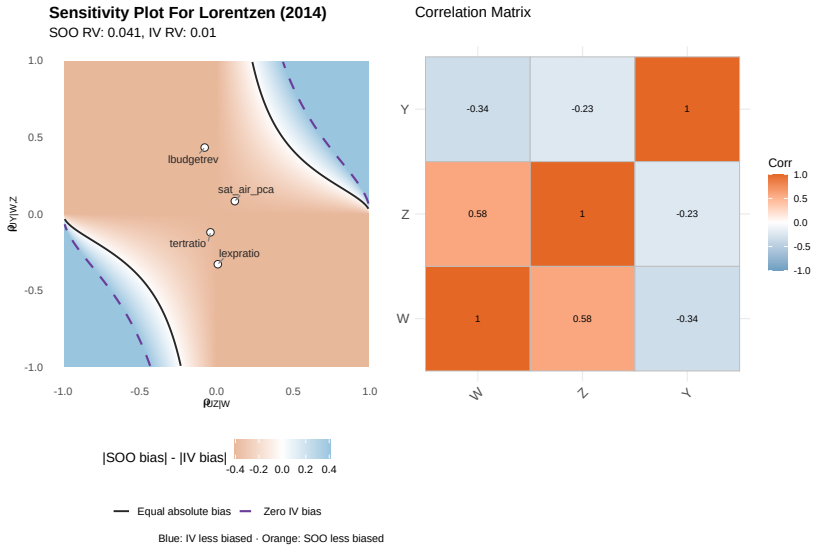

Sensitivity Plot For Kuipers (2022)
SOO RV: 0.052, IV RV: 0.002
Correlation Matrix
pop
-0.05
-0.06
1
0.15
1
-0.06
1
0.15
-0.05
Y
Z
W
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

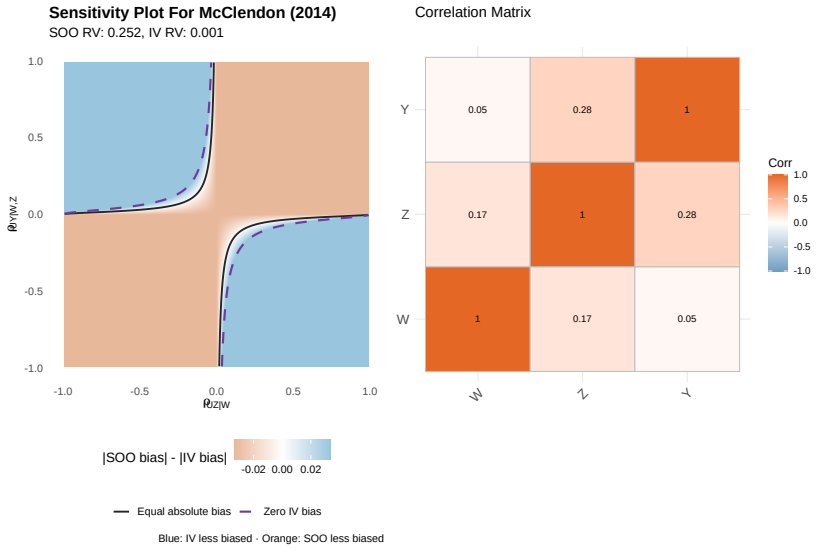

Sensitivity Plot For Laitin (2016)
SOO RV: 0.439, IV RV: 0.037
Correlation Matrix
xconst
ln_GDP_lndp
edes1975
America
cdf2003
-0.32
-0.57
1
0.55
1
-0.57
1
0.55
-0.32
Y
Z
W
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

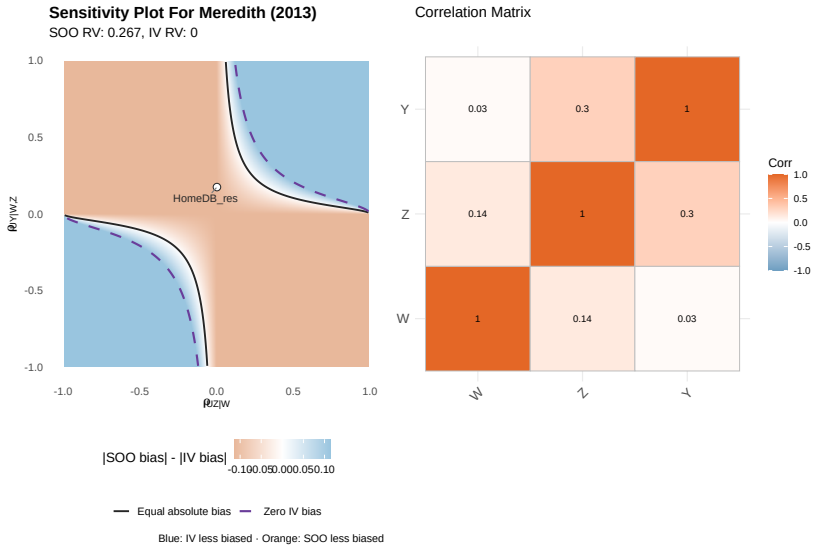

Sensitivity Plot For Lei (2022)
SOO RV: 0, IV RV: 0.007
Correlation Matrix
Per_pop_2
iv1_int
Y
Z
W
0.14
0.11
1
0.76
1
0.11
1
0.76
0.14
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

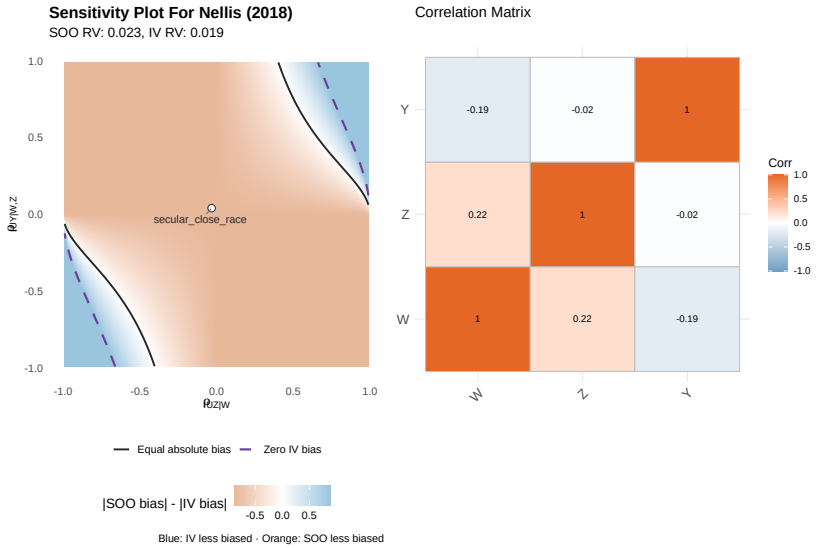

Sensitivity Plot For Lelkes (2017)
SOO RV: 0.003, IV RV: 0
Correlation Matrix
density_FIHHNC_log
Y
Z
W
0.01
0.01
1
0.28
1
0.01
1
0.28
0.01
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

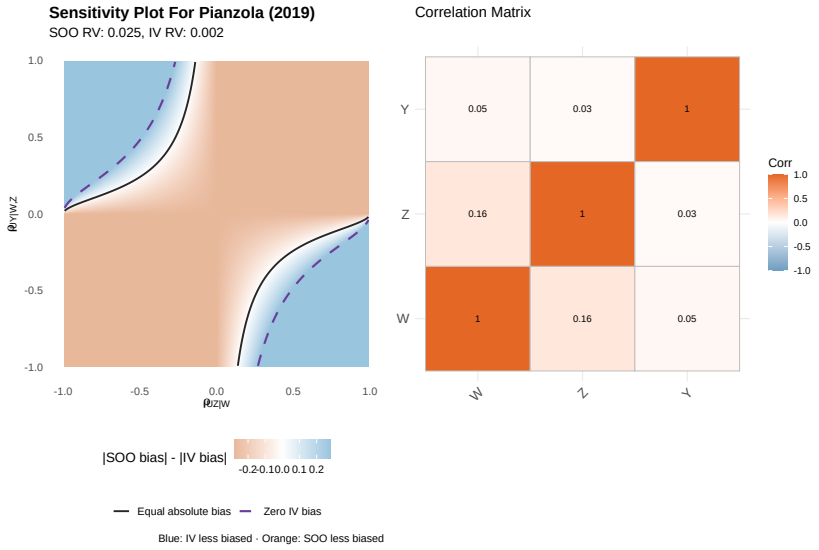

Sensitivity Plot For Lerman (2017)
SOO RV: 0.002, IV RV: 0.001
Correlation Matrix
con
ind
rep
Y
Z
W
0.03
0.01
1
0.48
1
0.01
1
0.48
0.03
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

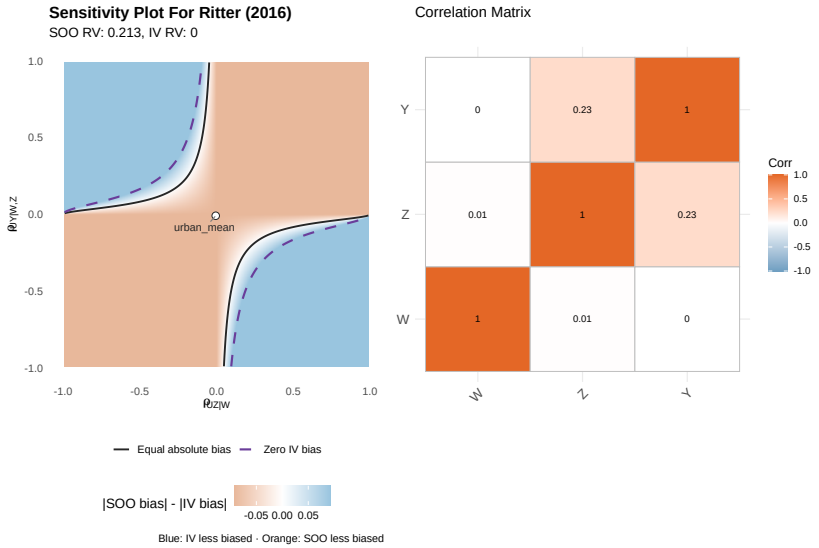

Sensitivity Plot For Lorentzen (2014)
SOO RV: 0.041, IV RV: 0.01
Correlation Matrix
lbudgetrev
sat_air_pca
tertratio
lexpratio
Y
Z
W
-0.34
-0.23
1
0.58
1
-0.23
1
0.58
-0.34
Corr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

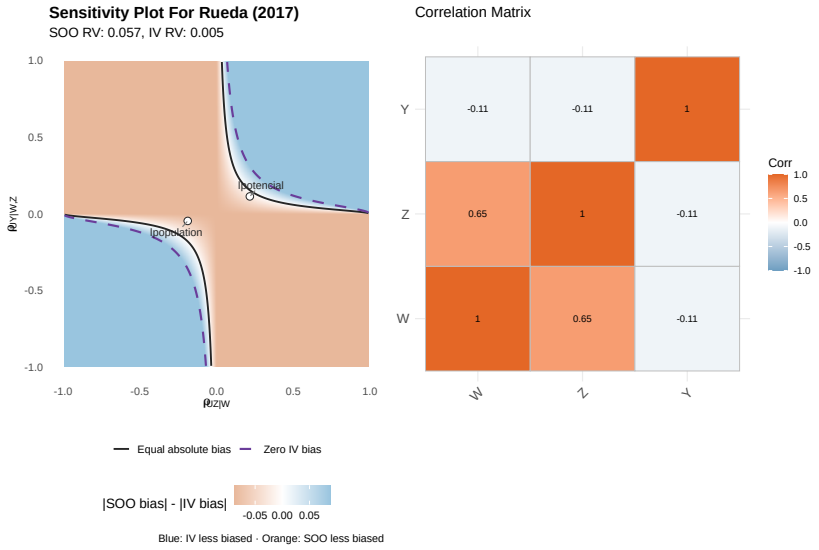

Sensitivity Plot For McClendon (2014)
SOO RV: 0.252, IV RV: 0.001
Correlation Matrix
1.0
0.5
0.0
-0.5
-1.0
ρY,Z|W
ρZ|W
-1.0
-0.5
0.0
0.5
1.0
Y
Z
W
0.05
0.28
1
0.17
1
0.28
1
0.17
0.05
Corr
1.0
0.5
0.0
-0.5
-1.0
|SOO bias| - |IV bias|
-0.02 0.00 0.02
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

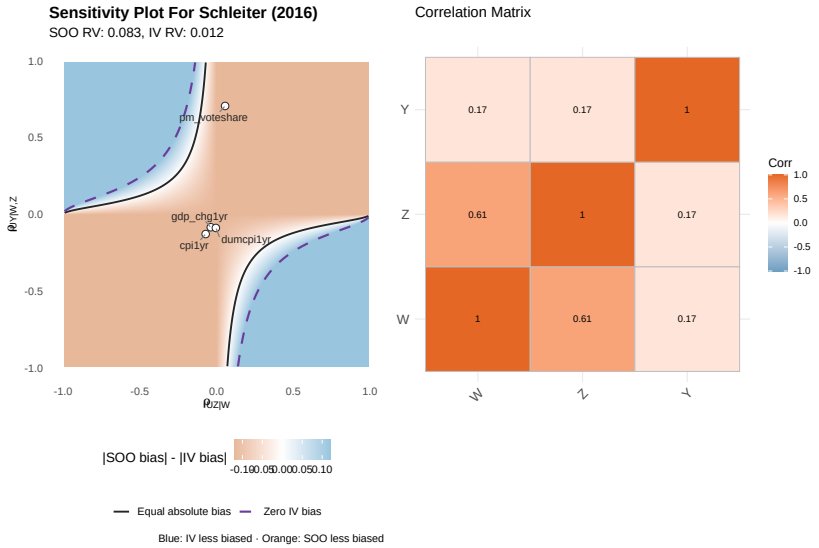

Sensitivity Plot For Meredith (2013)
SOO RV: 0.267, IV RV: 0
Correlation Matrix
HomeDB_res
1.0
0.5
0.0
-0.5
-1.0
ρY,Z|W
ρZ|W
-1.0
-0.5
0.0
0.5
1.0
Y
Z
W
0.03
0.3
1
0.14
1
0.3
1
0.14
0.03
Corr
1.0
0.5
0.0
-0.5
-1.0
|SOO bias| - |IV bias|
-0.10 -0.05 0.00 0.05 0.10
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased

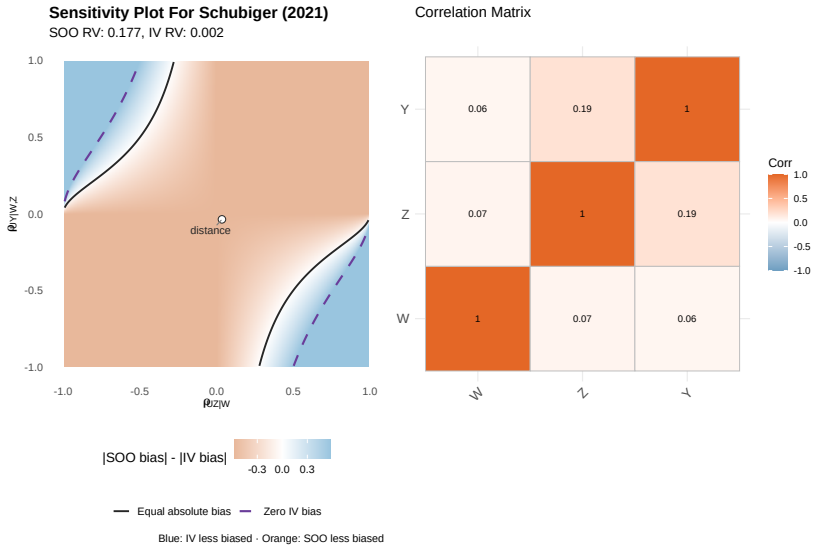

Sensitivity Plot For Nellis (2018)
SOO RV: 0.023, IV RV: 0.019
secular_close_race
Correlation Matrix
Y
Z
W
-0.19
-0.02
1
0.22
1
-0.02
1
0.22
-0.19
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

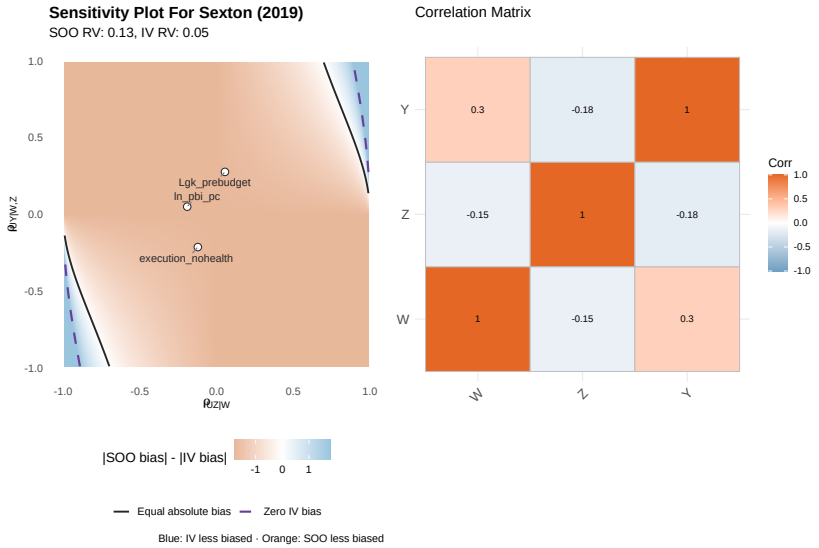

Sensitivity Plot For Pianzola (2019)
SOO RV: 0.025, IV RV: 0.002
Correlation Matrix
Y
Z
W
0.05
0.03
1
0.16
1
0.03
1
0.16
0.05
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

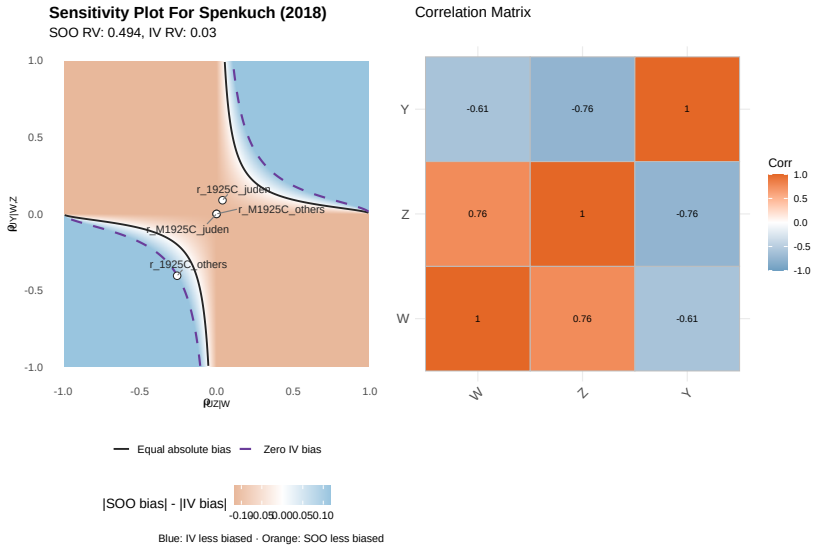

Sensitivity Plot For Ritter (2016)
SOO RV: 0.213, IV RV: 0
Correlation Matrix
urban_mean
Y
Z
W
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

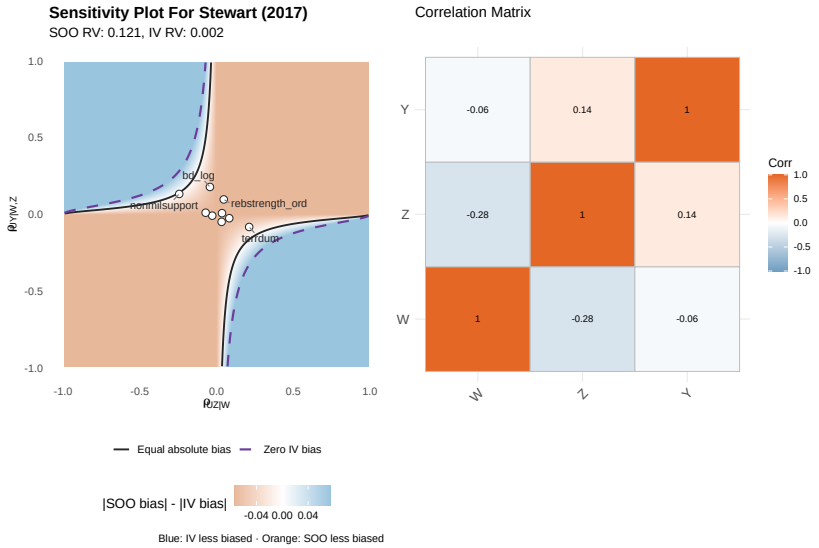

Sensitivity Plot For Rueda (2017)
SOO RV: 0.057, IV RV: 0.005
Correlation Matrix
Ipotensial
Ipopulation
Y
Z
W
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

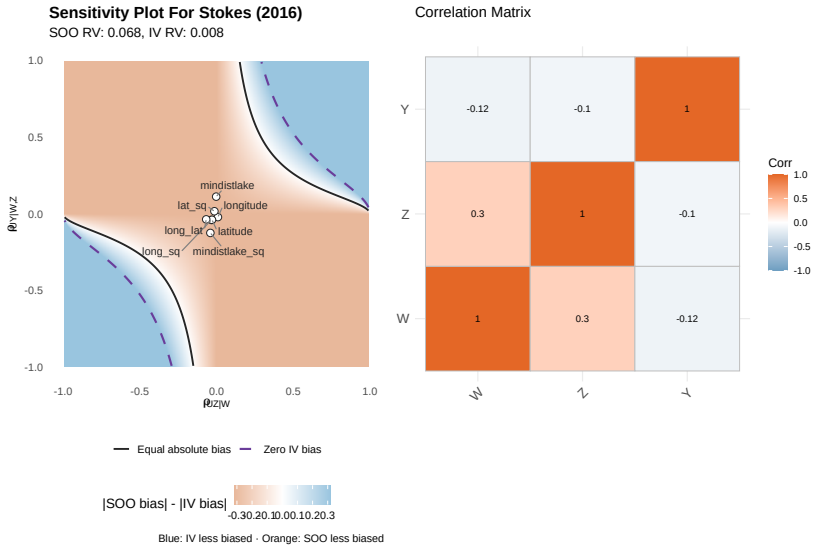

Sensitivity Plot For Schleiter (2016)
SOO RV: 0.083, IV RV: 0.012
pm_voteshare
gdp_chg1yr
cpi1yr
dumcpi1yr
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
0.17
0.17
1
0.61
1
0.17
1
0.61
0.17

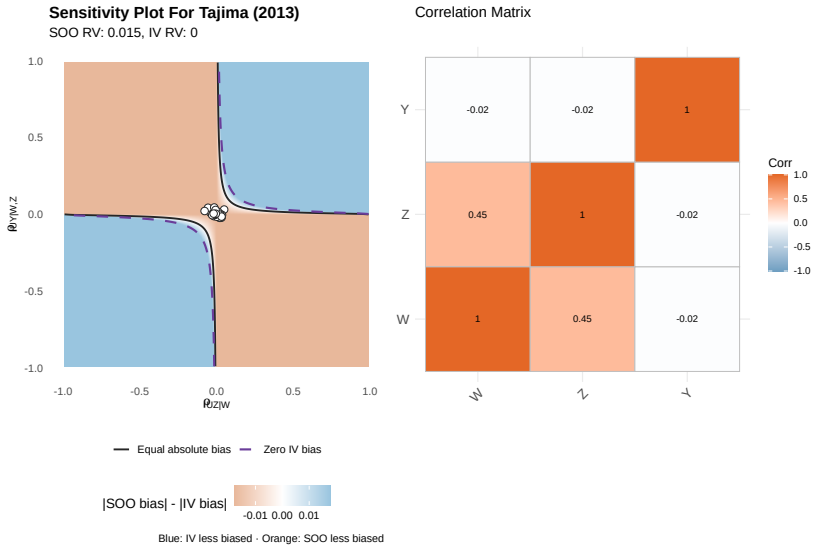

Sensitivity Plot For Schubiger (2021)
SOO RV: 0.177, IV RV: 0.002
distance
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr
Y
Z
W
0.06
0.19
1
0.07
1
0.19
1
0.07
0.06

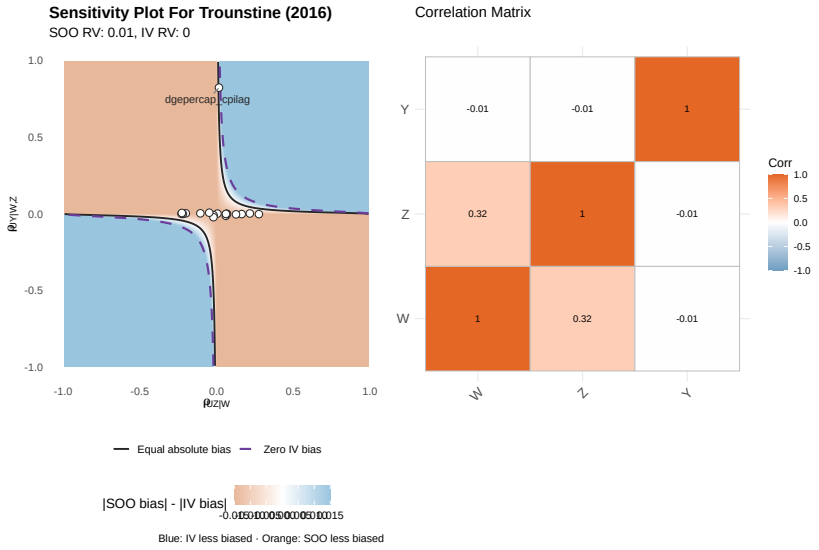

Sensitivity Plot For Sexton (2019)
SOO RV: 0.13, IV RV: 0.05
Correlation Matrix
Lgk_prebudget
ln_pbi_pc
execution_nohealth
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Corr
Y
Z
W

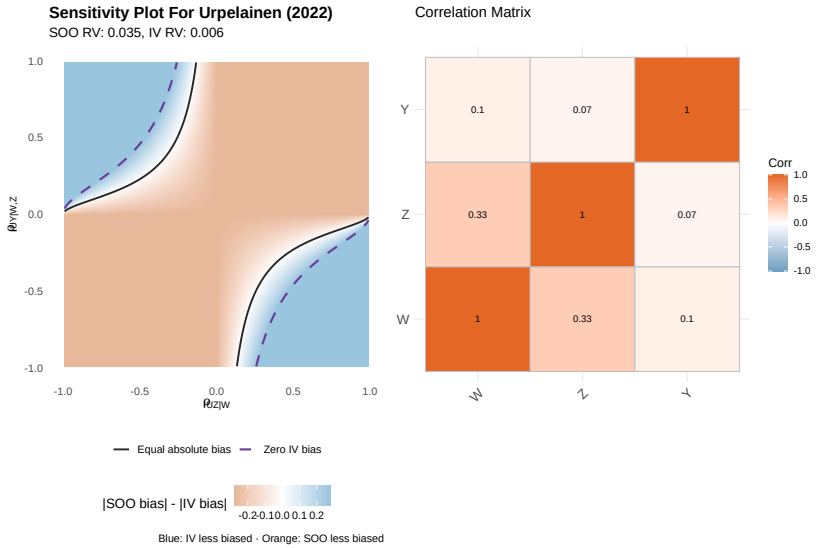

Sensitivity Plot For Spenkuch (2018)
SOO RV: 0.494, IV RV: 0.03
Correlation Matrix
r_1925C_juden
r_M1925C_others
r_M1925C_juden
r_1925C_others
|SOO bias| - |IV bias|
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Corr
Y
Z
W

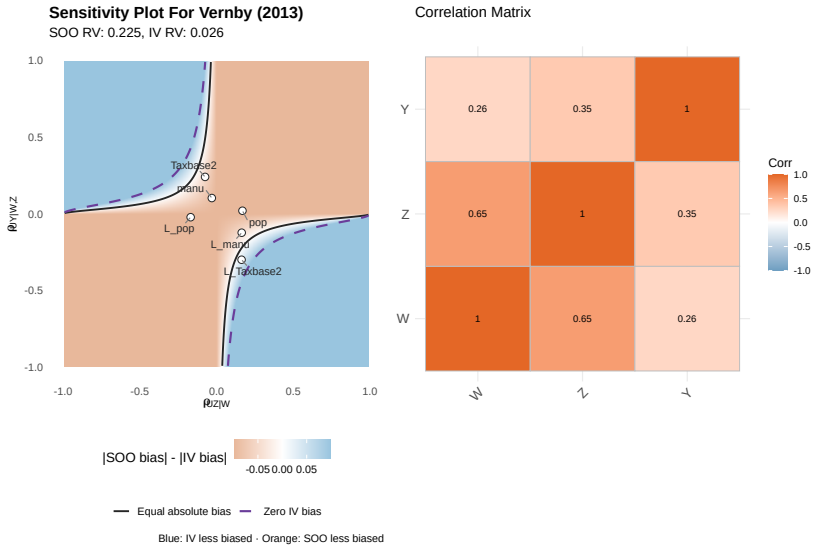

Sensitivity Plot For Stewart (2017)
SOO RV: 0.121, IV RV: 0.002
bd_log
rebstrength_ord
nonmilsupport
terrdum
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Y
Z
W
-0.06
0.14
1
-0.28
1
0.14
1
-0.28
-0.06
Corr

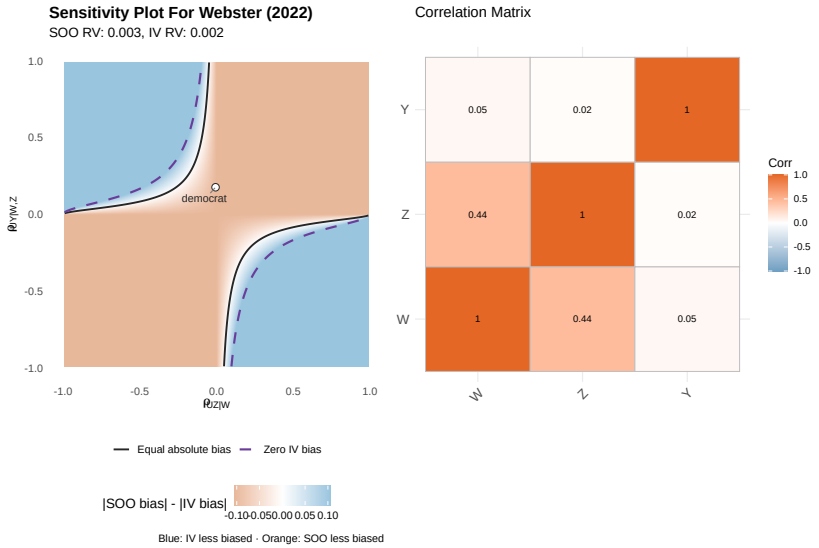

Sensitivity Plot For Stokes (2016)
SOO RV: 0.068, IV RV: 0.008
mindistlake
lat_sq
longitude
long_lat
latitude
long_sq
mindistlake_sq
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Y
Z
W
-0.12
-0.1
1
0.3
1
-0.1
1
0.3
-0.12
Corr

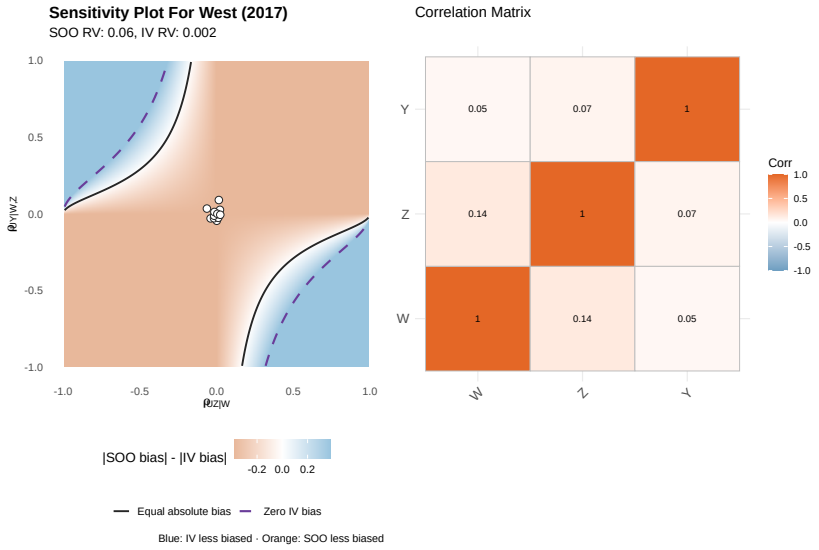

Sensitivity Plot For Tajima (2013)
SOO RV: 0.015, IV RV: 0
Correlation Matrix
Y
Z
W
-0.02
-0.02
1
0.45
1
-0.02
1
0.45
-0.02
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-0.01 0.00 0.01
Blue: IV less biased · Orange: SOO less biased

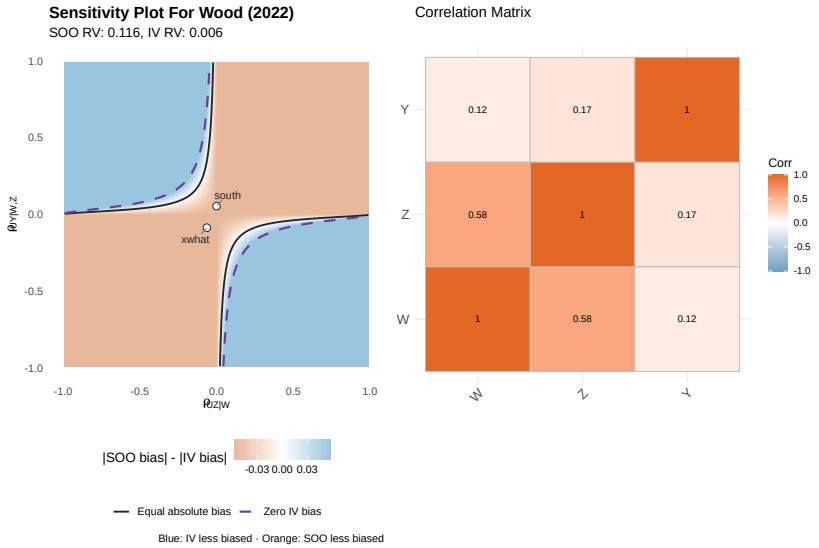

Sensitivity Plot For Trounstine (2016)
SOO RV: 0.01, IV RV: 0
Correlation Matrix
dgepercap, lcpilag
Y
Z
W
-0.01
-0.01
1
0.32
1
-0.01
1
0.32
-0.01
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

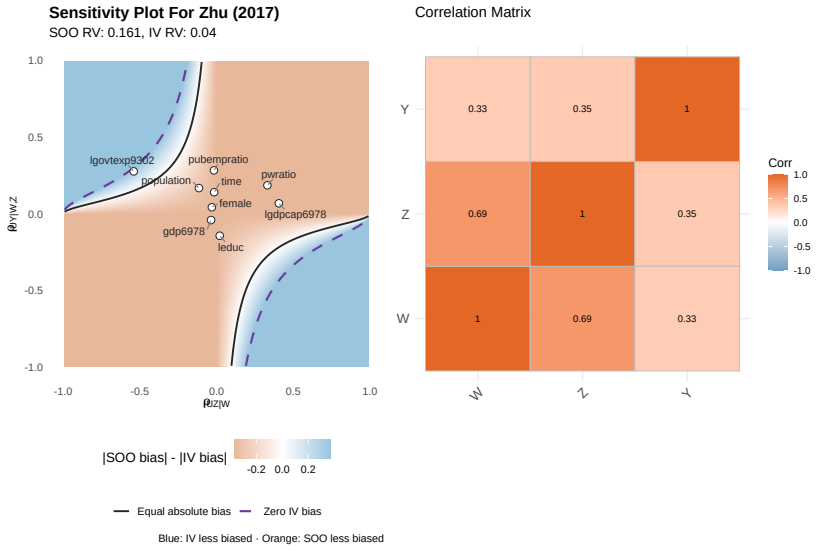

Sensitivity Plot For Urpelainen (2022)
SOO RV: 0.035, IV RV: 0.006
Correlation Matrix
Y
Z
W
0.1
0.07
1
0.33
1
0.07
1
0.33
0.1
Corr
1.0
0.5
0.0
-0.5
-1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased

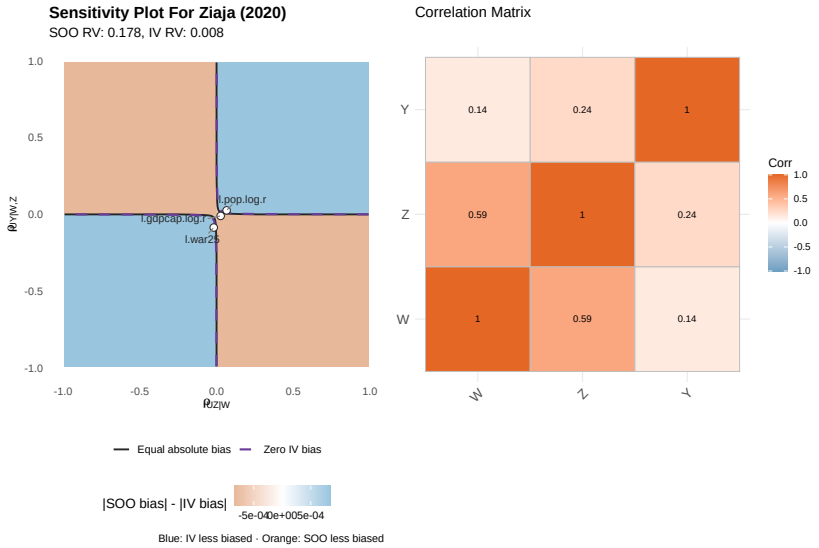

Sensitivity Plot For Vernby (2013)
SOO RV: 0.225, IV RV: 0.026
Correlation Matrix
Taxbase2
manu
L_pop
pop
L_manu
L_Taxbase2
Y
Z
W
0.26
0.35
1
0.65
1
0.35
1
0.65
0.26
Corr
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased


Sensitivity Plot For Webster (2022)
SOO RV: 0.003, IV RV: 0.002
democrat
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr


Sensitivity Plot For West (2017)
SOO RV: 0.06, IV RV: 0.002
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Corr

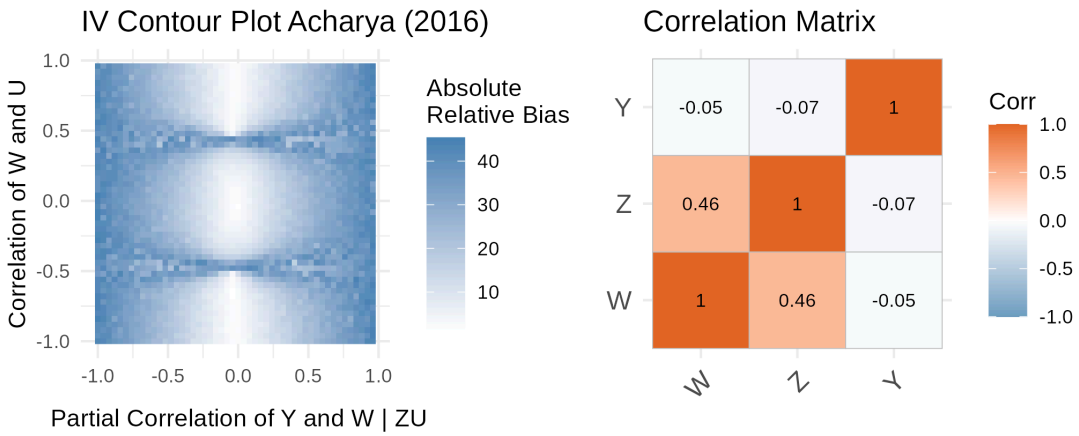

Sensitivity Plot For Wood (2022)
SOO RV: 0.116, IV RV: 0.006
south
xwhat
|SOO bias| - |IV bias|
-0.03 0.00 0.03
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Y
Z
W
0.12
0.17
1
0.58
1
0.17
1
0.58
0.12
Corr
1.0
0.5
0.0
-0.5
-1.0

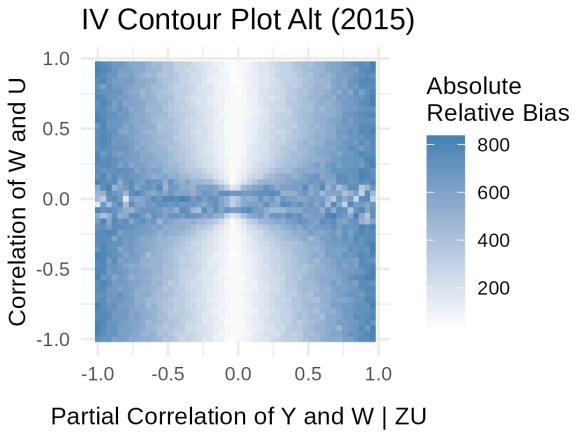

Sensitivity Plot For Zhu (2017)
SOO RV: 0.161, IV RV: 0.04
lgovtexp9302
pubempratio
population
time
pwratio
female
lgdpcap6978
gdp6978
leduc
|SOO bias| - |IV bias|
-0.2 0.0 0.2
Equal absolute bias
Zero IV bias
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Y
Z
W
0.33
0.35
1
0.69
1
0.35
1
0.69
0.33
Corr
1.0
0.5
0.0
-0.5
-1.0

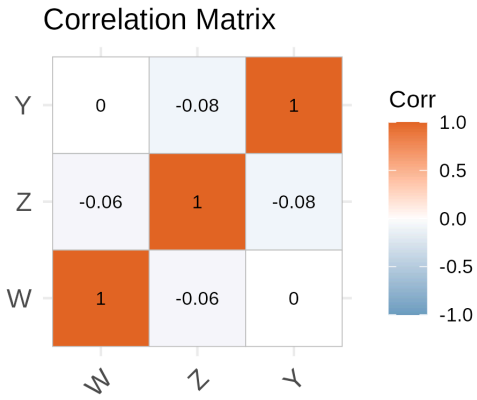

Sensitivity Plot For Ziaja (2020)
SOO RV: 0.178, IV RV: 0.008
1.0
0.5
0.0
-0.5
-1.0
l.pop.log.r
l.gdpcap.log.r
l.war25
-1.0
-0.5
0.0
0.5
1.0
Equal absolute bias
Zero IV bias
|SOO bias| - |IV bias|
-5e-04
0e+00
5e-04
Blue: IV less biased · Orange: SOO less biased
Correlation Matrix
Y
Z
W
0.14
0.24
1
0.59
1
0.24
1
0.59
0.14
Corr
1.0
0.5
0.0
-0.5
-1.0

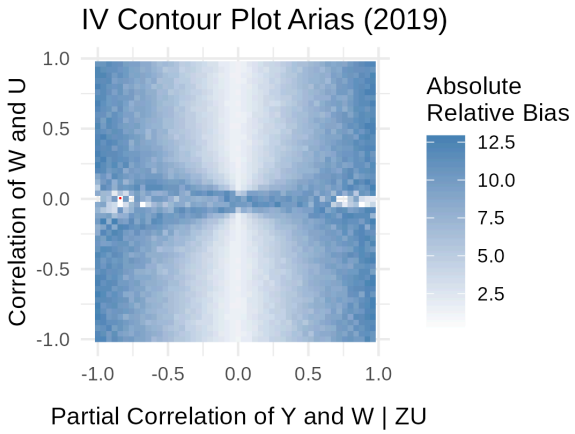
IV Contour Plots

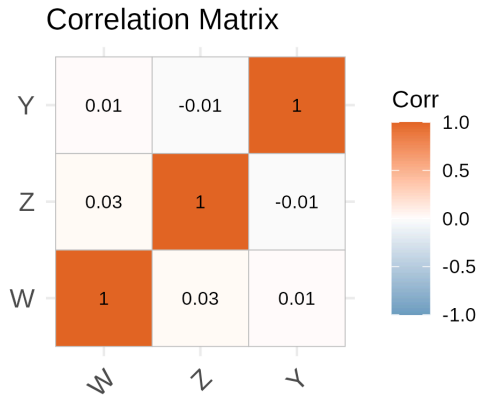
This section reports IV contour plots for the 69 analyses we re-analyze.

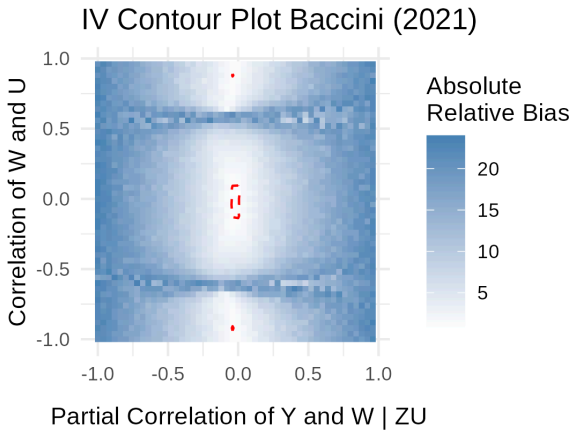


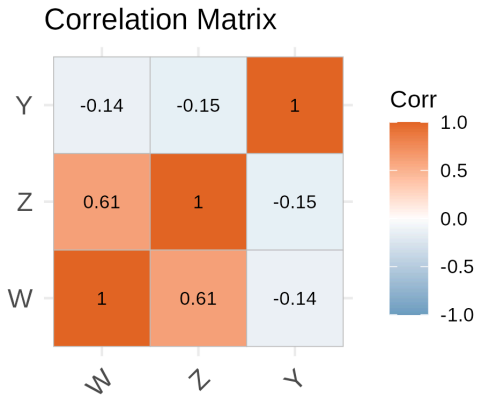

IV Contour Plot Alt (2015)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
800
600
400
200

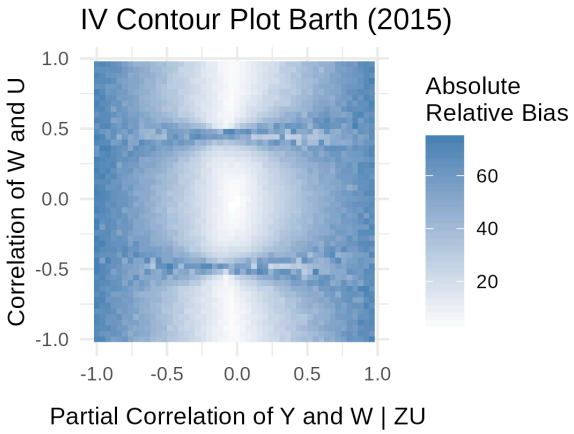

Correlation Matrix
Y
Z
W
0
-0.08
1
-0.06
1
-0.08
1
-0.06
0
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

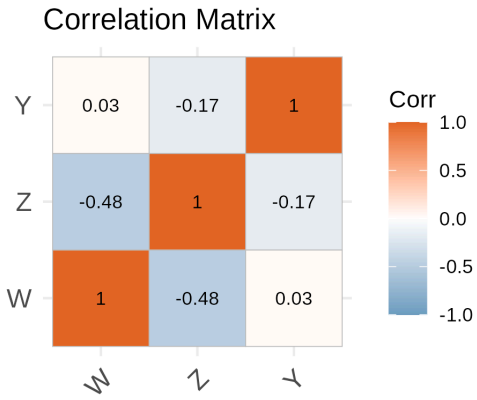

IV Contour Plot Arias (2019)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
12.5
10.0
7.5
5.0
2.5

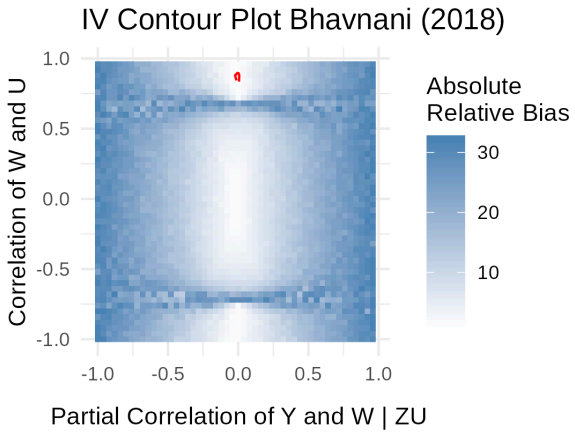

Correlation Matrix
Y
Z
W
0.01
-0.01
1
0.03
1
-0.01
1
0.03
0.01
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

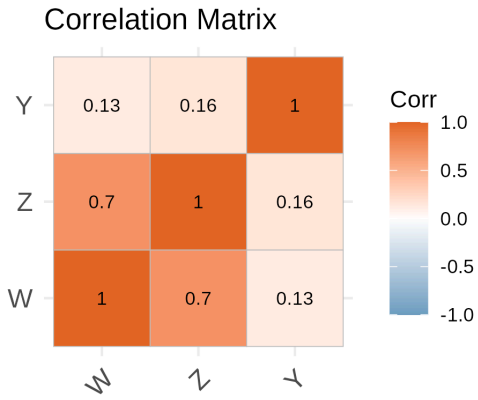

IV Contour Plot Baccini (2021)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
20
15
10
5

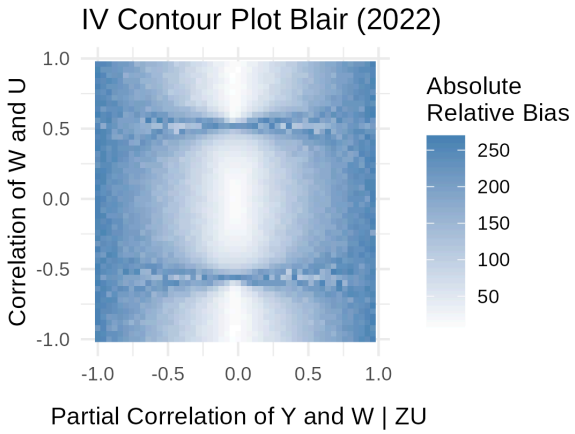

Correlation Matrix
Y
Z
W
-0.14
-0.15
1
0.61
1
-0.15
1
0.61
-0.14
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

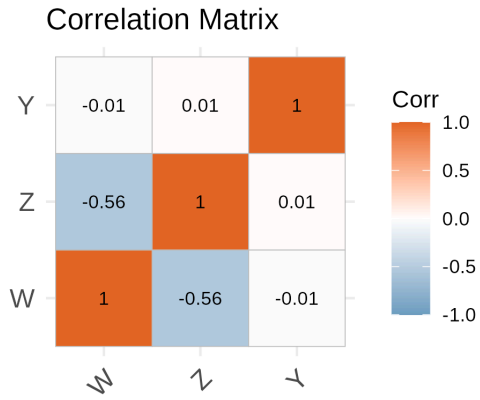

IV Contour Plot Barth (2015)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
60
40
20

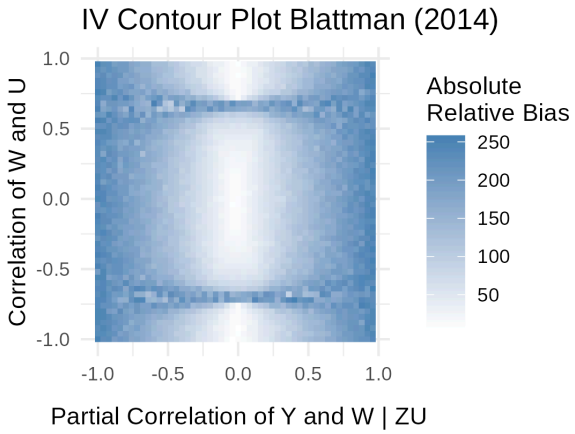

Correlation Matrix
Y
Z
W
0.03
-0.17
1
-0.48
1
-0.17
1
-0.48
0.03
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

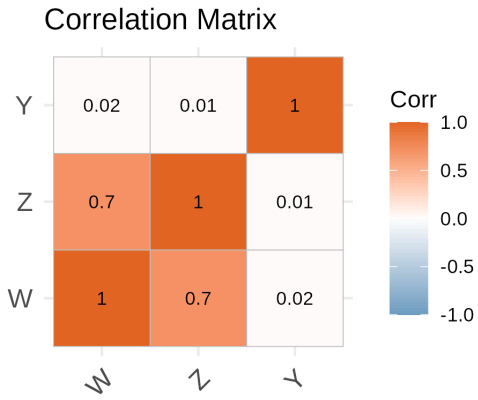

IV Contour Plot Bhavnani (2018)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
30
20
10

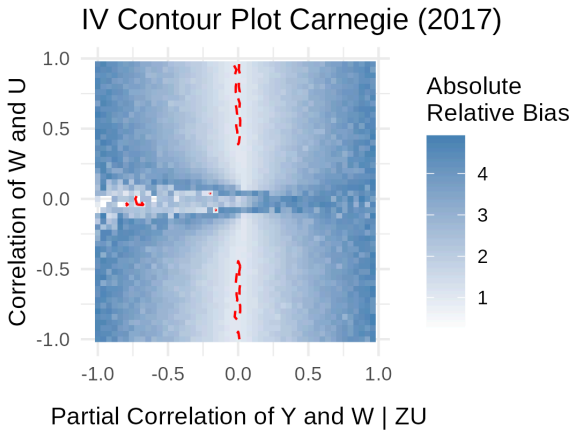

Correlation Matrix
Y
Z
W
0.13
0.16
1
0.7
1
0.16
1
0.7
0.13
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

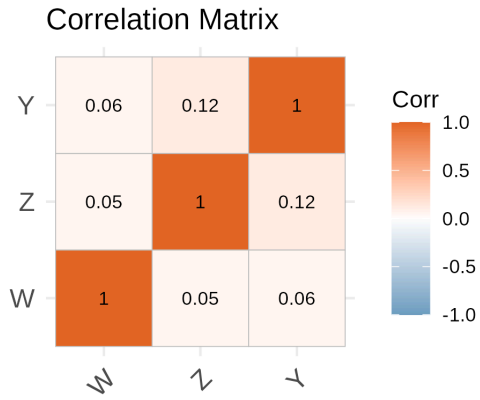

IV Contour Plot Blair (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
250
200
150
100
50

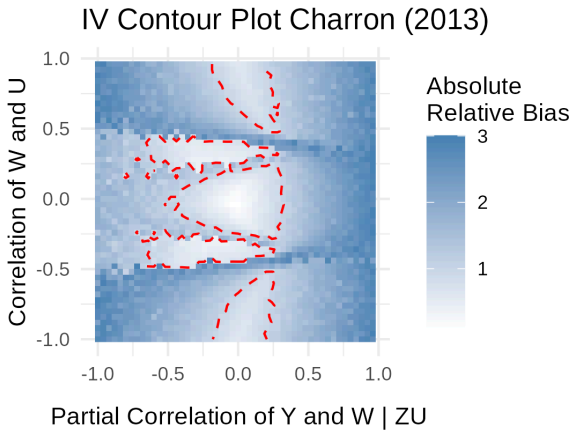

Correlation Matrix
Y
Z
W
-0.01
0.01
1
-0.56
1
0.01
1
-0.56
-0.01
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

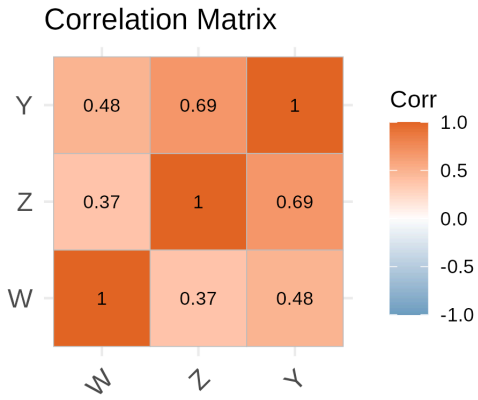

IV Contour Plot Blattman (2014)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
250
200
150
100
50

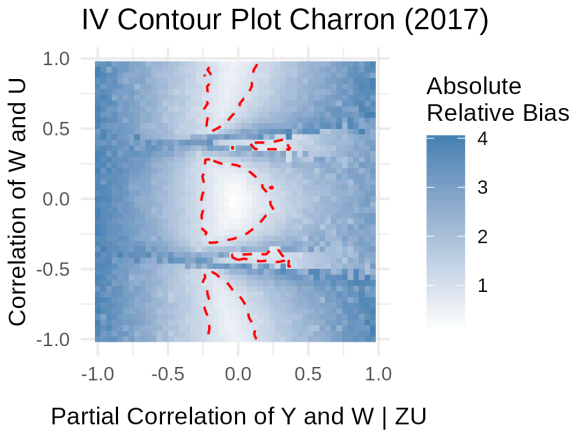

Correlation Matrix
Y
Z
W
0.02
0.01
1
0.7
1
0.01
1
0.7
0.02
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

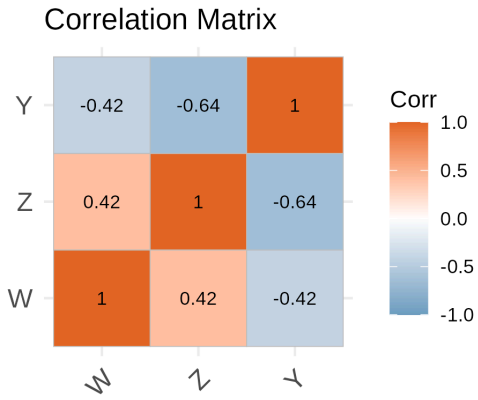

IV Contour Plot Carnegie (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
4
3
2
1

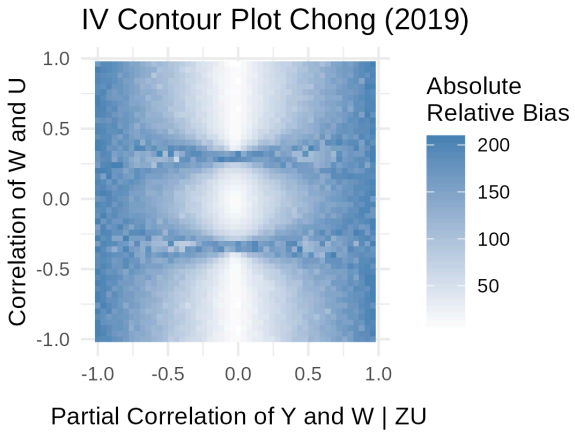

Correlation Matrix
Y
Z
W
0.06
0.12
1
0.05
1
0.12
1
0.05
0.06
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

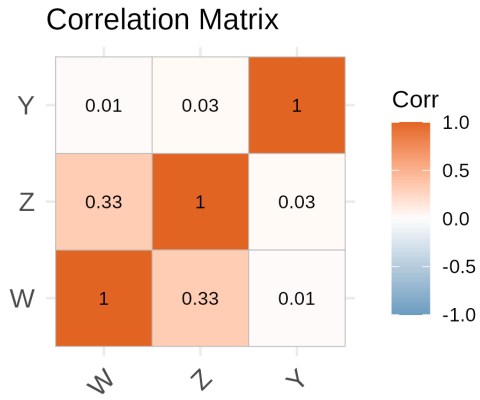

IV Contour Plot Charron (2013)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
3
2
1

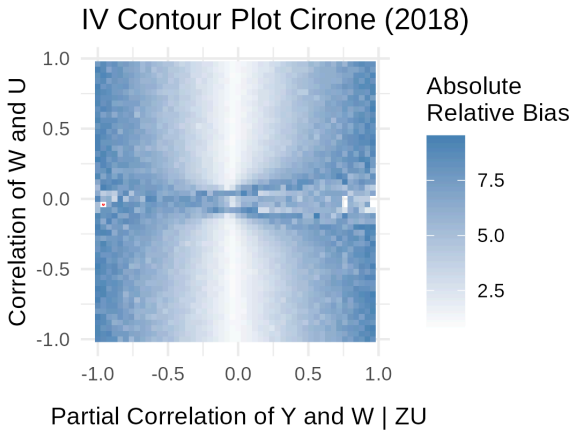

Correlation Matrix
Y
Z
W
0.48
0.69
1
0.37
1
0.69
1
0.37
0.48
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

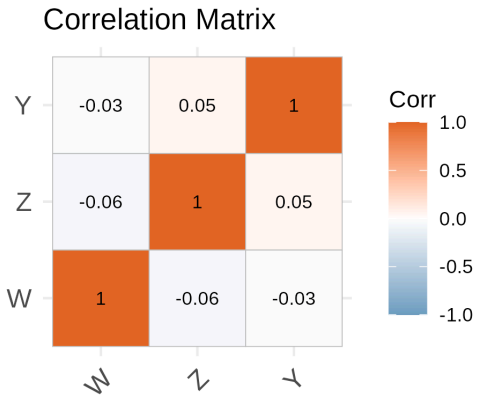

IV Contour Plot Charron (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
4
3
2
1

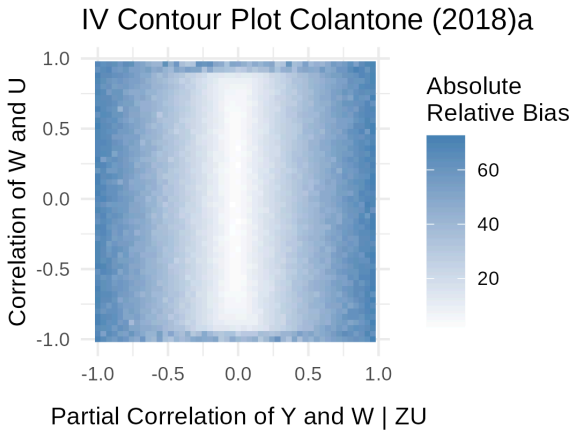

Correlation Matrix
Y
Z
W
-0.42
-0.64
1
0.42
1
-0.64
1
0.42
-0.42
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

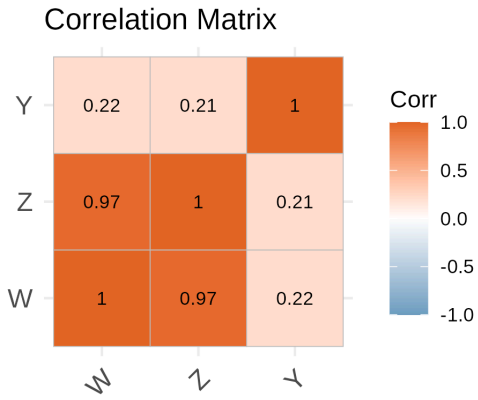

IV Contour Plot Chong (2019)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
200
150
100
50

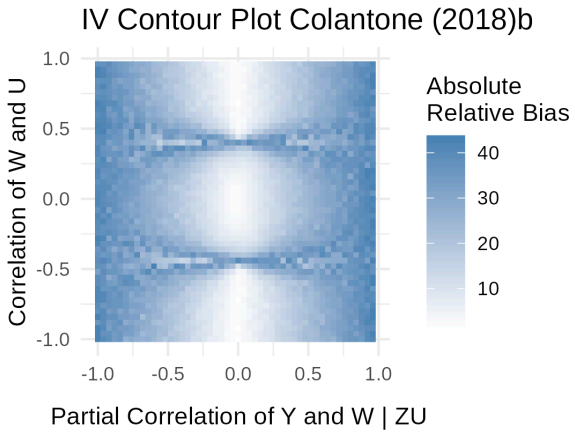

Correlation Matrix
Y
Z
W
0.01
0.03
1
0.33
1
0.03
1
0.33
0.01
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

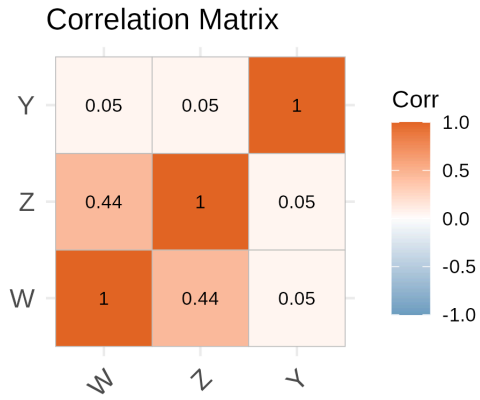

IV Contour Plot Cirone (2018)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
7.5
5.0
2.5

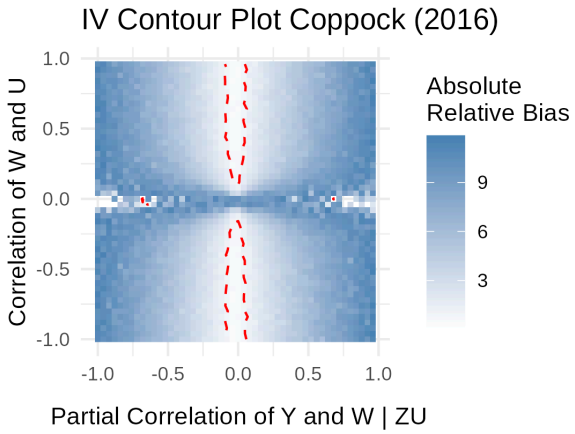

Correlation Matrix
Y
Z
W
-0.03
0.05
1
-0.06
1
0.05
1
-0.06
-0.03
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

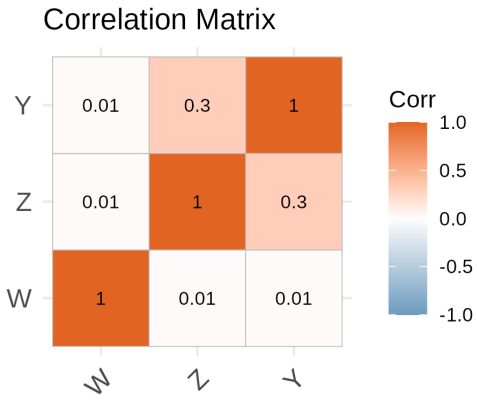

IV Contour Plot Colantone (2018)a
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
60
40
20

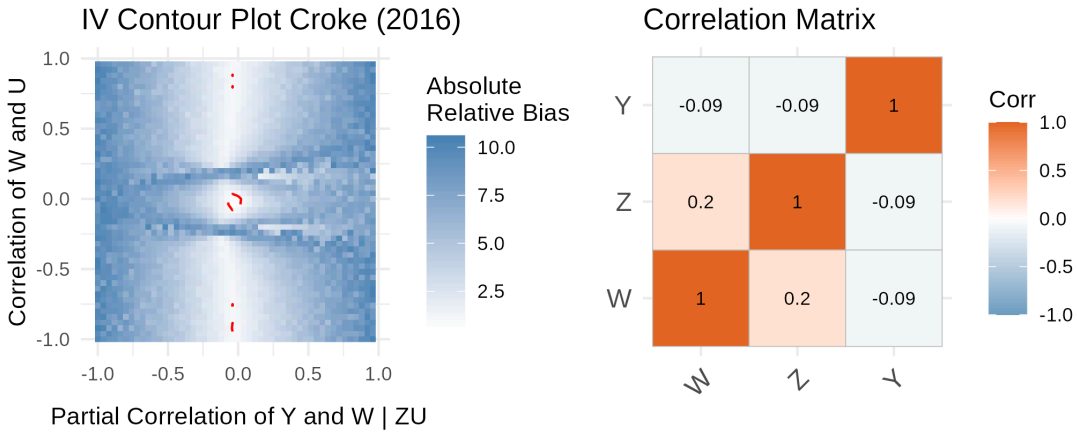

Correlation Matrix
Y
Z
W
0.22
0.21
1
0.97
1
0.21
1
0.97
0.22
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

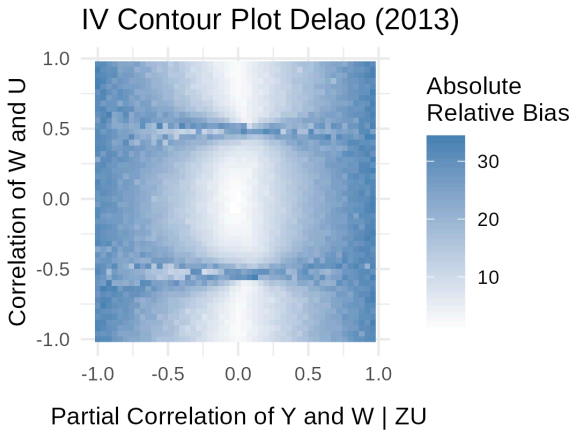

IV Contour Plot Colantone (2018)b
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
40
30
20
10

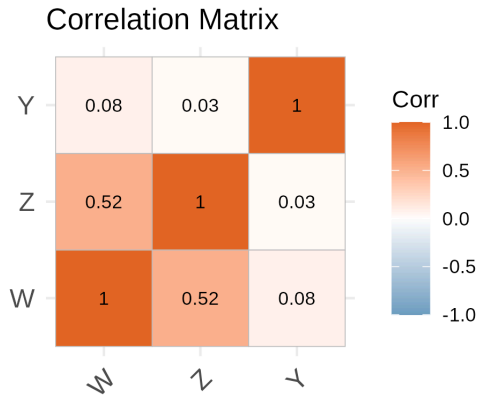

Correlation Matrix
Y
Z
W
0.05
0.05
1
0.44
1
0.05
1
0.44
0.05
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

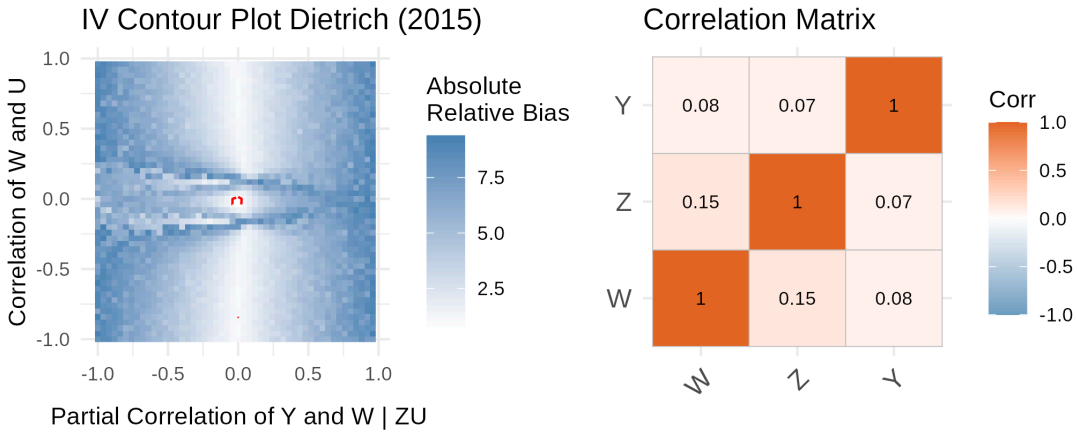

IV Contour Plot Coppock (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
9
6
3

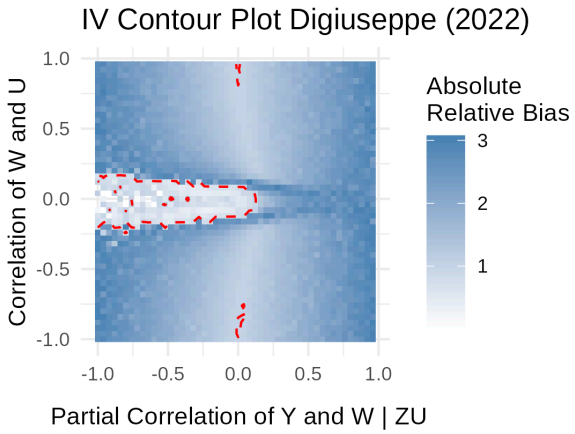

Correlation Matrix
Y
Z
W
0.01
0.3
1
0.01
1
0.3
1
0.01
0.01
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

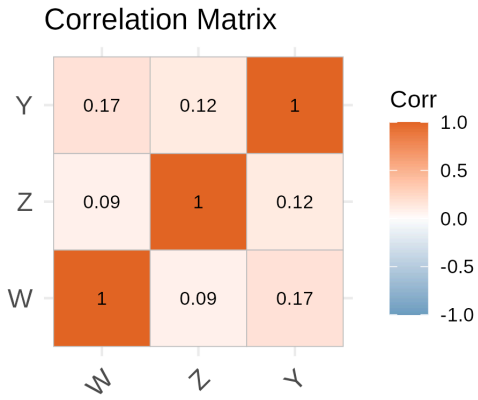

IV Contour Plot Croke (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
10.0
7.5
5.0
2.5
Correlation Matrix
Y
Z
W
-0.09
-0.09
1
0.2
1
-0.09
1
0.2
-0.09
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

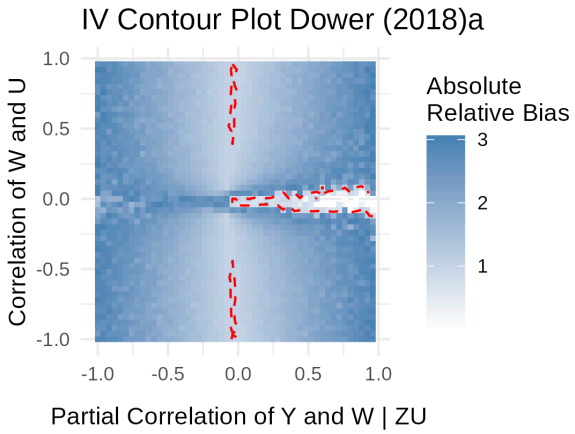

IV Contour Plot Delao (2013)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
30
20
10

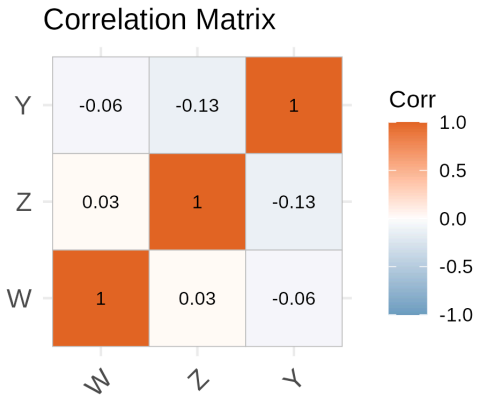

Correlation Matrix
Y
Z
W
0.08
0.03
1
0.52
1
0.03
1
0.52
0.08
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

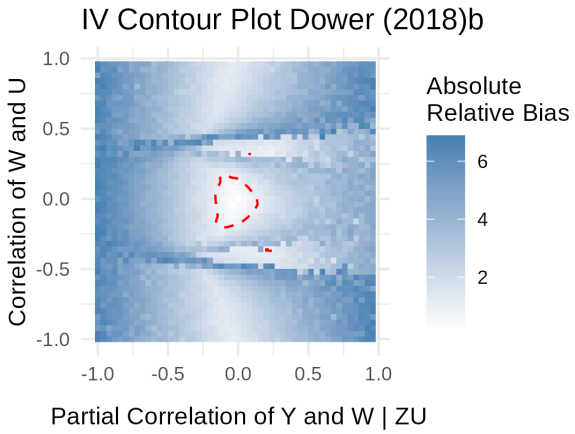

IV Contour Plot Dietrich (2015)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
7.5
5.0
2.5
Correlation Matrix
Y
Z
W
0.08
0.07
1
0.15
1
0.07
1
0.15
0.08
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

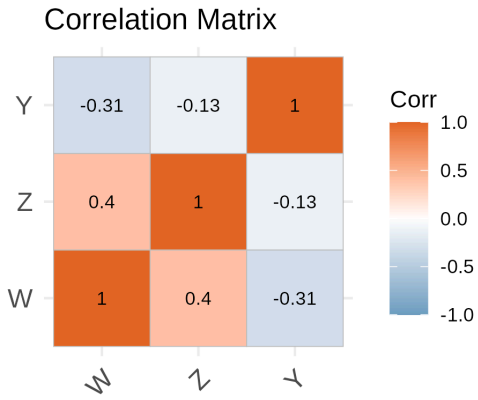

IV Contour Plot Digiuseppe (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
3
2
1

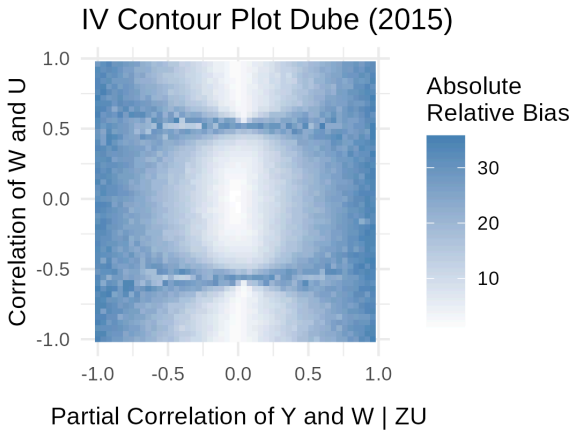

Correlation Matrix
Y
Z
W
0.17
0.12
1
0.09
1
0.12
1
0.09
0.17
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

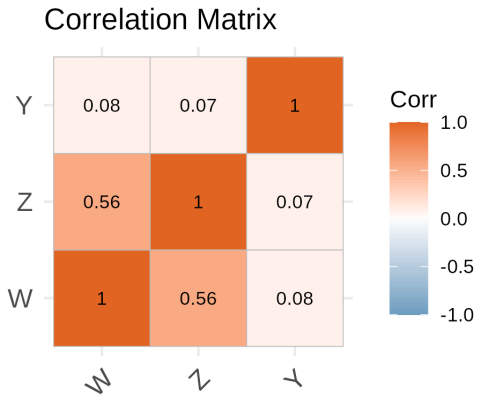

IV Contour Plot Dower (2018)a
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
3
2
1

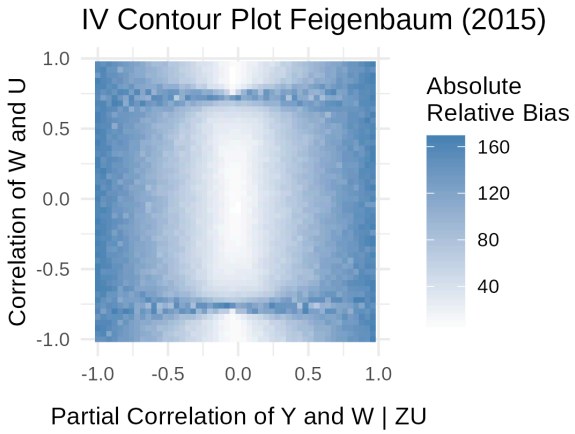

Correlation Matrix
Y
Z
W
-0.06
-0.13
1
0.03
1
-0.13
1
0.03
-0.06
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

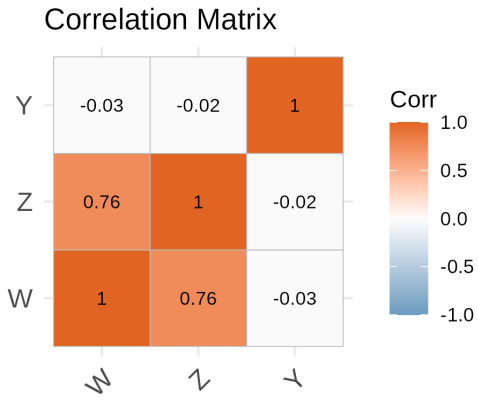

IV Contour Plot Dower (2018)b
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
6
4
2

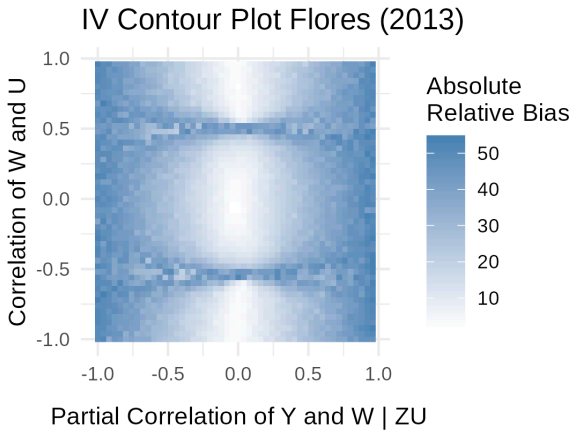

Correlation Matrix
Y
Z
W
-0.31
-0.13
1
0.4
1
-0.13
1
0.4
-0.31
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

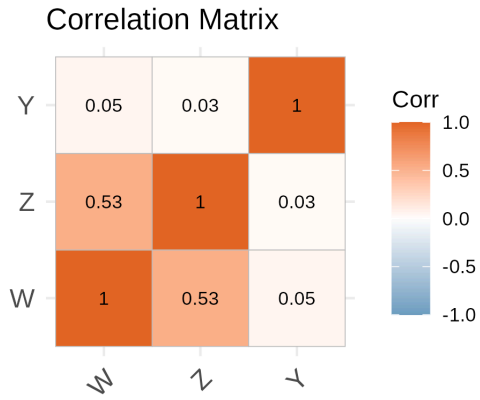

IV Contour Plot Dube (2015)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
30
20
10

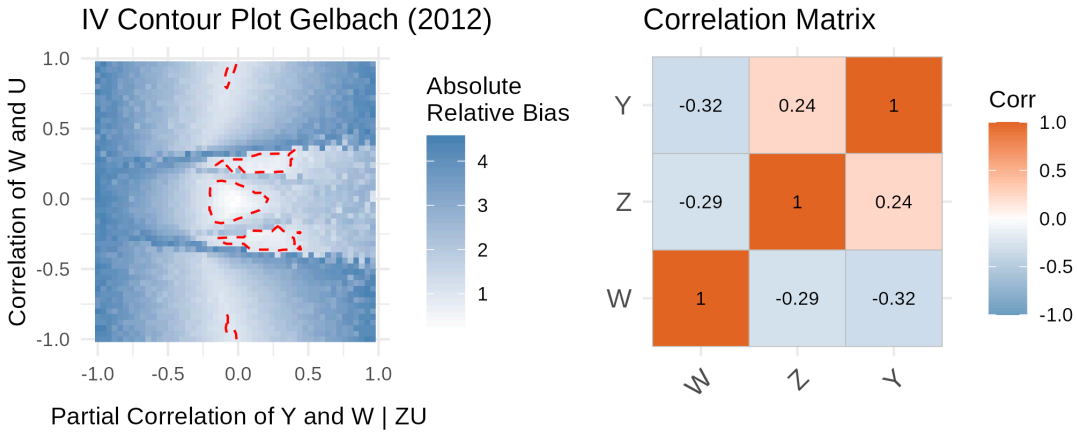

Correlation Matrix
Y
Z
W
0.08
0.07
1
0.56
1
0.07
1
0.56
0.08
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

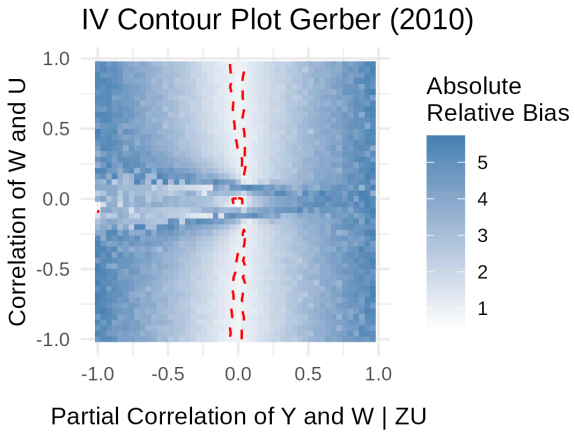

IV Contour Plot Feigenbaum (2015)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
160
120
80
40

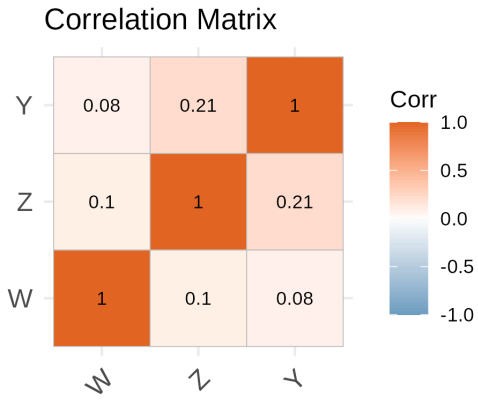

Correlation Matrix
Y
Z
W
-0.03
-0.02
1
0.76
1
-0.02
1
0.76
-0.03
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

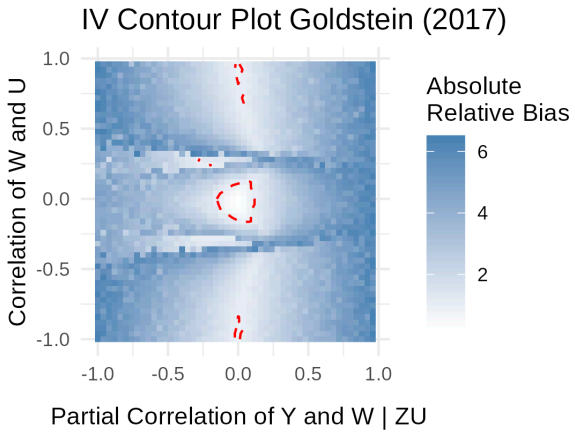

IV Contour Plot Flores (2013)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
50
40
30
20
10

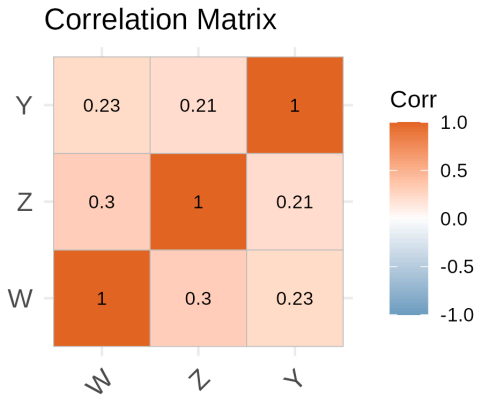

Correlation Matrix
Y
Z
W
0.05
0.03
1
0.53
1
0.03
1
0.53
0.05
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

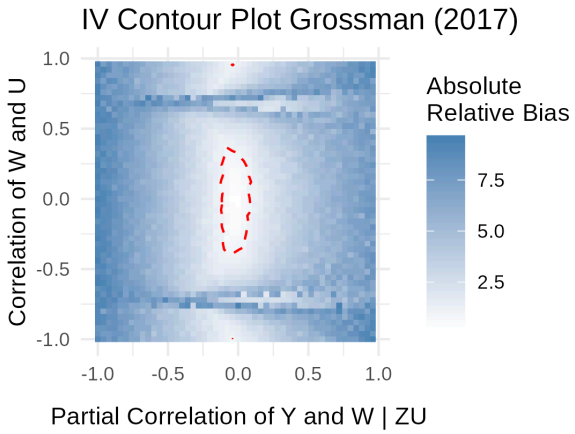

IV Contour Plot Gelbach (2012)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
4
3
2
1
Correlation Matrix
Y
Z
W
-0.32
0.24
1
-0.29
1
0.24
1
-0.29
-0.32
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

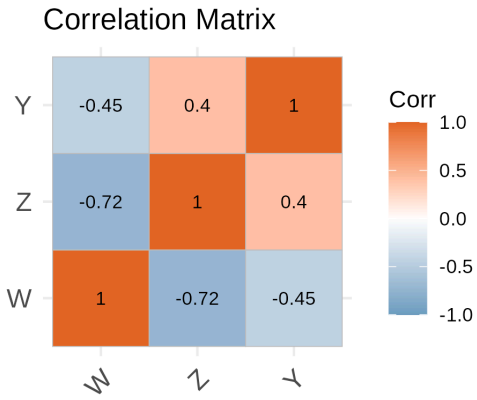

IV Contour Plot Gerber (2010)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
5
4
3
2
1

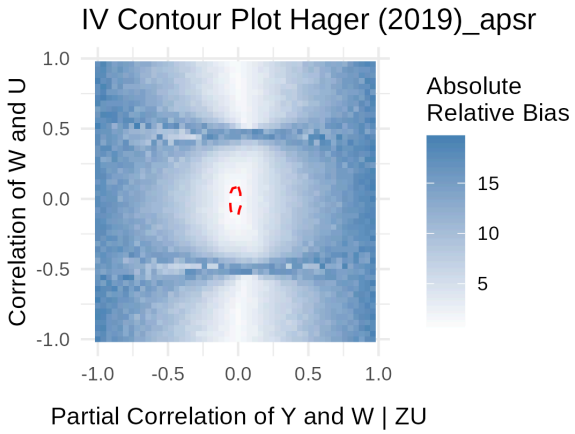

Correlation Matrix
Y
Z
W
0.08
0.21
1
0.1
1
0.21
1
0.1
0.08
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

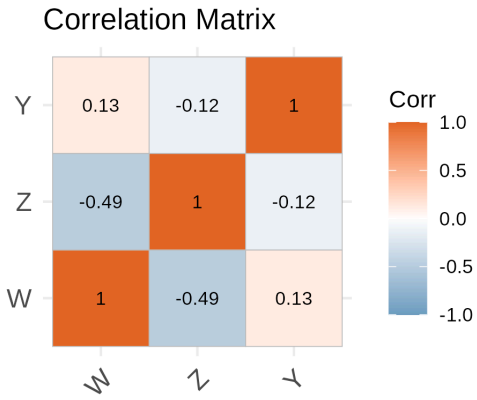

IV Contour Plot Goldstein (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
6
4
2

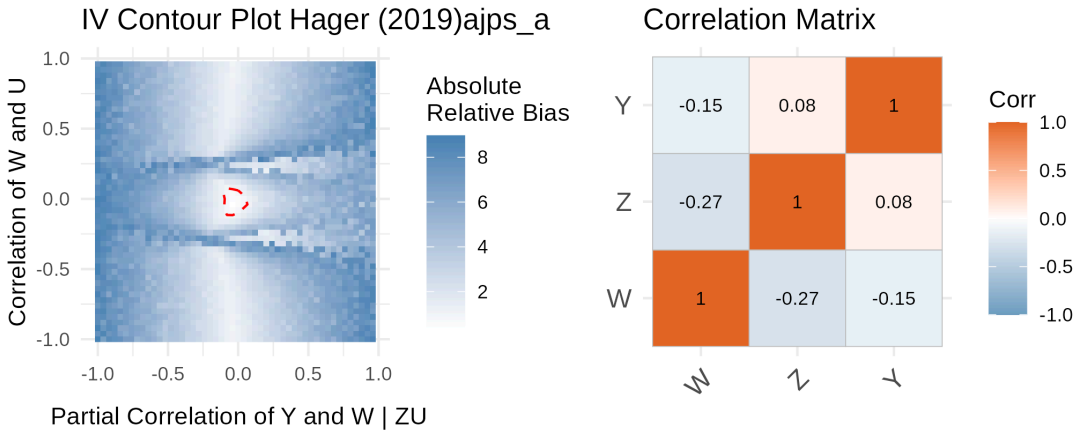

Correlation Matrix
Y
Z
W
0.23
0.21
1
0.3
1
0.21
1
0.3
0.23
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

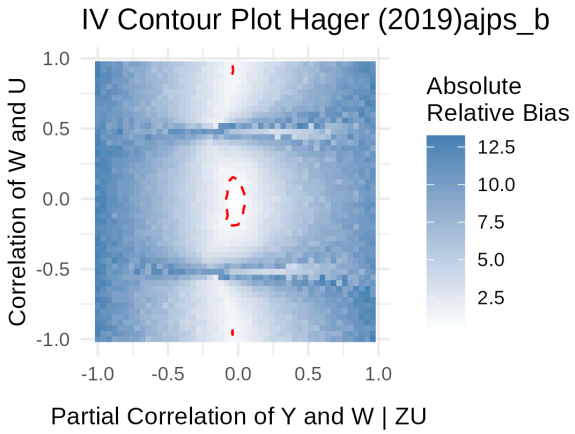

IV Contour Plot Grossman (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
7.5
5.0
2.5

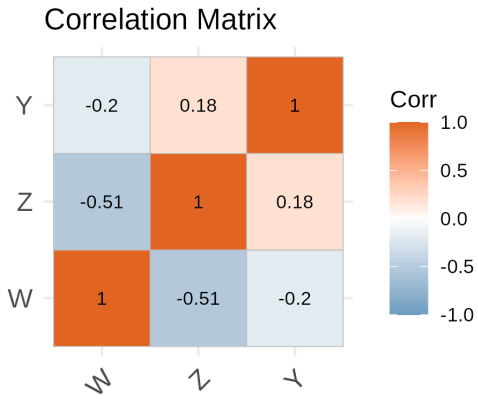

Correlation Matrix
Y
Z
W
-0.45
0.4
1
-0.72
1
0.4
1
-0.72
-0.45
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

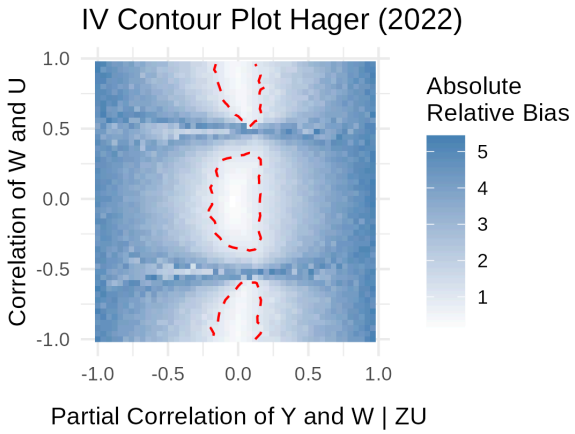

IV Contour Plot Hager (2019)_apsr
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
15
10
5

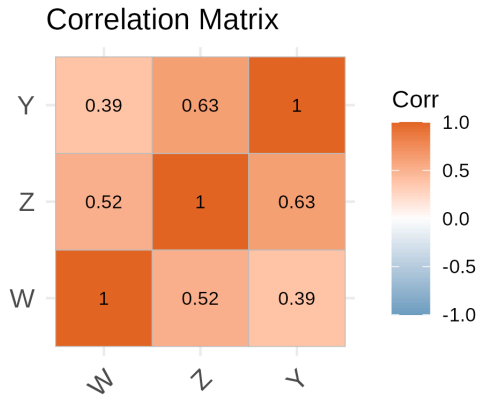

Correlation Matrix
Y
Z
W
0.13
-0.12
1
-0.49
1
-0.12
1
-0.49
0.13
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

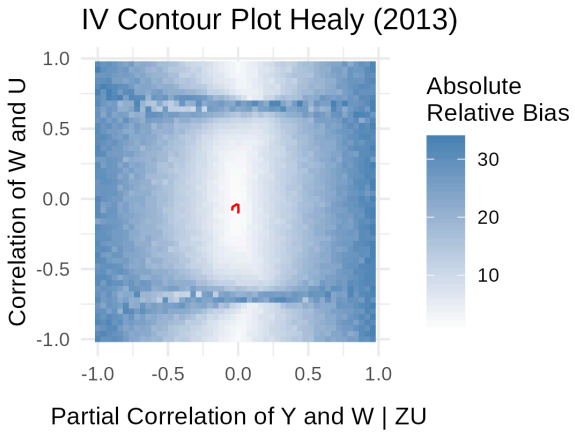

IV Contour Plot Hager (2019)ajps_a
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
8
6
4
2
Correlation Matrix
Y
Z
W
-0.15
0.08
1
-0.27
1
0.08
1
-0.27
-0.15
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

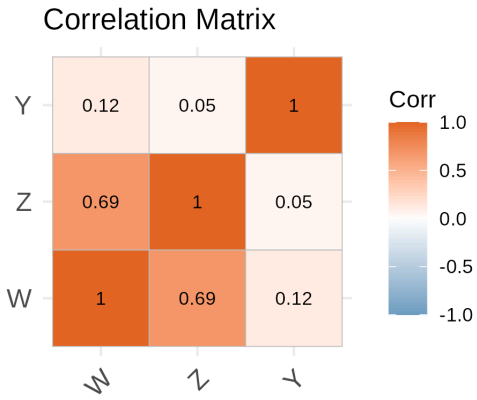

IV Contour Plot Hager (2019)ajps_b
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
12.5
10.0
7.5
5.0
2.5

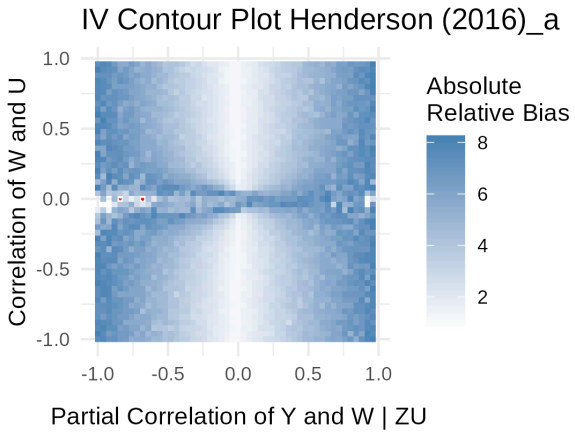

Correlation Matrix
Y
Z
W
-0.2
0.18
1
-0.51
1
0.18
1
-0.51
-0.2
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

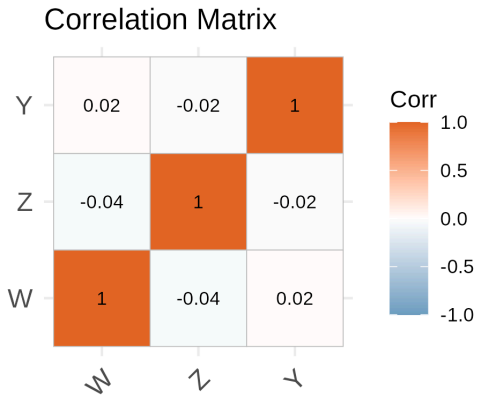

IV Contour Plot Hager (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
5
4
3
2
1

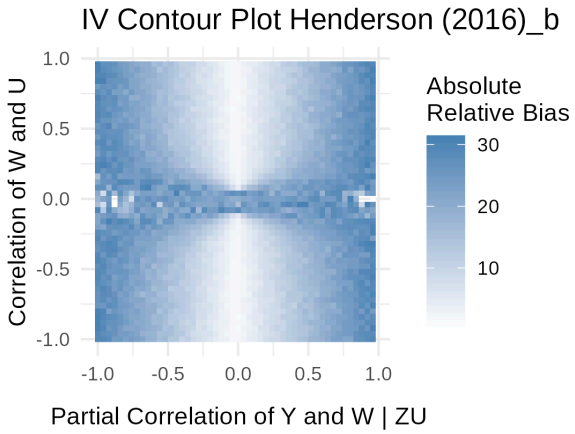

Correlation Matrix
Y
Z
W
0.39
0.63
1
0.52
1
0.63
1
0.52
0.39
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

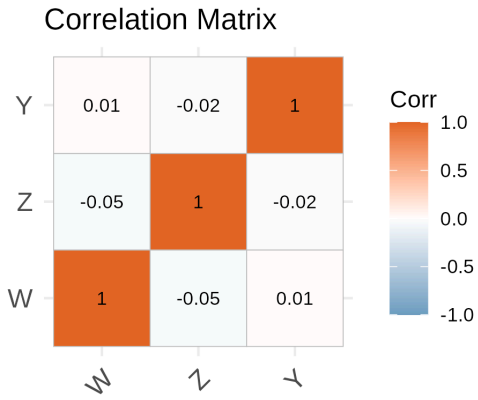

IV Contour Plot Healy (2013)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
30
20
10

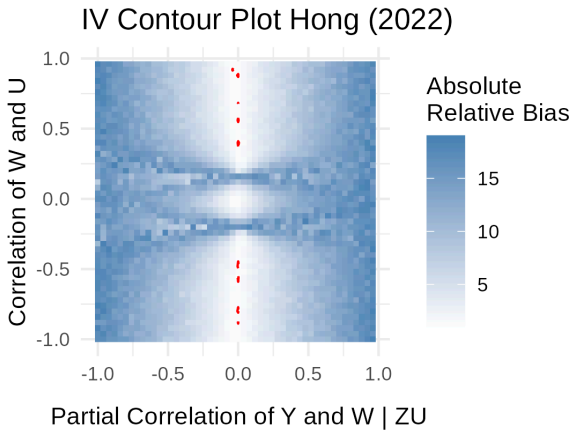

Correlation Matrix
Y
Z
W
0.12
0.05
1
0.69
1
0.05
1
0.69
0.12
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

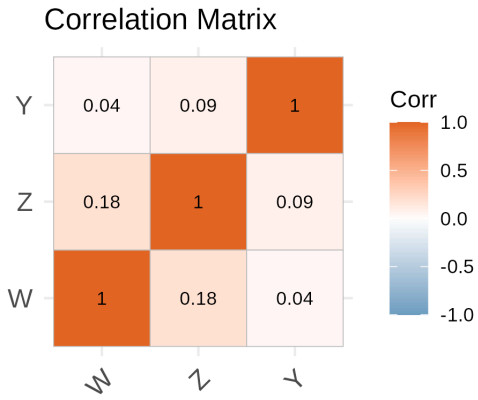

IV Contour Plot Henderson (2016)_a
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
8
6
4
2

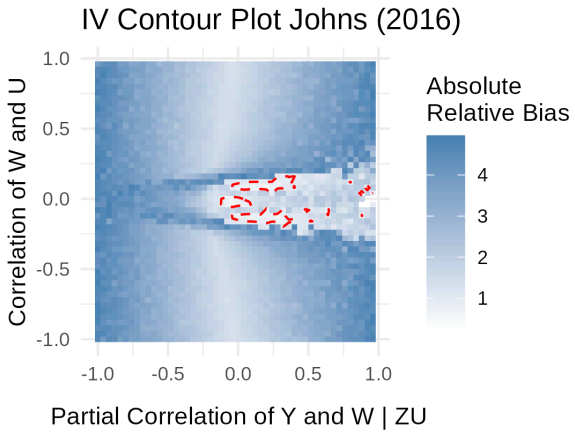

Correlation Matrix
Y
Z
W
0.02
-0.02
1
-0.04
1
-0.02
1
-0.04
0.02
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

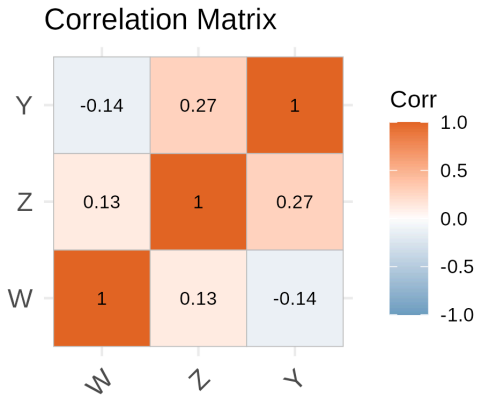

IV Contour Plot Henderson (2016)_b
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
30
20
10

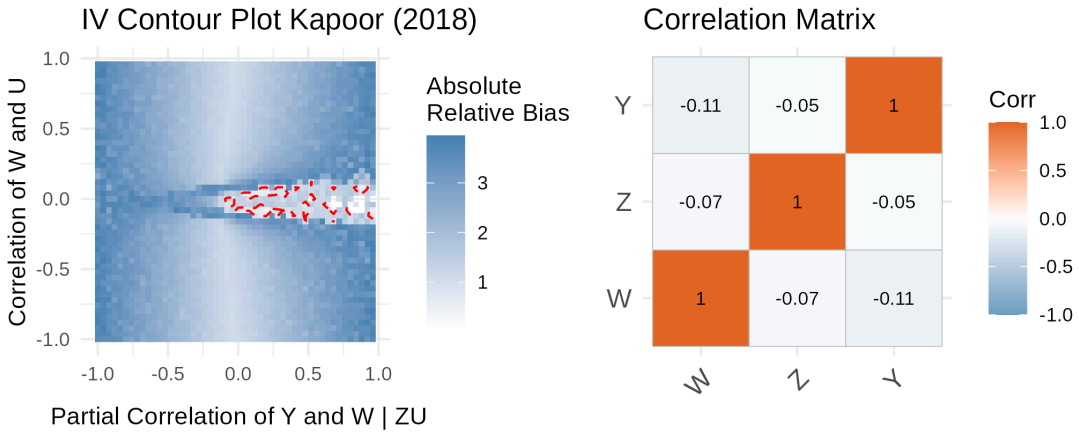

Correlation Matrix
Y
Z
W
0.01
-0.02
1
-0.05
1
-0.02
1
-0.05
0.01
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

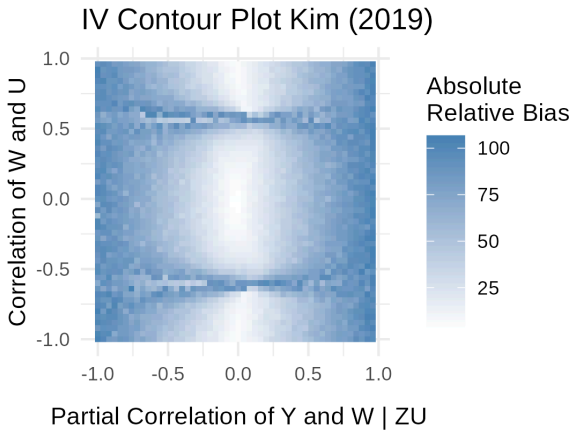

IV Contour Plot Hong (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
15
10
5

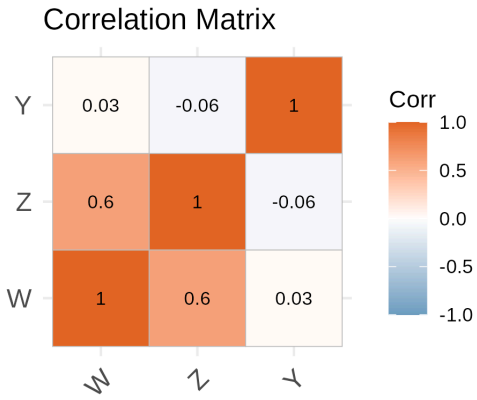

Correlation Matrix
Y
Z
W
0.04
0.09
1
0.18
1
0.09
1
0.18
0.04
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

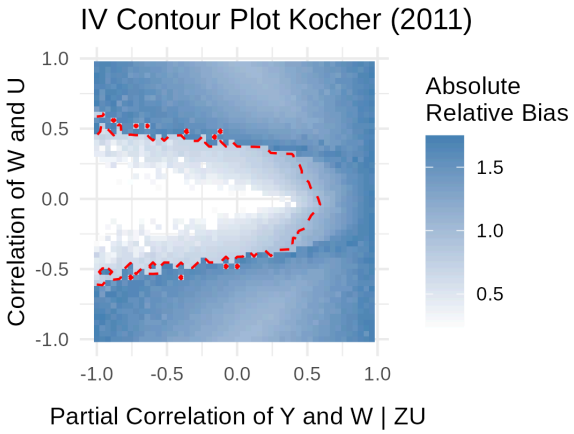

IV Contour Plot Johns (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
4
3
2
1

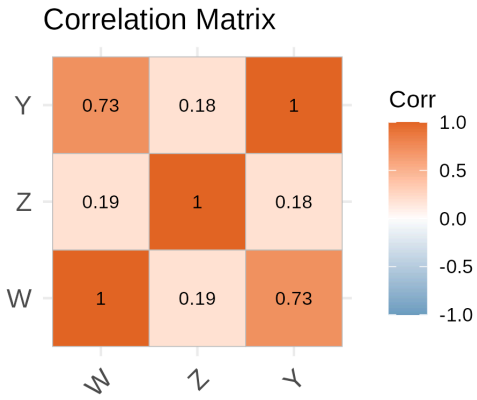

Correlation Matrix
Y
Z
W
-0.14
0.27
1
0.13
1
0.27
1
0.13
-0.14
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

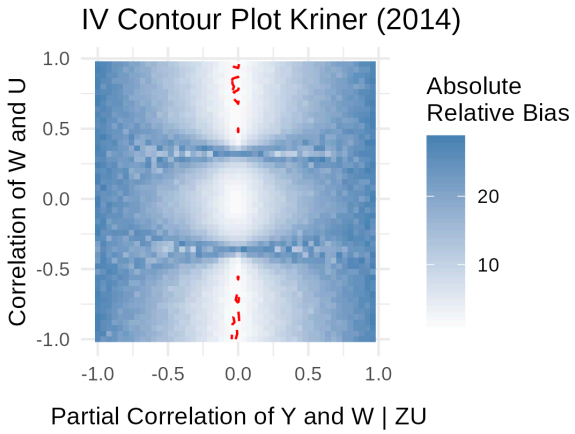

IV Contour Plot Kapoor (2018)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
3
2
1
Correlation Matrix
Y
Z
W
-0.11
-0.05
1
-0.07
1
-0.05
1
-0.07
-0.11
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

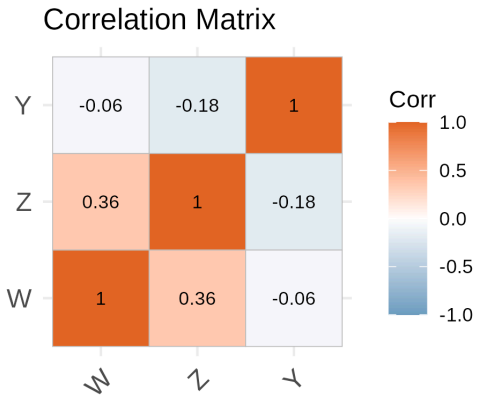

IV Contour Plot Kim (2019)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
100
75
50
25

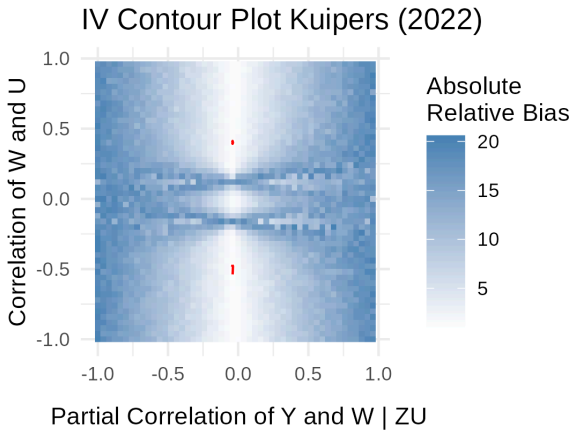

Correlation Matrix
Y
Z
W
0.03
-0.06
1
0.6
1
-0.06
1
0.6
0.03
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

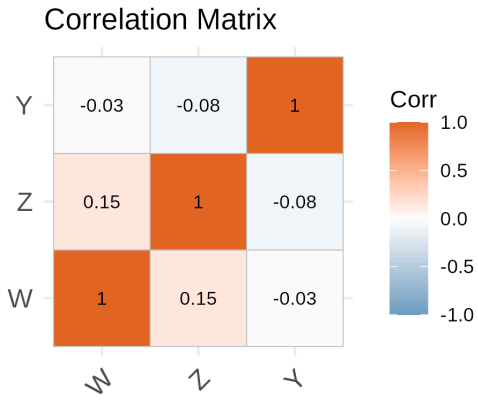

IV Contour Plot Kocher (2011)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
1.5
1.0
0.5

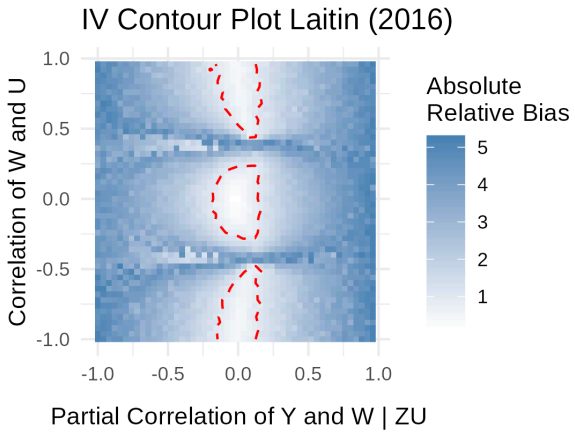

Correlation Matrix
Y
Z
W
0.73
0.18
1
0.19
1
0.18
1
0.19
0.73
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

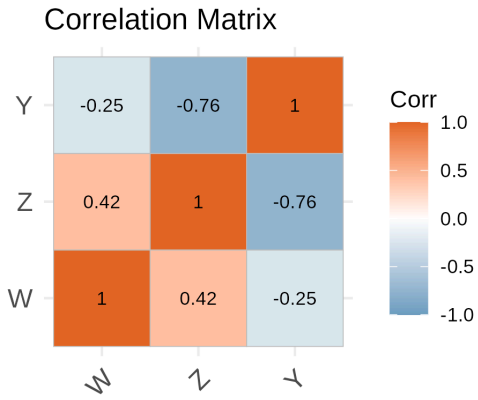

IV Contour Plot Kriner (2014)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
20
10

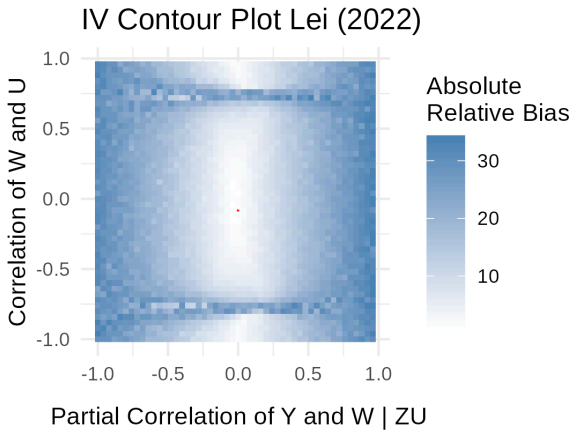

Correlation Matrix
Y
Z
W
-0.06
-0.18
1
0.36
1
-0.18
1
0.36
-0.06
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

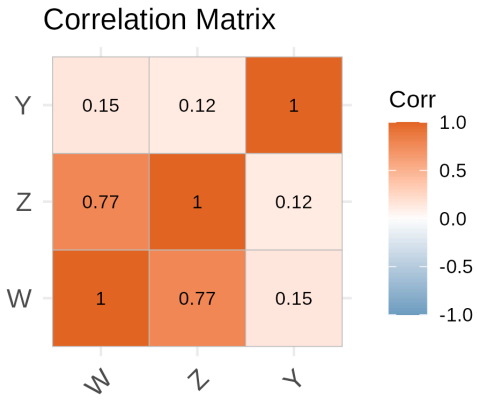

IV Contour Plot Kuipers (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
20
15
10
5

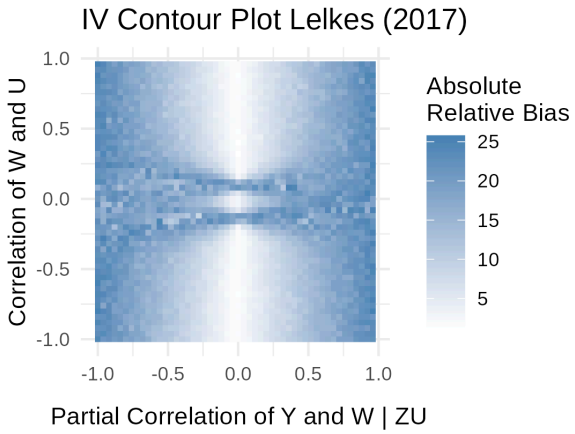

Correlation Matrix
Y
Z
W
-0.03
-0.08
1
0.15
1
-0.08
1
0.15
-0.03
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

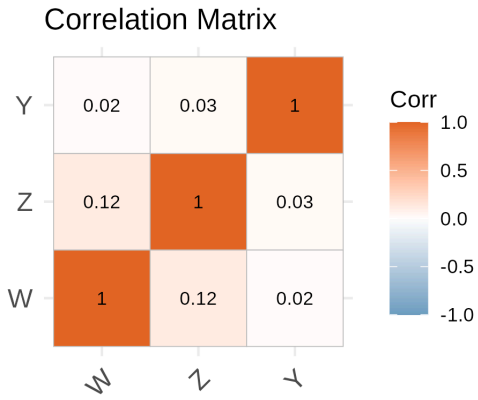

IV Contour Plot Laitin (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
5
4
3
2
1

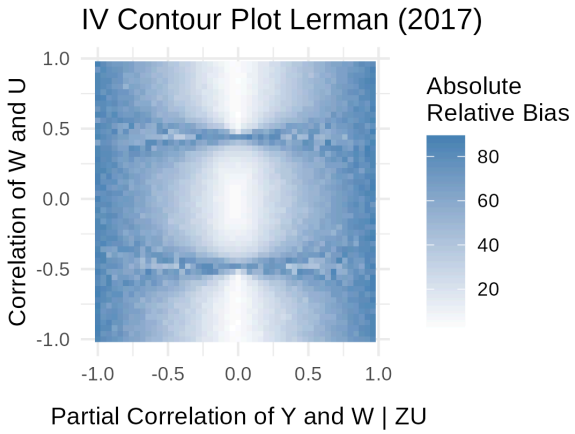

Correlation Matrix
Y
Z
W
-0.25
-0.76
1
0.42
1
-0.76
1
0.42
-0.25
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

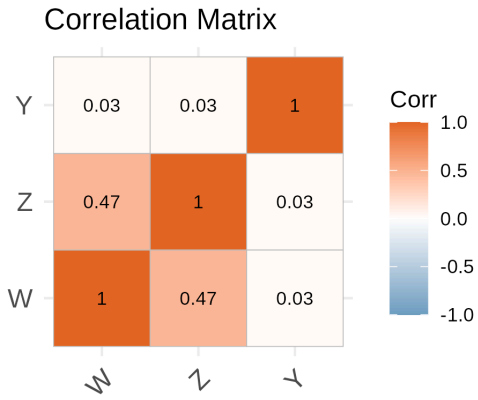

IV Contour Plot Lei (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
30
20
10

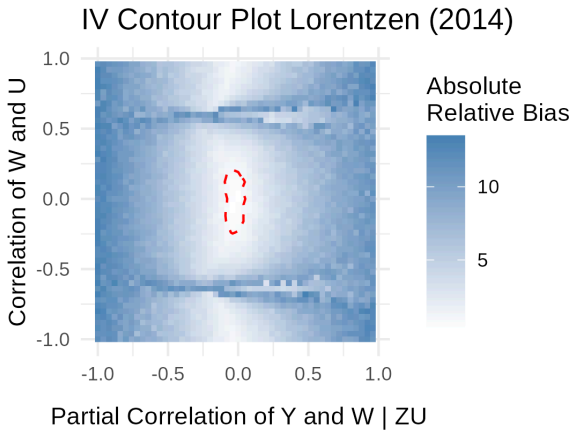

Correlation Matrix
Y
Z
W
0.15
0.12
1
0.77
1
0.12
1
0.77
0.15
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

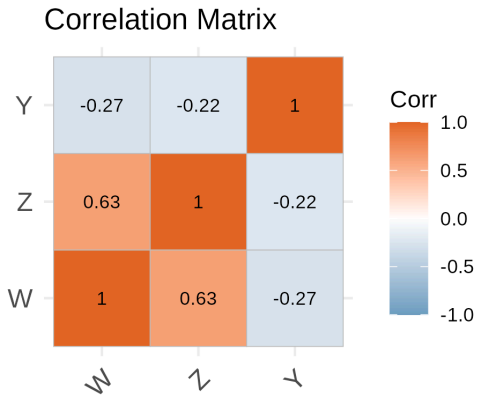

IV Contour Plot Lelkes (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
25
20
15
10
5

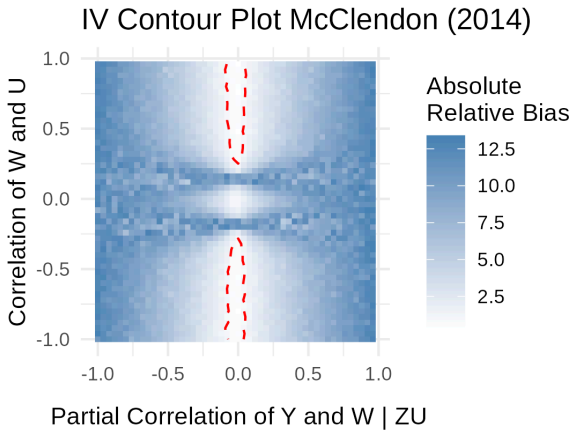

Correlation Matrix
Y
Z
W
0.02
0.03
1
0.12
1
0.03
1
0.12
0.02
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

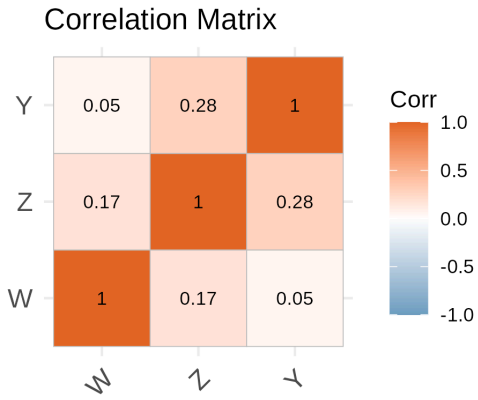

IV Contour Plot Lerman (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
80
60
40
20

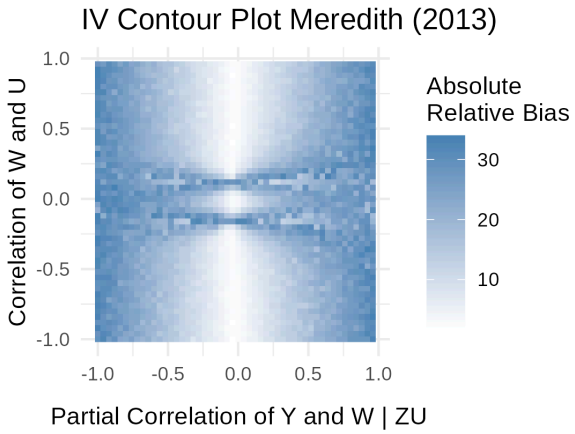

Correlation Matrix
Y
Z
W
0.03
0.03
1
0.47
1
0.03
1
0.47
0.03
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

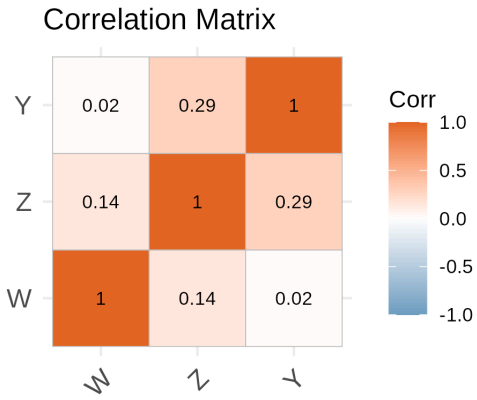

IV Contour Plot Lorentzen (2014)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
10
5

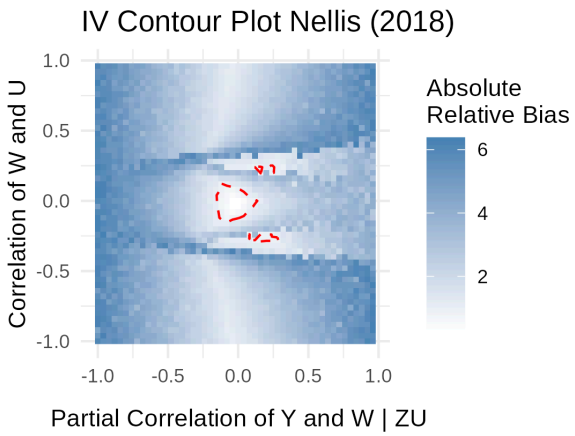

Correlation Matrix
Y
Z
W
-0.27
-0.22
1
0.63
1
-0.22
1
0.63
-0.27
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

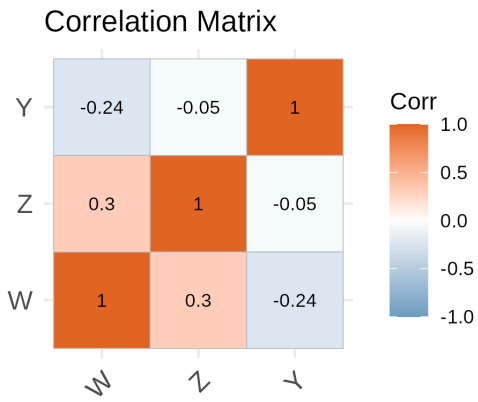

IV Contour Plot McClendon (2014)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
12.5
10.0
7.5
5.0
2.5

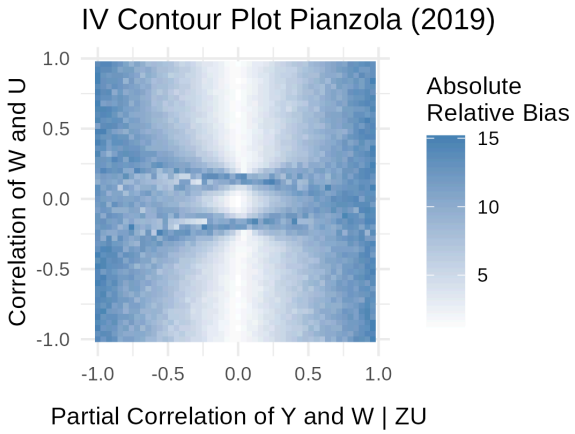

Correlation Matrix
Y
Z
W
0.05
0.28
1
0.17
1
0.28
1
0.17
0.05
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

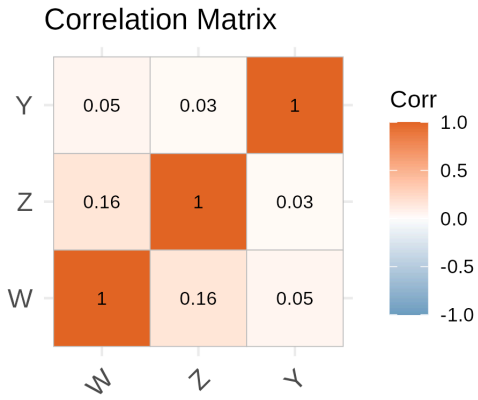

IV Contour Plot Meredith (2013)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
30
20
10

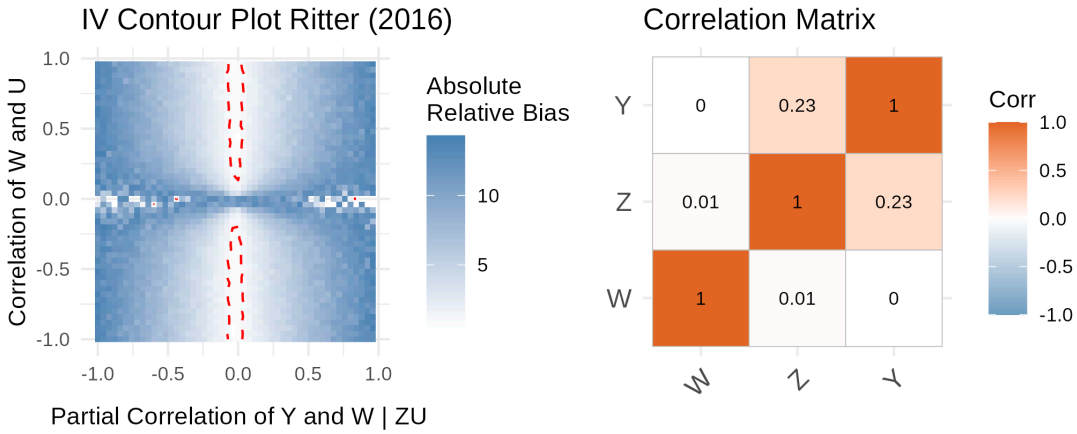

Correlation Matrix
Y
Z
W
0.02
0.29
1
0.14
1
0.29
1
0.14
0.02
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

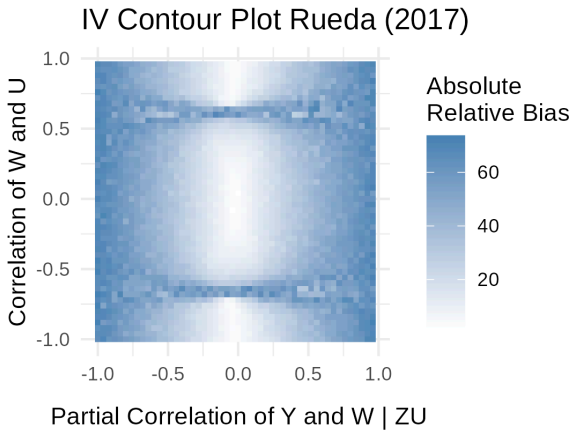

IV Contour Plot Nellis (2018)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
6
4
2

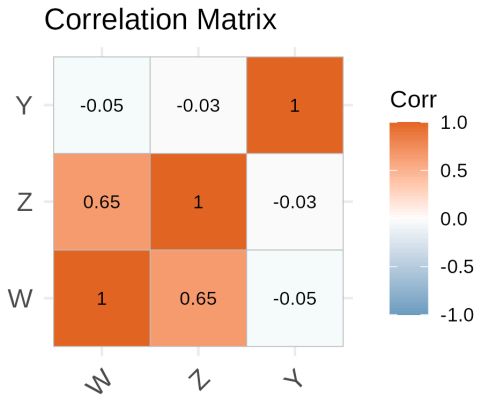

Correlation Matrix
Y
Z
W
-0.24
-0.05
1
0.3
1
-0.05
1
0.3
-0.24
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

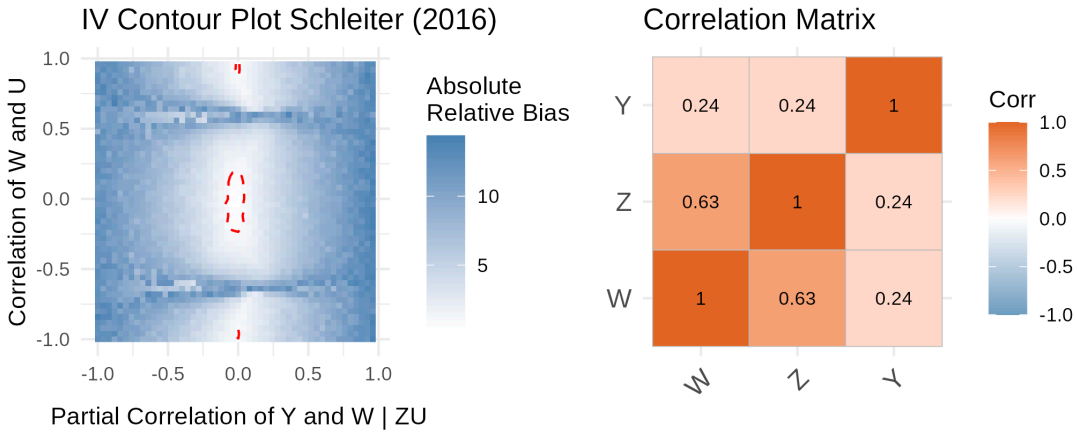

IV Contour Plot Pianzola (2019)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
15
10
5

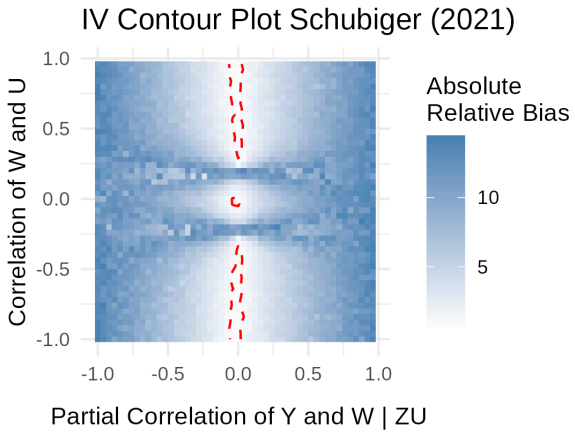

Correlation Matrix
Y
Z
W
0.05
0.03
1
0.16
1
0.03
1
0.16
0.05
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

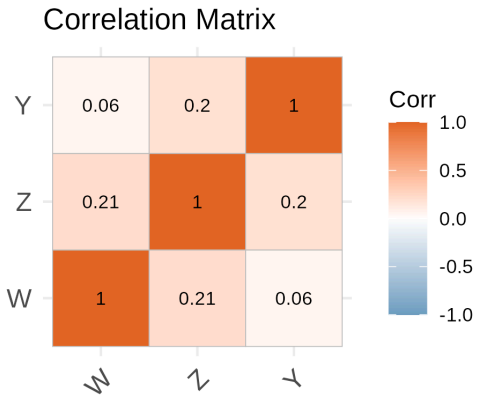

IV Contour Plot Ritter (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
10
5
Correlation Matrix
Y
Z
W
0
0.23
1
0.01
1
0.23
1
0.01
0
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

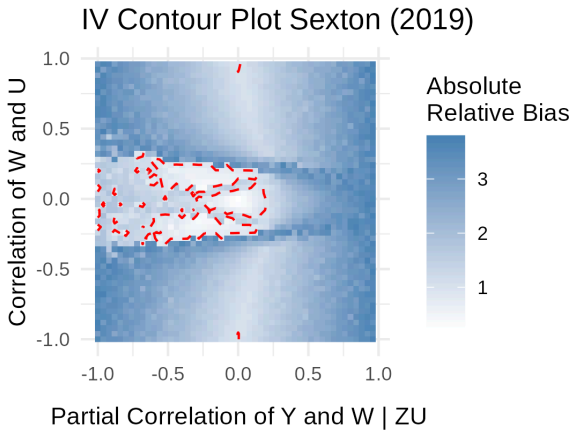

IV Contour Plot Rueda (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
60
40
20

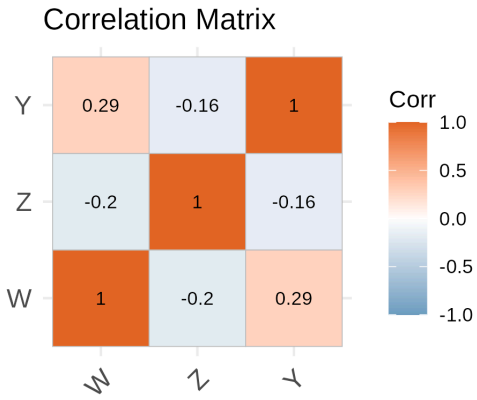

Correlation Matrix
Y
Z
W
-0.05
-0.03
1
0.65
1
-0.03
1
0.65
-0.05
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

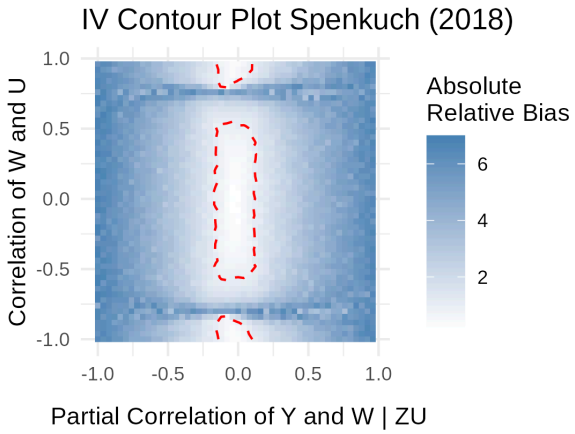

IV Contour Plot Schleiter (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
10
5
Correlation Matrix
Y
Z
W
0.24
0.24
1
0.63
1
0.24
1
0.63
0.24
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

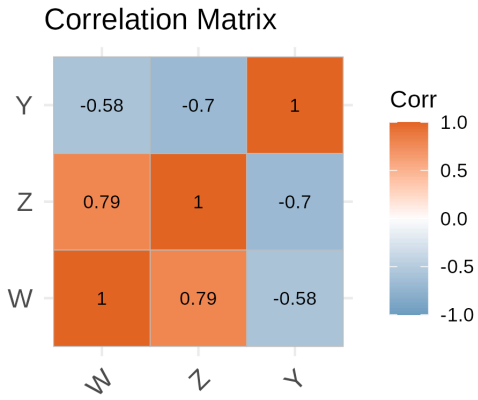

IV Contour Plot Schubiger (2021)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
10
5

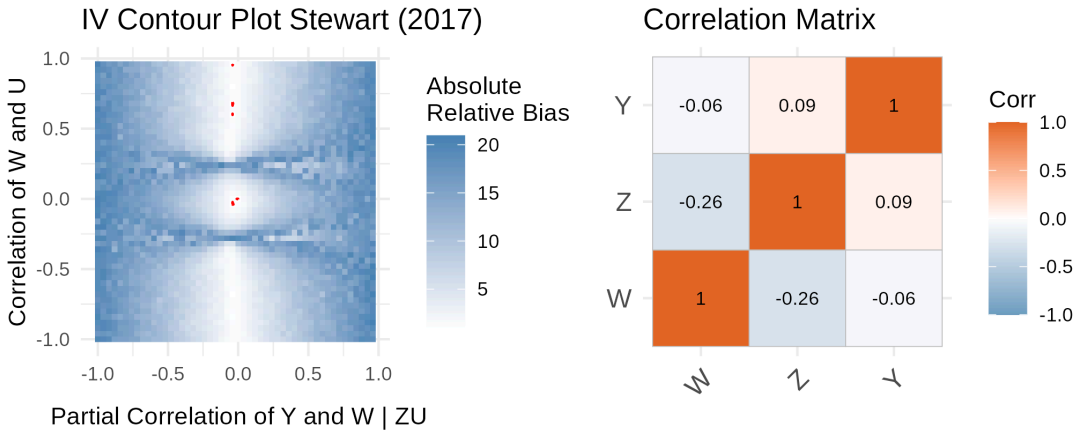

Correlation Matrix
Y
Z
W
0.06
0.2
1
0.21
1
0.2
1
0.21
0.06
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

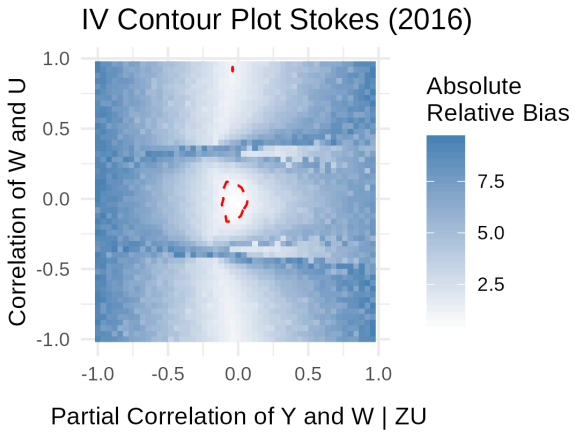

IV Contour Plot Sexton (2019)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
3
2
1

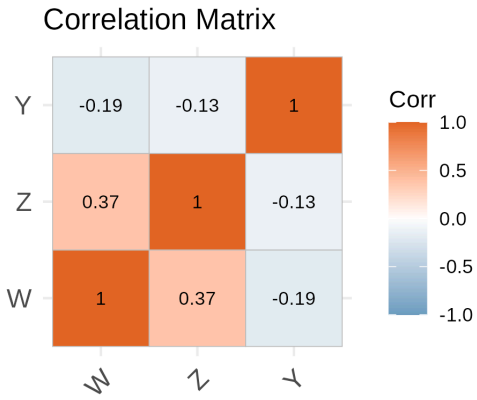

Correlation Matrix
Y
Z
W
0.29
-0.16
1
-0.2
1
-0.16
1
-0.2
0.29
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

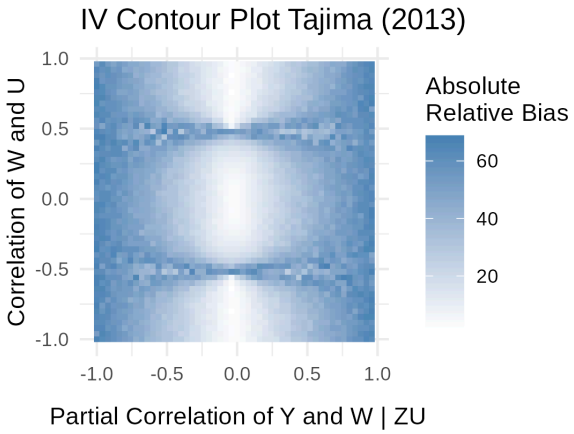

IV Contour Plot Spenkuch (2018)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
6
4
2

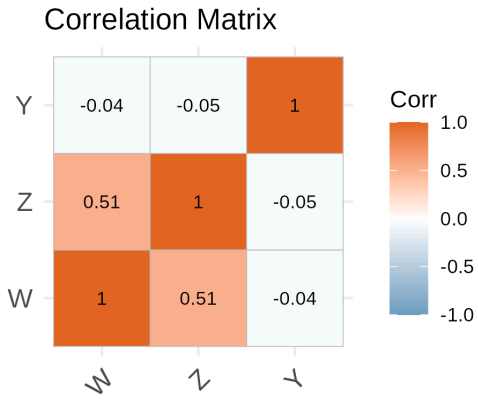

Correlation Matrix
Y
Z
W
-0.58
-0.7
1
0.79
1
-0.7
1
0.79
-0.58
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

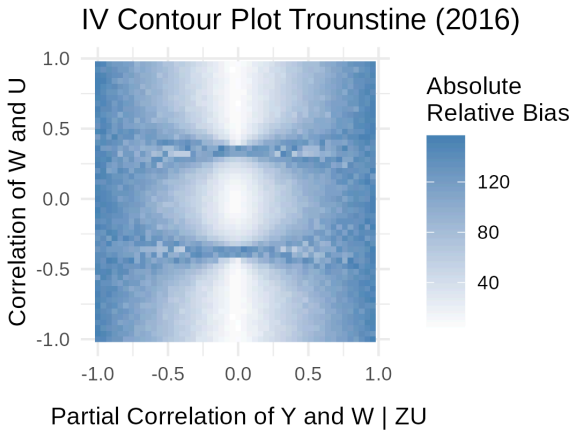

IV Contour Plot Stewart (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
20
15
10
5
Correlation Matrix
Y
Z
W
-0.06
0.09
1
-0.26
1
0.09
1
-0.26
-0.06
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

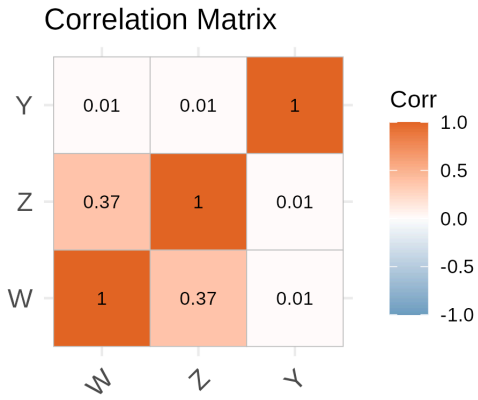

IV Contour Plot Stokes (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
7.5
5.0
2.5

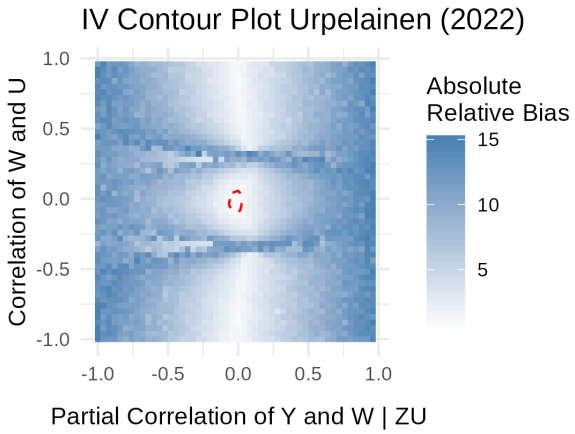

Correlation Matrix
Y
Z
W
-0.19
-0.13
1
0.37
1
-0.13
1
0.37
-0.19
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

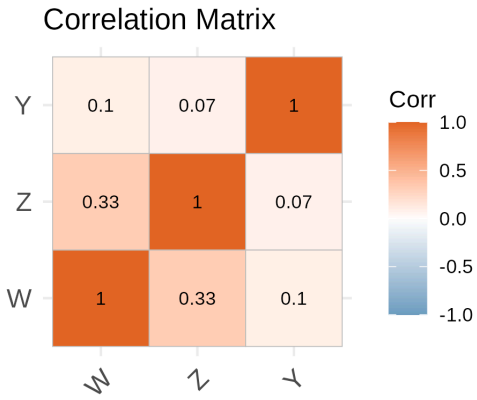

IV Contour Plot Tajima (2013)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
60
40
20

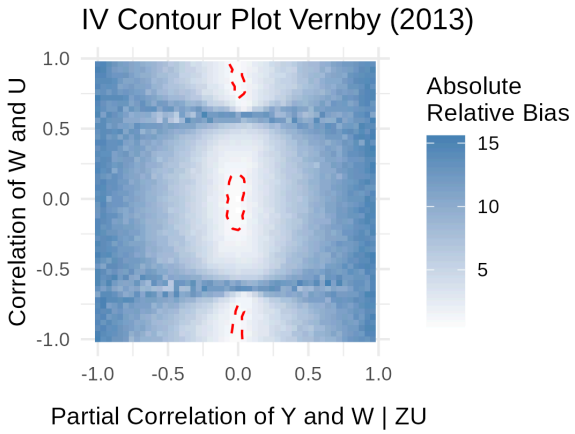

Correlation Matrix
Y
Z
W
-0.04
-0.05
1
0.51
1
-0.05
1
0.51
-0.04
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

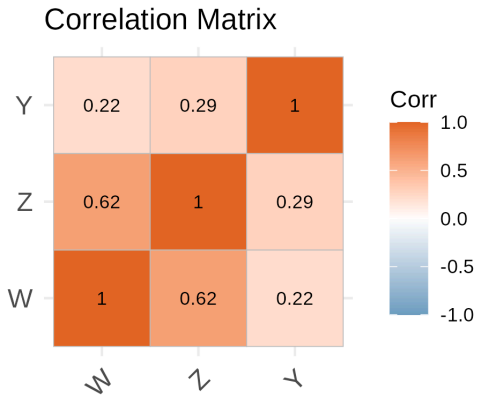

IV Contour Plot Trounstine (2016)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
120
80
40

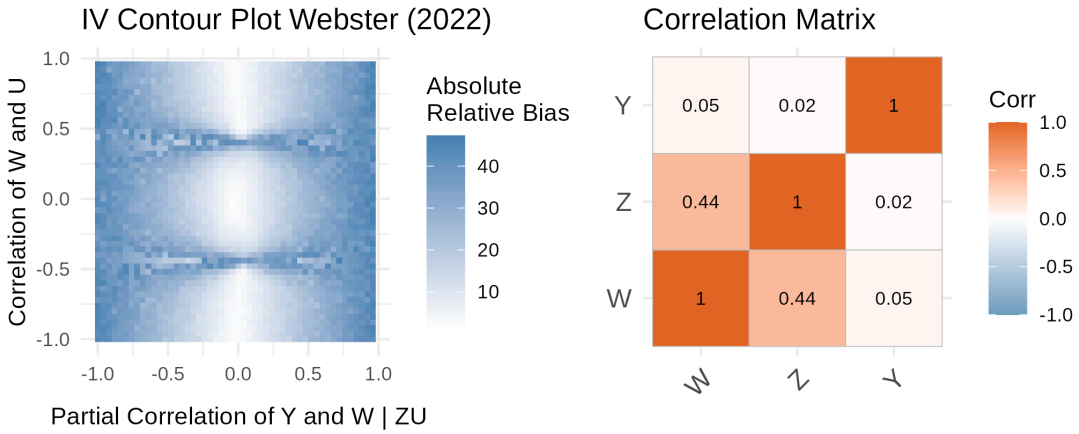

Correlation Matrix
Y
Z
W
0.01
0.01
1
0.37
1
0.01
1
0.37
0.01
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

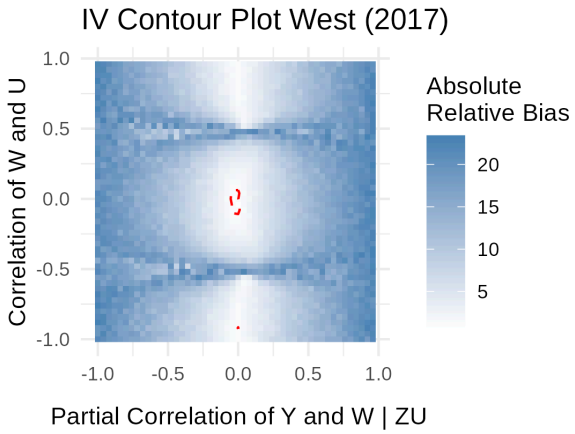

IV Contour Plot Urpelainen (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
15
10
5

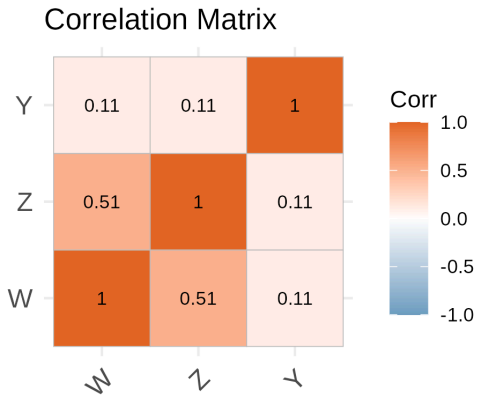

Correlation Matrix
Y
Z
W
0.1
0.07
1
0.33
1
0.07
1
0.33
0.1
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

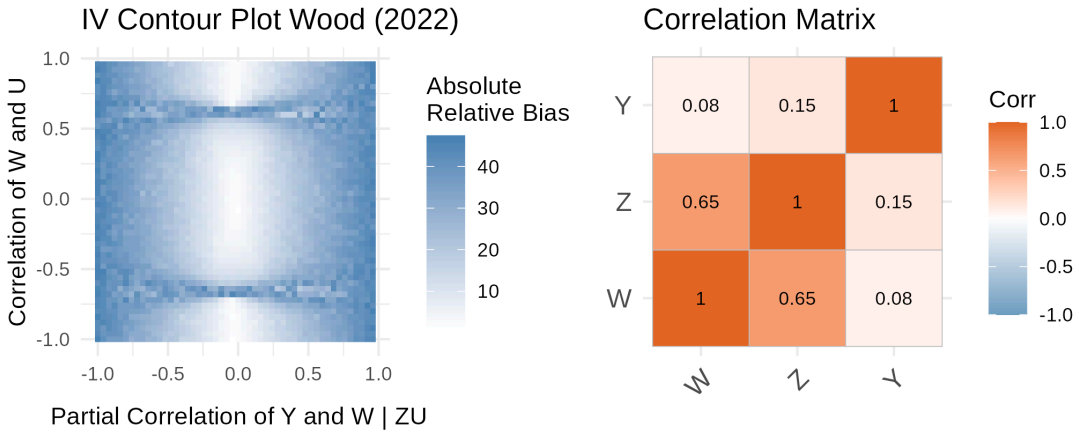

IV Contour Plot Vernby (2013)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
15
10
5

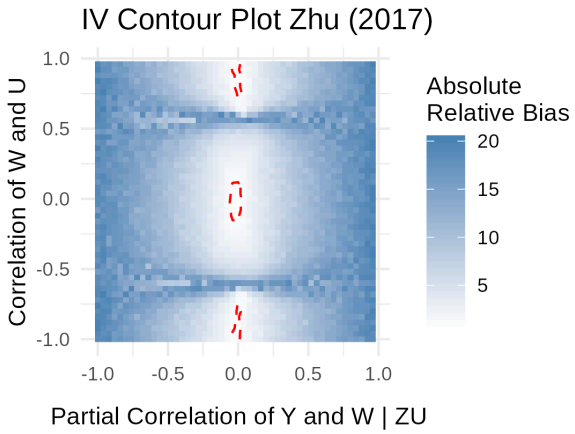

Correlation Matrix
Y
Z
W
0.22
0.29
1
0.62
1
0.29
1
0.62
0.22
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

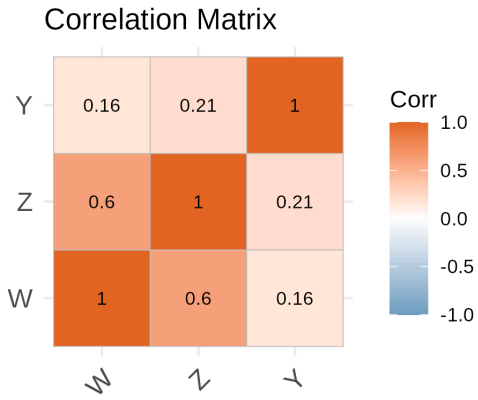

IV Contour Plot Webster (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
40
30
20
10
Correlation Matrix
Y
Z
W
0.05
0.02
1
0.44
1
0.02
1
0.44
0.05
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0

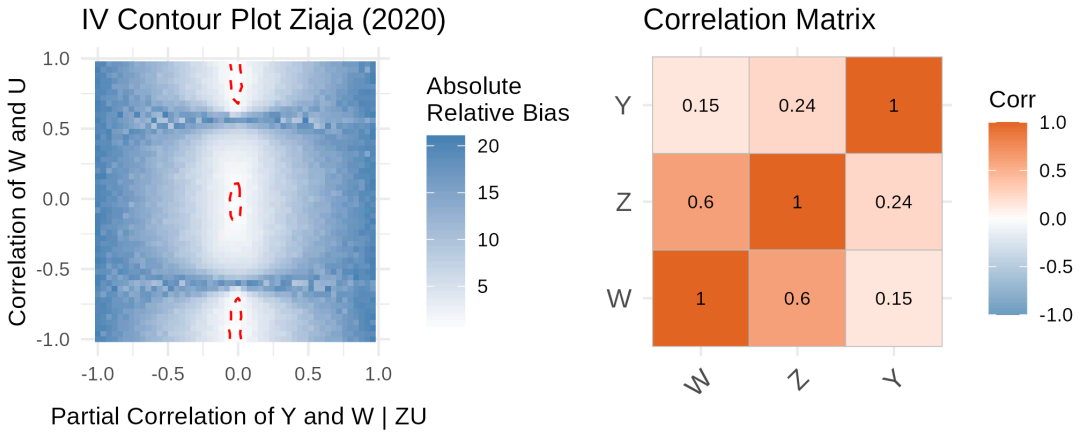

IV Contour Plot West (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
20
15
10
5


Correlation Matrix
Y
Z
W
0.11
0.11
1
0.51
1
0.11
1
0.51
0.11
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0


IV Contour Plot Wood (2022)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
40
30
20
10
Correlation Matrix
Y
Z
W
0.08
0.15
1
0.65
1
0.15
1
0.65
0.08
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0


IV Contour Plot Zhu (2017)
Correlation of W and U
1.0
0.5
0.0
-0.5
-1.0
-1.0
-0.5
0.0
0.5
1.0
Partial Correlation of Y and W | ZU
Absolute
Relative Bias
20
15
10
5


Correlation Matrix
Y
Z
W
0.16
0.21
1
0.6
1
0.21
1
0.6
0.16
W
Z
Y
Corr
1.0
0.5
0.0
-0.5
-1.0



Bibliography

[1] A. Lal, M. Lockhart, Y. Xu, and Z. Zu, "How Much Should We Trust Instrumental Variable Estimates in Political Science? Practical Advice Based on 67 Replicated Studies," *Political Analysis*, vol. 32, no. 4, pp. 521–540, May 2024, doi: 10.1017/pan.2024.2.

[2] M. Huang and C. McCartan, "Relative Bias Under Imperfect Identification in Observational Causal Inference." [Online]. Available: https://arxiv.org/abs/2507.23743